

Created on....01.08.2018 Last changed on/by..... 14.05.2019 \$KPHUTANE
Author.....\$KPHUTANE

Title.....Honeywell Interface Block employees with invalid Certificate

Type.....1 Reporting
Status.....
Application.....
Class.....
Logical database.....PN
 from Application....P Human resources
Selection screen.....
Authorization group...
Package.....ZHR1

Start by Variants
Deactivate editor lock
X Fixed point arithmetic active

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1  *&-----*
2  *& Report ZHR_HONEYWELL_INTERFACE
3  *&-----*
4  *& Author       : Kundan K. Phutane
5  *&
6  *& Requirements : Specified Sishen employees without a Fire Fighting
7  *&                and/or First Aid qualification/s (none or expired),
8  *&                should be blocked from entering the mine premises,
9  *&                which poses risks in terms of safety according to the
10 *&                Mine Health and Safety Act: #all supervisors under the
11 *&                age of 50, must have a valid first aid certificate,
12 *&                within a year of his/her appointment in the supervisor
13 Y
14 *&                role, at the mine" (Refer to Regulation 24.7).
15 *& Charm        : Charm2000002362
16 *& TR           : DS1K9A0B7J
17 *& Date         : 01.08.2018
18
19 *&-----*
20 REPORT zrppa0088.
21
22 TABLES: pernr.
23
24 INFOTYPES: 0001,0002,0008.
25
26 DATA : lt_final TYPE TABLE OF zpapa0088,
27         ls_final TYPE zpapa0088.
28
29 DATA: lt_obj  LIKE  hrsobid OCCURS 0 WITH HEADER LINE,
30       lt_prof LIKE hrpe_profq OCCURS 0 WITH HEADER LINE,
31       ls_prof TYPE hrpe_profq.
32
33 TYPES : BEGIN OF ly_hrp1001,
34         objid TYPE hrp1001-objid,
35         sobid TYPE hrp1001-sobid,
36         begda TYPE hrp1001-begda,
37         endda TYPE hrp1001-endda,
38         END OF ly_hrp1001.
39
40 DATA : lt_hrp1001 TYPE TABLE OF ly_hrp1001,
41       ls_hrp1001 TYPE ly_hrp1001.
42
43 DATA: lv_begda TYPE sy-datum,
44       lv_endda TYPE sy-datum,
45       lv_age   TYPE  CMP_NOYRS,
46       lv_year  TYPE sy-datum,
47       lv_pernr TYPE p0001-pernr.
48
49 DATA: lf_saferep(1) TYPE c,
50       lf_faid(1)    TYPE c,
51       lf_firef(1)   TYPE c,
52       lf_40days(1)  TYPE c,
53       lf_del(1)      TYPE c.
54
55 DATA : lv_40date TYPE sy-datum.
56
57 DATA: lt_fcat  TYPE slis_t_fieldcat_alv,
58       ls_fcat  TYPE slis_fieldcat_alv,
59       ls_layout TYPE slis_layout_alv.
60

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61  ls_layout-zebra = 'X' .
62  ls_layout-colwidth_optimize = 'X' .
63
64  "IDOC data
65  DATA:
66      ls_final2          TYPE zppa0088,
67      ls_idoc_control    LIKE edidc,
68      lc_invrev_segname(9) TYPE c VALUE 'Z1PP004',
69      lt_idoc_data       TYPE STANDARD TABLE OF edidd,
70      ls_idoc_data       TYPE edidd,
71      lt_comm_control    TYPE STANDARD TABLE OF edidc,
72      ls_comm_control    TYPE edidc,
73      lv_count          TYPE i.
74
75  TYPES : BEGIN OF ly_qual_date,
76          tbjid(8) TYPE c, ".TBJID,
77          datum    TYPE datum,
78          reason(1) TYPE c,
79      END OF ly_qual_date.
80
81  DATA : ls_qual_date TYPE ly_qual_date,
82          lt_qual_date TYPE TABLE OF ly_qual_date.
83
84  DATA : lv_records TYPE cmp_nodys.
85  DATA: lv_date TYPE sy-datum.
86
87  "For BADI Qualifcation
88  TYPES: BEGIN OF ly_exp_new,
89          objid TYPE objid,
90          begda TYPE begda,
91          endda TYPE endda,
92      END OF ly_exp_new.
93
94  DATA : ls_exp_new TYPE ly_exp_new,
95          lt_exp_new TYPE TABLE OF ly_exp_new.
96
97  "For BADI IT0007
98
99  TYPES : BEGIN OF ly_exp_0007,
100      pernr TYPE pernr-pernr,
101      begda TYPE begda,
102      endda TYPE endda,
103      data(250) TYPE c,
104      END OF ly_exp_0007.
105
106  DATA : ls_exp_0007 TYPE ly_exp_0007,
107          lt_exp_0007 TYPE TABLE OF ly_exp_0007.
108
109  TYPES : BEGIN OF ly_exp_2003,
110      pernr TYPE pernr-pernr,
111      subty TYPE subty,
112      begda TYPE begda,
113      endda TYPE endda,
114      data(250) TYPE c,
115      END OF ly_exp_2003.
116
117  DATA : ls_exp_2003 TYPE ly_exp_2003,
118          lt_exp_2003 TYPE TABLE OF ly_exp_2003.
119
120  TYPES : BEGIN OF ly_exp_2004,
121      pernr TYPE pernr-pernr,

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122          subty      TYPE subty,
123          begda      TYPE begda,
124          enddda     TYPE enddda,
125          data(250) TYPE c,
126          END OF ly_exp_2004.
127
128 DATA : ls_exp_2004 TYPE ly_exp_2004,
129          lt_exp_2004 TYPE TABLE OF ly_exp_2004.
130
131 TYPES : BEGIN OF ly_exp_2002,
132          pernr      TYPE pernr-pernr,
133          subty      TYPE subty,
134          begda      TYPE begda,
135          enddda     TYPE enddda,
136          data(250) TYPE c,
137          END OF ly_exp_2002.
138
139 DATA : ls_exp_2002 TYPE ly_exp_2002,
140          lt_exp_2002 TYPE TABLE OF ly_exp_2002.
141
142
143 TYPES : BEGIN OF ly_exp_2007,
144          pernr      TYPE pernr-pernr,
145          subty      TYPE subty,
146          begda      TYPE begda,
147          enddda     TYPE enddda,
148          data(250) TYPE c,
149          END OF ly_exp_2007.
150
151 DATA : ls_exp_2007 TYPE ly_exp_2007,
152          lt_exp_2007 TYPE TABLE OF ly_exp_2007.
153
154 TYPES : BEGIN OF ly_exp_2001,
155          pernr      TYPE pernr-pernr,
156          subty      TYPE subty,
157          begda      TYPE begda,
158          enddda     TYPE enddda,
159          data(250) TYPE c,
160          END OF ly_exp_2001.
161
162 DATA : ls_exp_2001 TYPE ly_exp_2001,
163          lt_exp_2001 TYPE TABLE OF ly_exp_2001.
164
165
166 DATA : lf_info(4) TYPE c.
167
168 *****
169 """"""time
170 DATA: lv_begda1 TYPE sy-datum,
171          lv_endda1 TYPE sy-datum.
172
173 DATA : lv_bdate TYPE sy-datum.
174
175 DATA : lv_def_beg TYPE sy-uzeit,
176          lv_def_end TYPE sy-uzeit.
177
178 lv_def_beg = '000000'.
179 lv_def_end = '235959'.
180
181 DATA: lv_seconds TYPE i.
182 lv_seconds = 2 * 3600."Before 2 hours
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183
184 DATA : lf_night(1) TYPE c.
185
186 TYPES :BEGIN OF ly_list_pernr,
187         pernr TYPE pernr-pernr,
188         status(1) TYPE c,
189         eg TYPE p0001-persg,
190         psg TYPE p0001-persk,
191     END OF ly_list_pernr.
192
193 DATA : lt_list_pernr TYPE TABLE OF ly_list_pernr,
194
195         ls_list_pernr TYPE ly_list_pernr.
196
197 TYPES: BEGIN OF ly_position,
198         pernr TYPE pernr-pernr,
199         plans TYPE p0001-plans,
200         eg TYPE p0001-persg,
201     END OF ly_position.
202
203 DATA: lt_position TYPE TABLE OF ly_position,
204         ls_position TYPE ly_position.
205 *data: lt_temp TYPE TABLE OF ZPPA0088,
206 *      ls_temp TYPE ZPPA0088.
207
208 DATA : lt_time_t TYPE TABLE OF zppa0088,
209         ls_time_t TYPE zppa0088.
210
211 DATA : ls_final_avail TYPE pdpsp.
212
213
214 DATA : lv_def_beg1 TYPE sy-uzeit,
215         lv_def_end1 TYPE sy-uzeit.
216
217 DATA : lv_day_before TYPE sy-datum.
218
219
220 *****
221
222 DATA : lv_line TYPE i.
223 DATA :lv_index TYPE sy-index.
224 DATA : lf_before(1) TYPE c,
225         lf_after(1) TYPE c.
226 TYPES : BEGIN OF ty_temp_time,
227         pernr TYPE pernr-pernr,
228         begda TYPE begda,
229         endda TYPE endda,
230         begty TYPE tims,
231         endty TYPE tims,
232     END OF ty_temp_time.
233 DATA : lt_temp_time TYPE TABLE OF ty_temp_time,
234         ls_temp_time TYPE ty_temp_time.
235
236
237 *****
238 DATA : lt_pa2007 TYPE TABLE OF pa2007,
239         ls_pa2007 TYPE pa2007.
240 *****
241 ** ** PERFORM check_2001."Check for leave 10170
242 DATA : lt_pa2001 TYPE TABLE OF pa2001,
243         ls_pa2001 TYPE pa2001.

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244
245 "*****"
246
247 SELECTION-SCREEN : BEGIN OF BLOCK b1 WITH FRAME TITLE TEXT-001 ##TEXT_PO
OL.
248 "SELECTION SCREEN NEEDS TO BE.... 1ST IN PROG
249 PARAMETERS : p_check AS CHECKBOX DEFAULT 'X',
250               p_rad1 RADIOBUTTON GROUP rb1 DEFAULT 'X' USER-COMMAND usr,
251               p_rad2 RADIOBUTTON GROUP rb1.
252
253 SELECTION-SCREEN END OF BLOCK b1.
254
255
256 START-OF-SELECTION.
257
258     PERFORM select_queries.
259
260 GET pernr.
261
262     PERFORM select_data."Valid employees
263
264 END-OF-SELECTION.
265
266     PERFORM badi_call.
267
268     IF lf_info IS INITIAL."Not from IT
269         PERFORM fill_data."Blocked/Unblocked As per Qualification
270
271 ***   Stage II
272 *     PERFORM CHECK_TIME."0007 ,2003 ,2007
273 *     PERFORM CHECK_2004."Availability Standby
274 *     PERFORM CHECK_2002."Attendances Callout
275 *     PERFORM ADD_QUAL_TIME."Block and Unblock
276
277     ENDIF.
278
279     CASE lf_info.
280
281         "Badi Changes Qualification
282         WHEN '0024'.
283             PERFORM fill_badi_data.
284
285         "Badi Changes 0007
286         *     WHEN '0007'.
287         *         PERFORM BADI_0007.
288         *
289         *     WHEN '2003'.
290         *         PERFORM badi_2003.
291         **
292         *     WHEN '2004'.
293         *         PERFORM BADI_2004.
294         *
295         *     WHEN '2002'.
296         *         PERFORM badi_2002.
297         *
298         *     WHEN '2007'.
299         *         PERFORM badi_2007.
300         *
301         *     WHEN '2001'.
302         *         PERFORM badi_2001.
303
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304     ENDCASE.
305
306
307
308     IF p_rad1 EQ 'X'.
309         PERFORM field_cat."Field Catalog
310         PERFORM disp_data."ALV
311
312     ELSEIF p_rad2 EQ 'X'.
313         IF p_check NE 'X'.
314             PERFORM send_idoc.
315
316         ELSEIF p_check EQ 'X'.
317             CLEAR : lv_count.
318             lv_count = lines( lt_final ).
319             WRITE: 'Number of Employees will be blocked : ' ##NO_TEXT,lv_count
320
321         ENDIF.
322
323     ENDIF.
324
325
326 *-----*
327 *      Form  SELECT_DATA
328 *-----*
329 *      text
330 *-----*
331 *  —>  p1      text
332 *  <—  p2      text
333 *-----*
334 FORM select_data .
335
336     rp-provide-from-last p0001 space pn-begda pn-endda.
337     IF pnp-sw-found NE 1. CLEAR p0001. ENDIF.
338
339     rp-provide-from-last p0008 space pn-begda pn-endda.
340     IF pnp-sw-found NE 1. CLEAR p0008. ENDIF.
341
342     ls_position-pernr = p0001-pernr.
343     ls_position-plans = p0001-plans.
344     ls_position-eg = p0001-persg.
345     APPEND ls_position TO lt_position.
346     CLEAR:ls_position.
347
348     IF p0001-persg = '1' .
349
350         IF p0008-trfgr = 'GBF4' OR p0008-trfgr = 'GBF5' OR p0008-trfgr = 'GB
351 F6'.
352             "Exclude
353             ELSE.
354                 ls_list_pernr-pernr = pernr-pernr.
355                 ls_list_pernr-eg = p0001-persg.
356                 ls_list_pernr-psg = p0001-persk.
357                 ls_list_pernr-status = 'U'.
358                 APPEND ls_list_pernr TO lt_list_pernr.
359             ENDIF.
360
361         ELSEIF p0001-persg = '2'.
362             ls_list_pernr-pernr = pernr-pernr.

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363     ls_list_pernr-eg = p0001-persg.
364     ls_list_pernr-psg = p0001-persk.
365     ls_list_pernr-status = 'U'.
366     APPEND ls_list_pernr TO lt_list_pernr.
367
368     ELSEIF p0001-persg = '3'.
369
370         IF p0008-trfgr = 'GBF4' OR p0008-trfgr = 'GBF5'.
371         ELSE.
372             ls_final-zz_pernr = pernr-pernr.
373             ls_final-zz_eg = p0001-persg.
374             ls_final-zz_psg = p0001-persk.
375         ENDIF.
376
377     ELSEIF p0001-persg = '4'.
378
379     *   READ TABLE LT_HRP1001 INTO LS_HRP1001 WITH KEY OBJID = P0001-PLANS.
380     "Jaisy
381     "A012 relationship if maintained
382     *   IF P0008-TRFGR = 'GBF4' OR P0008-TRFGR = 'GBF5'.
383     *   else.
384     *-Added by K.Jaisy Augustine under TR-DS1K9A0CNX
385     *   IF SY-SUBRC = '0' and P0008-TRFGR ne 'GBF4' and P0008-TRFGR ne 'GBF
386     5' and P0008-TRFGR ne 'GBF6'.
387     IF p0008-trfgr NE 'GBF4' AND p0008-trfgr NE 'GBF5' AND p0008-trfgr N
388     E 'GBF6'.
389         ls_final-zz_pernr = pernr-pernr.
390         ls_final-zz_eg = p0001-persg.
391         ls_final-zz_psg = p0001-persk.
392     ELSE.
393         IF p0008-trfgr = 'GBF7' OR p0008-trfgr = 'GBF8'." OR P0008-TRFGR =
394         'GBF6'.
395             ls_list_pernr-pernr = pernr-pernr.
396             ls_list_pernr-eg = p0001-persg.
397             ls_list_pernr-psg = p0001-persk.
398             ls_list_pernr-status = 'U'.
399             APPEND ls_list_pernr TO lt_list_pernr.
400         ELSE.
401             "Exclude
402         ENDIF.
403     ENDIF.
404
405     ELSEIF p0001-persg = '8' OR p0001-persg = '9'.
406     ENDIF.
407
408     lv_begda = pn-begda.
409     lv_endda = pn-endda.
410
411     IF ls_final-zz_pernr IS NOT INITIAL.
412
413         CALL FUNCTION 'HR_99S_INTERVAL_BETWEEN_DATES'
414             EXPORTING
415                 begda = lv_begda
416                 endda = lv_endda
417             IMPORTING
418                 days = lv_count.
419

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420     lv_date = lv_begda.
421
422     DO lv_count TIMES.
423
424         ls_final-zz_begda = lv_date.
425         ls_final-zz_endda = lv_date.
426         lv_date = lv_date + 1.
427         ls_final-zz_begtym = '000000'.
428         ls_final-zz_endtym = '235959'.
429         lv_count = lv_count - 1.
430         APPEND ls_final TO lt_final .
431
432     ENDDO.
433
434     ENDIF.
435
436     CLEAR : ls_final.
437     lv_pernr = pernr-pernr.
438
439     ENDFORM.
440     *&-----*
441     *&      Form  FILL_DATA
442     *&-----*
443     *      text
444     *-----*
445     *  —>  p1      text
446     *  <—  p2      text
447     *-----*
448     FORM fill_data .
449
450     CALL FUNCTION 'RP_CALC_DATE_IN_INTERVAL'
451         EXPORTING
452             date      = sy-datum
453             days      = '40'
454             months    = '00'
455     *   SIGNUM      = '+'
456             years     = '00'
457         IMPORTING
458             calc_date = lv_40date.
459
460     LOOP AT lt_final INTO ls_final.
461
462         REFRESH: lt_obj, lt_prof.
463         lt_obj-plvar = '01'.
464         lt_obj-otype = 'P'.
465         lt_obj-sobid = ls_final-zz_pernr.
466         APPEND lt_obj.
467
468         CLEAR: lv_begda1,lv_endda1.
469         lv_begda1 = ls_final-zz_begda.
470         lv_endda1 = ls_final-zz_endda.
471
472         CLEAR :lf 40days.
473         CALL FUNCTION 'ZRHPP_Q_PROFILE_READ'
474             EXPORTING
475                 begda      = lv_begda1
476                 endda      = lv_endda1
477             TABLES
478                 objects    = lt_obj
479                 profile     = lt_prof
480             EXCEPTIONS

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481         no_authority      = 1
482         wrong_otype       = 2
483         object_not_found  = 3
484         undefined         = 4
485         OTHERS             = 5.
486
487     IF sy-subrc = 0.
488         CLEAR :lf_saferep,lf_faaid,lf_firef.
489
490         CLEAR:ls_prof.
491         READ TABLE lt_prof INTO ls_prof WITH KEY tbjid = '80010932' ."TRAN
SPORTING NO FIELDS .
492         IF sy-subrc = 0.
493             lf_saferep = 'X'."Safety Representatives qualification
494
495             ls_final-zz_80010932 = ls_prof-vendd.
496             *      IF LS_PROF-VENDD LE LV_40DATE.
497             *      LF_40DAYS = '4'.
498             *      ls_qual_date-TBJID = LS_PROF-TBJID.
499             *      ls_qual_date-DATUM = LS_PROF-VENDD.
500             *      ls_qual_date-REASON = '4'.
501             *      APPEND ls_qual_date to lt_qual_date.
502             *      CLEAR : ls_qual_date.
503             *      ENDIF.
504         ENDIF.
505
506         CLEAR:ls_prof.
507         READ TABLE lt_prof INTO ls_prof WITH KEY tbjid = '60002065' ."TRAN
SPORTING NO FIELDS.
508         IF sy-subrc = 0.
509             lf_faaid = 'X'."Fire fighting
510
511             ls_final-zz_60002065 = ls_prof-vendd.
512             *      IF LS_PROF-VENDD LE LV_40DATE.
513             *      LF_40DAYS = '5'.
514             *      ls_qual_date-TBJID = LS_PROF-TBJID.
515             *      ls_qual_date-DATUM = LS_PROF-VENDD.
516             *      ls_qual_date-REASON = '5'.
517             *      APPEND ls_qual_date to lt_qual_date.
518             *      CLEAR : ls_qual_date.
519             *      ENDIF.
520         ENDIF.
521
522         CLEAR:ls_prof.
523         READ TABLE lt_prof INTO ls_prof WITH KEY tbjid = '60002787' ."TRAN
SPORTING NO FIELDS."Changing to 60002787 # First Aid level 1
524         IF sy-subrc = 0.
525             lf_firef = 'X'."Basic first aid
526             ls_final-zz_60002787 = ls_prof-vendd.
527             *      IF LS_PROF-VENDD LE LV_40DATE.
528             *      LF_40DAYS = '6'.
529             *      ls_qual_date-TBJID = LS_PROF-TBJID.
530             *      ls_qual_date-DATUM = LS_PROF-VENDD.
531             *      ls_qual_date-REASON = '6'.
532             *      APPEND ls_qual_date to lt_qual_date.
533             *      CLEAR : ls_qual_date.
534             *      ENDIF.
535         ENDIF.
536
537         CLEAR: ls_prof.
538         READ TABLE lt_prof INTO ls_prof WITH KEY tbjid = '60014576' ."BASI

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C FIRST AID COURSE - 1 DAY (60014576
539     IF sy-subrc = 0.
540         lf_firef = 'X'."Basic first aid
541
542         ls_final-zz_60014576 = ls_prof-vendd.
543     *         IF LS_PROF-VENDD LE LV_40DATE.
544     *         LF_40DAYS = '6'.
545     *         ls_qual_date-TBJID = LS_PROF-TBJID.
546     *         ls_qual_date-DATUM = LS_PROF-VENDD.
547     *         ls_qual_date-REASON = '6'.
548     *         APPEND ls_qual_date to lt_qual_date.
549     *         CLEAR : ls_qual_date.
550     *     ENDIF.
551     ENDIF.
552 *     ENDIF.
553
554     "Mine mangers ticket
555     READ TABLE lt_prof WITH KEY tbjid = '60011242' TRANSPORTING NO FIE
LDS.
556     IF sy-subrc = 0.
557     *         LF_minem = 'X'.
558     *         LS_FINAL-ZZ_STATUS = 'U'.
559     *         IF LS_PROF-VENDD LE LV_40DATE.
560     *         LF_40DAYS = 'X'.
561     *         ENDIF.
562         DELETE lt_final WHERE zz_pernr = ls_final-zz_pernr AND zz_begda
= ls_final-zz_begda.
563     ENDIF.
564
565     "GCC qualification
566     READ TABLE lt_prof WITH KEY tbjid = '80002235' TRANSPORTING NO FIE
LDS.
567     IF sy-subrc = 0.
568     *         LF_gcc = 'X'.
569     *         LS_FINAL-ZZ_STATUS = 'U'.
570     *         IF LS_PROF-VENDD LE LV_40DATE.
571     *         LF_40DAYS = 'X'.
572     *         ENDIF.
573         DELETE lt_final WHERE zz_pernr = ls_final-zz_pernr AND zz_begda
= ls_final-zz_begda.
574     ENDIF.
575     ENDIF.
576
577     IF ls_final-zz_eg = '3'.
578
579         IF lf_saferep = 'X' AND lf_faid = '' AND lf_firef = ''AND ls_final
-zz_status IS INITIAL.
580             ls_final-zz_status = 'B'##NO_TEXT.
581             ls_final-zz_reason = '3'##NO_TEXT.
582
583         ELSEIF lf_saferep = 'X' AND lf_faid = ''AND ls_final-zz_status IS
INITIAL.
584             ls_final-zz_status = 'B'##NO_TEXT.
585             ls_final-zz_reason = '1'##NO_TEXT.
586
587         ELSEIF lf_saferep = 'X' AND lf_firef = ''AND ls_final-zz_status IS
INITIAL.
588             ls_final-zz_status = 'B'##NO_TEXT.
589             ls_final-zz_reason = '2'##NO_TEXT.
590
591         ELSEIF lf_saferep = 'X' AND lf_faid = 'X' AND lf_firef = '' AND ls

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    _final-zz_status IS INITIAL.
592     ls_final-zz_status = 'U'##NO_TEXT.
593     ls_final-zz_reason = '9' ##NO_TEXT.
594
595     ELSEIF lf_saferep = 'X' AND lf_faaid = 'X' AND lf_firef = 'X' AND l
s_final-zz_status IS INITIAL.
596         ls_final-zz_status = 'U'##NO_TEXT.
597         ls_final-zz_reason = '9'.
598
599     ELSEIF lf_saferep = ''.
600         lf_del = 'X'.
601
602     ENDIF.
603
604     "Employee Group 4
605     ELSEIF ls_final-zz_eg = '4'.
606         "don't have qualifications First Aid and or Fire Fighting
607     *         IF ( LF_FAID = '' OR LF_FIREF = '' ) and
608
609         " Calculatig Age if less than 50
610         CLEAR :lv_pernr.
611         lv_pernr = ls_final-zz_pernr.
612     *         CALL FUNCTION 'HR_AUPBS_AGE'
613     *             EXPORTING
614     *                 PERNR             = LV_PERNR
615     *                 BSDTE             = SY-DATUM
616     **                REACTION         = ' '
617     *             IMPORTING
618     *                 VALUE             = LV_AGE
619     *             EXCEPTIONS
620     *                 RECORD_NOT_FOUND = 1
621     *                 OTHERS           = 2.
622
623         SELECT SINGLE gbdat FROM pa0002 INTO lv_bdate
624             WHERE pernr EQ ls_final-zz_pernr
625             AND begda LE ls_final-zz_endda
626             AND endda GE ls_final-zz_begda .
627
628         CALL FUNCTION 'HRCM_TIME_PERIOD_CALCULATE'
629             EXPORTING
630                 begda         = lv_bdate
631                 endda         = sy-datum
632             IMPORTING
633                 noyrs         = lv_age
634             EXCEPTIONS
635                 invalid_dates = 1
636                 overflow      = 2
637                 OTHERS        = 3.
638         IF sy-subrc <> 0.
639     * Implement suitable error handling here
640         ENDIF.
641
642
643     *         IF SY-SUBRC = 0 AND LV_AGE LT '50' AND LS_FINAL-ZZ_STATUS IS IN
ITIAL.
644
645     *         IF LF_FAID = '' AND LF_FIREF = '' .
646     *             LS_FINAL-ZZ_STATUS = 'B'##NO_TEXT.
647     *             LS_FINAL-ZZ_REASON = '3'##NO_TEXT.
648     *
649     *         ELSEIF LF_FAID = ''.60002065

```

```

650 *           LS_FINAL-ZZ_STATUS = 'B'##NO_TEXT.
651 *           LS_FINAL-ZZ_REASON = '1'##NO_TEXT.
652
653 "ELSE
654 IF lf_saferep = 'X'."60002065 or 60014576
655 *       IF LF_FIREF = ' ' .
656     IF lf_faïd = 'X' AND lf_firef = 'X' .
657       ls_final-zz_status = 'U'##NO_TEXT.
658       ls_final-zz_reason = '9'##NO_TEXT.
659
660     ELSE."IF LF_FIREF = 'X' and LF_FAID = 'X'.
661       ls_final-zz_status = 'B'.
662       ls_final-zz_reason = '2' ##NO_TEXT.
663 *       IF LF_40DAYS is not INITIAL ."= 'X'.
664 *       sort lt_qual_date by DATUM ASCENDING.
665 *       READ TABLE lt_qual_date INTO ls_qual_date INDEX 1.
666 *       LS_FINAL-ZZ_REASON = ls_qual_date-REASON.'"4' ##NO_TEXT.
667 *       CLEAR: ls_qual_date.
668 *       ENDIF.
669     ENDIF.
670
671 ELSE.
672   READ TABLE lt_position INTO ls_position WITH KEY pernr = lv_pern
673   r.
674   READ TABLE lt_hrp1001 INTO ls_hrp1001 WITH KEY objid = ls_positi
675   on-plans.
676   IF sy-subrc = 0 AND lv_age LT '50'.
677     IF lf_firef = ' ' .
678       ls_final-zz_status = 'B'##NO_TEXT.
679       ls_final-zz_reason = '2'##NO_TEXT.
680
681     ELSE."IF LF_FIREF = 'X' and LF_FAID = 'X'.
682       ls_final-zz_status = 'U'.
683       ls_final-zz_reason = '9' ##NO_TEXT.
684     ENDIF.
685   ELSE.
686     lf_del = 'X'. "If Age is >50
687   ENDIF.
688 ENDIF.
689 * ELSE.
690 *   LS_FINAL-ZZ_STATUS = 'U'##NO_TEXT.
691 *   LS_FINAL-ZZ_REASON = '9' ##NO_TEXT.
692 * ENDIF.
693
694 * ELSE.
695 **   ELSEIF LF_FIREF = 'X' and LF_FAID = 'X'.
696 *     LS_FINAL-ZZ_STATUS = 'U'.
697 *     LS_FINAL-ZZ_REASON = '9' ##NO_TEXT.
698 *     IF LF_40DAYS is not INITIAL ."= 'X'.
699 *     sort lt_qual_date by DATUM ASCENDING.
700 *     READ TABLE lt_qual_date INTO ls_qual_date INDEX 1.
701 *     LS_FINAL-ZZ_REASON = ls_qual_date-REASON.'"4' ##NO_TEXT.
702 *     CLEAR: ls_qual_date.
703 *     ENDIF.
704
705 *   ENDIF.
706
707 ENDIF.
708

```

```

709
710     MODIFY lt_final FROM ls_final TRANSPORTING zz_status zz_reason
711                                           zz_60002065 zz_60014576
712                                           zz_60002787 zz_80010932
713     WHERE zz_pernr = ls_final-zz_pernr
714     AND zz_begda = ls_final-zz_begda
715     AND zz_endda = ls_final-zz_endda
716     AND zz_begtym = ls_final-zz_begtym
717     AND zz_endtym = ls_final-zz_endtym.
718 *     and ZZ_60002065 = LS_FINAL-ZZ_60002065
719 *     and ZZ_60002787 = LS_FINAL-ZZ_60002787
720 *     and ZZ_60014576 = LS_FINAL-ZZ_60014576
721 *     and ZZ_80010932 = LS_FINAL-ZZ_80010932.
722
723
724
725     IF ls_final-zz_status = 'U'.
726         ls_list_pernr-pernr = ls_final-zz_pernr.
727         ls_list_pernr-status = 'U'.
728         APPEND ls_list_pernr TO lt_list_pernr.
729     ENDIF.
730
731
732
733     ENDLOOP.
734
735     IF lf_del = 'X'. "Excluding if A012 relationship is not present or Age
is >50
736         DELETE lt_final WHERE zz_status IS INITIAL.
737     ENDIF.
738     DELETE ADJACENT DUPLICATES FROM lt_list_pernr.
739 *     DELETE LT_FINAL WHERE ZZ_REASON IS INITIAL."Delete for Stage II
740 *     DELETE LT_FINAL WHERE ZZ_STATUS EQ 'U' AND ZZ_REASON NE '4' ##NO_TEXT
." do not delete stage II changes
741
742     ENDFORM.
743 *-----*
744 *      Form  SELECT_QUERIES
745 *-----*
746 *      text
747 *-----*
748 *  —>  p1      text
749 *  <—  p2      text
750 *-----*
751     FORM select_queries .
752
753     CLEAR :lv_year.
754     CALL FUNCTION 'RP_CALC_DATE_IN_INTERVAL'
755         EXPORTING
756             date      = sy-datum
757             days      = '00'
758             months    = '00'
759             signum    = '-'
760             years     = '1'
761         IMPORTING
762             calc_date = lv_year.
763
764     IF lv_year IS NOT INITIAL.
765
766         SELECT objid sobid begda endda FROM hrp1001 INTO TABLE lt_hrp1001
767         WHERE begda LE lv_year

```

```

768      AND endda GE sy-datum
769      AND plvar = '01'
770      AND rsign = 'A'
771      AND relat = '012'.
772
773      IF lt_hrp1001 IS NOT INITIAL.
774          SORT lt_hrp1001 BY objid.
775          DELETE ADJACENT DUPLICATES FROM lt_hrp1001 COMPARING objid.
776      ENDIF.
777  ENDIF.
778
779  ENDFORM.
780  *&-----*
781  *&      Form  DISP_DATA
782  *&-----*
783  *      text
784  *-----*
785  *  →  p1      text
786  *  ←  p2      text
787  *-----*
788  FORM disp_data .
789
790      IF lt_final IS NOT INITIAL.
791
792          CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
793              EXPORTING
794                  is_layout      = ls_layout
795                  it_fieldcat    = lt_fcat
796              TABLES
797                  t_outtab      = lt_final
798              EXCEPTIONS
799                  program_error = 1
800                  OTHERS       = 2 ##FM_SUBRC_OK.
801          IF sy-subrc <> 0.
802
803              ENDIF.
804
805      ELSE.
806          WRITE: 'No Employees to block matching selection criteria'##NO_TEXT.
807      ENDIF.
808
809  ENDFORM.
810  *&-----*
811  *&      Form  FIELD_CAT
812  *&-----*
813  *      text
814  *-----*
815  *  →  p1      text
816  *  ←  p2      text
817  *-----*
818  FORM field_cat .
819
820      CLEAR ls_fcat.
821      ls_fcat-fieldname = 'ZZ_PERNR'.
822      ls_fcat-tabname   = 'IT_FINAL'.
823      ls_fcat-seltext_m = 'Personal Number'##NO_TEXT.
824      APPEND ls_fcat TO lt_fcat. "1
825
826      CLEAR ls_fcat.
827      ls_fcat-fieldname = 'ZZ_BEGDA'.
828      ls_fcat-tabname   = 'IT_FINAL'.

```

```
829  ls_fcat-seltext_m = 'Start Date'##NO_TEXT.
830  APPEND ls_fcat TO lt_fcat."2
831
832  CLEAR ls_fcat.
833  ls_fcat-fieldname = 'ZZ_BEGTYM'.
834  ls_fcat-tabname = 'IT_FINAL'.
835  ls_fcat-seltext_m = 'Start Time'##NO_TEXT.
836  APPEND ls_fcat TO lt_fcat."3
837
838  CLEAR ls_fcat.
839  ls_fcat-fieldname = 'ZZ_ENDDA'.
840  ls_fcat-tabname = 'IT_FINAL'.
841  ls_fcat-seltext_m = 'End Date'##NO_TEXT.
842  APPEND ls_fcat TO lt_fcat."4
843
844  CLEAR ls_fcat.
845  ls_fcat-fieldname = 'ZZ_ENDTYM'.
846  ls_fcat-tabname = 'IT_FINAL'.
847  ls_fcat-seltext_m = 'End Time'##NO_TEXT.
848  APPEND ls_fcat TO lt_fcat."5
849
850  CLEAR ls_fcat.
851  ls_fcat-fieldname = 'ZZ_EG'.
852  ls_fcat-tabname = 'IT_FINAL'.
853  ls_fcat-seltext_m = 'EG'##NO_TEXT.
854  APPEND ls_fcat TO lt_fcat."6
855
856  CLEAR ls_fcat.
857  ls_fcat-fieldname = 'ZZ_PSG'.
858  ls_fcat-tabname = 'IT_FINAL'.
859  ls_fcat-seltext_m = 'PSG'##NO_TEXT.
860  APPEND ls_fcat TO lt_fcat."7
861
862  CLEAR ls_fcat.
863  ls_fcat-fieldname = 'ZZ_STATUS'.
864  ls_fcat-tabname = 'IT_FINAL'.
865  ls_fcat-seltext_m = 'Block/Unblock'##NO_TEXT.
866  APPEND ls_fcat TO lt_fcat."8
867
868  CLEAR ls_fcat.
869  ls_fcat-fieldname = 'ZZ_REASON'.
870  ls_fcat-tabname = 'IT_FINAL'.
871  ls_fcat-seltext_m = 'Block/Unblock Reason'##NO_TEXT.
872  APPEND ls_fcat TO lt_fcat."9
873
874  CLEAR ls_fcat.
875  ls_fcat-fieldname = 'ZZ_80010932'.
876  ls_fcat-tabname = 'IT_FINAL'.
877  ls_fcat-seltext_m = 'Q80010932'##NO_TEXT.
878  APPEND ls_fcat TO lt_fcat."10
879
880  CLEAR ls_fcat.
881  ls_fcat-fieldname = 'ZZ_60002065'.
882  ls_fcat-tabname = 'IT_FINAL'.
883  ls_fcat-seltext_m = 'Q60002065'##NO_TEXT.
884  APPEND ls_fcat TO lt_fcat."11
885
886  CLEAR ls_fcat.
887  ls_fcat-fieldname = 'ZZ_60014576'.
888  ls_fcat-tabname = 'IT_FINAL'.
889  ls_fcat-seltext_m = 'Q60014576'##NO_TEXT.
```



```

890 APPEND ls_fcat TO lt_fcat."12
891
892 CLEAR ls_fcat.
893 ls_fcat-fieldname = 'ZZ_60002787'.
894 ls_fcat-tabname = 'IT_FINAL'.
895 ls_fcat-seltext_m = 'Q60002787'##NO_TEXT.
896 APPEND ls_fcat TO lt_fcat."13
897
898
899 ENDFORM.
900 *&-----*
901 *&      Form  SEND_IDOC
902 *&-----*
903 *      text
904 *-----*
905 *  —>  p1      text
906 *  <—  p2      text
907 *-----*
908 FORM send_idoc .
909
910 CLEAR lv_count.
911 ls_idoc_control-mestyp = 'Z1PP004'.
912                                "Bulding the Control record
913 ls_idoc_control-idoctp = 'Z1PP004'.
914 ls_idoc_control-rcvprn = 'AAHWELL'.
915 ls_idoc_control-rcvpert = 'LS'.
916
917 LOOP AT lt_final INTO ls_final.
918     MOVE-CORRESPONDING ls_final TO ls_final2.
919     ls_idoc_data-segnam = lc_invrev_segname.
920     ls_idoc_data-sdata = ls_final2.
921     APPEND ls_idoc_data TO lt_idoc_data.
922     CLEAR ls_idoc_data.
923
924 ENDLOOP.
925
926 CALL FUNCTION 'MASTER_IDOC_DISTRIBUTE'
927     EXPORTING
928         master_idoc_control      = ls_idoc_control
929     TABLES
930         communication_idoc_control = lt_comm_control
931         master_idoc_data          = lt_idoc_data
932     EXCEPTIONS
933         error_in_idoc_control      = 1
934         error_writing_idoc_status = 2
935         error_in_idoc_data        = 3
936         sending_logical_system_unknown = 4
937         OTHERS                    = 5.
938
939 IF sy-subrc = 0 AND lt_comm_control IS NOT INITIAL.
940
941     CALL FUNCTION 'DB_COMMIT'.
942     CALL FUNCTION 'DEQUEUE_ALL'.
943     COMMIT WORK.
944
945     "Changing the status of the Employees data in Staging Table once sen
946 t
947 *      delete ZPPA0088 WHERE
948 LOOP AT lt_final INTO ls_final.
949     DELETE FROM zppa0088 WHERE zz_pernr = ls_final-zz_pernr

```

```

949          AND zz_begda LE ls_final-zz_endda
950          AND zz_endda GE ls_final-zz_begda.
951      ENDLOOP.
952
953      LOOP AT lt_final INTO ls_final.
954          lv_count = lv_count + 1.
955          INSERT zppa0088 FROM ls_final.
956          CLEAR ls_final.
957      ENDLOOP.
958      WRITE : 'ZPPA0088 is modified'##NO_TEXT.
959
960      WRITE : 'Data successfully sent for : ' ##NO_TEXT, lv_count , ' : Em
ployees'##NO_TEXT.
961      WRITE : /'IDOC number:'##NO_TEXT.
962
963      LOOP AT lt_comm_control INTO ls_comm_control.
964          WRITE: /15 ls_comm_control-docnum.
965      ENDLOOP.
966
967      ELSE.
968      *-Added by K.Jaisy Augustine under TR-DS1K9A0CNX
969          WRITE: 'No Employees to block matching selection criteria'##NO_TEXT.
970
971      ENDIF.
972
973
974      ENDFORM.
975      *&-----*
976      *&          Form  BADI_CALL*
977      *&-----*
978      *          text*
979      *-----*
980      *  ->  p1          text*
981      *  <-  p2          text*
982      *-----*
983      FORM badi_call .
984          " if Qualification infotype changed
985          CONSTANTS: c_memid_text(60) TYPE c VALUE '%%ZTEXT%%'.
986
987          IMPORT objid TO lt_exp_new
988                  begda TO lt_exp_new
989                  endda TO lt_exp_new
990                  FROM MEMORY ID c_memid_text.
991          IF lt_exp_new IS NOT INITIAL.
992              lf_info = '0024'.
993          ENDIF.
994
995          " if IT0007 (Planned working time) changed
996          CONSTANTS: c_memid_0007(60) TYPE c VALUE '%%Z0007%%'.
997
998          IMPORT pernr TO lt_exp_0007
999                  begda TO lt_exp_0007
1000                  endda TO lt_exp_0007
1001                  data TO lt_exp_0007
1002                  FROM MEMORY ID c_memid_0007.
1003          IF lt_exp_0007 IS NOT INITIAL.
1004              lf_info = '0007'.
1005          ENDIF.
1006
1007
1008          CONSTANTS: c_memid_2003(60) TYPE c VALUE '%%Z2003%%'.

```

"IT2003

```

1009  IMPORT objid TO lt_exp_2003
1010          subty TO lt_exp_2003
1011          begda TO lt_exp_2003
1012          endda TO lt_exp_2003
1013          data TO lt_exp_2003
1014          FROM MEMORY ID c_memid_2003.
1015  IF lt_exp_2003 IS NOT INITIAL.
1016      lf_info = '2003'.
1017  ENDIF.
1018
1019
1020  CONSTANTS: c_memid_2004(60) TYPE c VALUE '%%Z2004%%'.
1021  IMPORT objid TO lt_exp_2004
1022          subty TO lt_exp_2004
1023          begda TO lt_exp_2004
1024          endda TO lt_exp_2004
1025          data TO lt_exp_2004
1026          FROM MEMORY ID c_memid_2004.
1027  IF lt_exp_2004 IS NOT INITIAL.
1028      lf_info = '2004'.
1029  ENDIF.
1030
1031  CONSTANTS: c_memid_2002(60) TYPE c VALUE '%%Z2002%%'.
1032  IMPORT objid TO lt_exp_2002
1033          subty TO lt_exp_2002
1034          begda TO lt_exp_2002
1035          endda TO lt_exp_2002
1036          data TO lt_exp_2002
1037          FROM MEMORY ID c_memid_2002.
1038  IF lt_exp_2002 IS NOT INITIAL.
1039      lf_info = '2002'.
1040  ENDIF.
1041
1042  CONSTANTS: c_memid_2007(60) TYPE c VALUE '%%Z2007%%'.
1043  IMPORT objid TO lt_exp_2007
1044          subty TO lt_exp_2007
1045          begda TO lt_exp_2007
1046          endda TO lt_exp_2007
1047          data TO lt_exp_2007
1048          FROM MEMORY ID c_memid_2007.
1049  IF lt_exp_2007 IS NOT INITIAL.
1050      lf_info = '2007'.
1051  ENDIF.
1052
1053  CONSTANTS: c_memid_2001(60) TYPE c VALUE '%%Z2001%%'.
1054  IMPORT objid TO lt_exp_2001
1055          subty TO lt_exp_2001
1056          begda TO lt_exp_2001
1057          endda TO lt_exp_2001
1058          data TO lt_exp_2001
1059          FROM MEMORY ID c_memid_2001.
1060  IF lt_exp_2001 IS NOT INITIAL.
1061      lf_info = '2001'.
1062  ENDIF.
1063
1064  ENDFORM.
1065  *&-----*
1066  *&      Form  FILL_BADI_DATA
1067  *&-----*
1068  *      text
1069  *-----*
```

"IT2003

```

1070 *  ->  p1      text
1071 *  <-  p2      text
1072 *
1073 FORM fill_badi_data .
1074
1075     CALL FUNCTION 'RP_CALC_DATE_IN_INTERVAL'
1076     EXPORTING
1077         date      = sy-datum
1078         days      = '40'
1079         months    = '00'
1080 *     SIGNUM      = '+'
1081         years     = '00'
1082     IMPORTING
1083         calc_date = lv_40date.
1084
1085     LOOP AT lt_final INTO ls_final.
1086
1087         CLEAR :lf_40days.
1088         CLEAR :lf_saferep,lf_faid,lf_firef.
1089         CLEAR:ls_prof.
1090         READ TABLE lt_exp_new INTO ls_exp_new WITH KEY objid = '80010932' ."
1091     TRANSPORTING NO FIELDS .
1092     IF sy-subrc = 0.
1093         lf_saferep = 'X'."Safety Representatives qualification
1094         ls_final-zz_80010932 = ls_exp_new-endda.
1095 *         IF LS_EXP_NEW-ENDDA LE LV_40DATE.
1096 *             LF_40DAYS = '4'.
1097 *             ls_qual_date-TBJID = LS_PROF-TBJID.
1098 *             ls_qual_date-DATUM = LS_PROF-VENDD.
1099 *             ls_qual_date-REASON = '4'.
1100 *             APPEND ls_qual_date to lt_qual_date.
1101 *             CLEAR : ls_qual_date.
1102 *         ENDIF.
1103     ENDIF.
1104
1105     CLEAR:ls_prof.
1106     READ TABLE lt_exp_new INTO ls_exp_new WITH KEY objid = '60002065' ."
1107     TRANSPORTING NO FIELDS.
1108     IF sy-subrc = 0.
1109         lf_faid = 'X'."First Aid
1110         ls_final-zz_60002065 = ls_exp_new-endda.
1111 *         IF LS_EXP_NEW-ENDDA LE LV_40DATE.
1112 *             LF_40DAYS = '5'.
1113 *             ls_qual_date-TBJID = LS_PROF-TBJID.
1114 *             ls_qual_date-DATUM = LS_PROF-VENDD.
1115 *             ls_qual_date-REASON = '5'.
1116 *             APPEND ls_qual_date to lt_qual_date.
1117 *             CLEAR : ls_qual_date.
1118 *         ENDIF.
1119     ENDIF.
1120
1121     CLEAR:ls_prof.
1122     READ TABLE lt_exp_new INTO ls_exp_new WITH KEY objid = '60002787' ."
1123     TRANSPORTING NO FIELDS.
1124     IF sy-subrc = 0.
1125         lf_firef = 'X'."Fire Fighting
1126         ls_final-zz_60002787 = ls_exp_new-endda.
1127 *         IF LS_EXP_NEW-ENDDA LE LV_40DATE.
1128 *             LF_40DAYS = '6'.
1129 *             ls_qual_date-TBJID = LS_PROF-TBJID.
1130 *             ls_qual_date-DATUM = LS_PROF-VENDD.

```

```

1128 *      ls_qual date-REASON = '6'.
1129 *      APPEND ls_qual date to lt_qual_date.
1130 *      CLEAR : ls_qual_date.
1131 *      ENDIF.
1132
1133      ENDIF.
1134
1135      CLEAR : ls_prof.
1136      READ TABLE lt_exp_new INTO ls_exp_new WITH KEY objid = '60014576' ."
BASIC FIRST AID COURSE - 1 DAY (60014576
1137      IF sy-subrc = 0.
1138          lf_firef = 'X'."Basic first aid
1139          ls_final-zz_60014576 = ls_exp_new-endda.
1140 *          IF LS_EXP_NEW-ENDDa LE LV_40DATE.
1141 *          LF_40DAYS = '6'.
1142 *          ls_qual_date-TBJID = LS_PROF-TBJID.
1143 *          ls_qual_date-DATUM = LS_PROF-VENDD.
1144 *          ls_qual_date-REASON = '6'.
1145 *          APPEND ls_qual_date to lt_qual_date.
1146 *          CLEAR : ls_qual_date.
1147 *          ENDIF.
1148      ENDIF.
1149
1150 *      ENDIF.
1151
1152      "Mine mangers ticket
1153      READ TABLE lt_exp_new WITH KEY objid = '60011242' TRANSPORTING NO FI
ELDS.
1154      IF sy-subrc = 0.
1155 *          LF_minem = 'X'.
1156 *          LS_FINAL-ZZ_STATUS = 'U'.
1157 *          IF LS_PROF-VENDD LE LV_40DATE.
1158 *          LF_40DAYS = 'X'.
1159 *          ENDIF.
1160      DELETE lt_final WHERE zz_pernr = ls_final-zz_pernr AND zz_begda =
ls_final-zz_begda.
1161      ENDIF.
1162
1163      "GCC qualification
1164      READ TABLE lt_exp_new WITH KEY objid = '80002235' TRANSPORTING NO FI
ELDS.
1165      IF sy-subrc = 0.
1166 **          LF_gcc = 'X'.
1167 *          LS_FINAL-ZZ_STATUS = 'U'.
1168 *          IF LS_PROF-VENDD LE LV_40DATE.
1169 *          LF_40DAYS = 'X'.
1170      DELETE lt_final WHERE zz_pernr = ls_final-zz_pernr AND zz_begda =
ls_final-zz_begda.
1171 *      ENDIF.
1172      ENDIF.
1173
1174
1175      IF ls_final-zz_eg = '3'.
1176
1177          IF lf_saferep = 'X' AND lf_faaid = '' AND lf_firef = ''AND ls_final
-  
zz_status IS INITIAL.
1178              ls_final-zz_status = 'B'##NO_TEXT.
1179              ls_final-zz_reason = '3'##NO_TEXT.
1180
1181          ELSEIF lf_saferep = 'X' AND lf_faaid = ''AND ls_final-zz_status IS
INITIAL.

```

```

1182         ls_final-zz_status = 'B'##NO_TEXT.
1183         ls_final-zz_reason = '1'##NO_TEXT.
1184
1185         ELSEIF lf_saferep = 'X' AND lf_firef = '' AND ls_final-zz_status IS
INITIAL.
1186         ls_final-zz_status = 'B'##NO_TEXT.
1187         ls_final-zz_reason = '2'##NO_TEXT.
1188
1189         ELSEIF lf_saferep = 'X' AND lf_faidd = 'X' AND lf_firef = '' AND ls
_final-zz_status IS INITIAL.
1190         ls_final-zz_status = 'U'##NO_TEXT.
1191         ls_final-zz_reason = '9' ##NO_TEXT.
1192
1193         ELSEIF lf_saferep = 'X' AND lf_faidd = 'X' AND lf_firef = 'X' AND l
s_final-zz_status IS INITIAL.
1194         ls_final-zz_status = 'U'##NO_TEXT.
1195         ls_final-zz_reason = '9'.
1196
1197         ELSEIF lf_saferep = ''.
1198         lf_del = 'X'.
1199
1200     ENDIF.
1201
1202     "Employee Group 4
1203     ELSEIF ls_final-zz_eg = '4'.
1204     "don't have qualifications First Aid and or Fire Fighting
1205     *     IF ( LF_FAID = '' OR LF_FIREF = '' ) and
1206     *     if LF_SAFEREP = 'X'.
1207     " Calculatig Age if less than 50
1208     CLEAR :lv_pernr.
1209     lv_pernr = ls_final-zz_pernr.
1210     *     CALL FUNCTION 'HR_AUPBS_AGE'
1211     *     EXPORTING
1212     *         PERNR                = LV_PERNR
1213     *         BSDTE                = SY-DATUM
1214     **         REACTION            = ' '
1215     *     IMPORTING
1216     *         VALUE                = LV_AGE
1217     *     EXCEPTIONS
1218     *         RECORD_NOT_FOUND = 1
1219     *         OTHERS            = 2.
1220
1221
1222     SELECT SINGLE gbdat FROM pa0002 INTO lv_bdate
1223             WHERE pernr EQ LS_FINAL-zz_pernr
1224             AND begda LE ls_final-zz_endda
1225             AND endda GE ls_final-zz_begda .
1226
1227     CALL FUNCTION 'HRCM_TIME_PERIOD_CALCULATE'
1228     EXPORTING
1229         begda        = lv_bdate
1230         endda        = sy-datum
1231     IMPORTING
1232         noyrs        = lv_age
1233     EXCEPTIONS
1234         invalid_dates = 1
1235         overflow      = 2
1236         OTHERS        = 3.
1237     IF sy-subrc <> 0.
1238     * Implement suitable error handling here
1239     ENDIF.

```

```

1240
1241
1242 *           IF SY-SUBRC = 0 AND LV_AGE LT '50' AND LS_FINAL-ZZ_STATUS IS IN
ITIAL.
1243
1244 *           IF LF_FAID = '' AND LF_FIREF = '' .
1245 *               LS_FINAL-ZZ_STATUS = 'B'##NO_TEXT.
1246 *               LS_FINAL-ZZ_REASON = '3'##NO_TEXT.
1247
1248 *           ELSEIF LF_FAID = '' .
1249 *               LS_FINAL-ZZ_STATUS = 'B'##NO_TEXT.
1250 *               LS_FINAL-ZZ_REASON = '1'##NO_TEXT.
1251
1252 *           ELSE
1253 *-Changed by Jaisy Augustine
1254     IF lf_saferep = 'X'. "For Safety Representative
1255 *
1256     IF lf_faaid = 'X' AND lf_firef = 'X' .
1257         ls_final-zz_status = 'U'##NO_TEXT.
1258         ls_final-zz_reason = '9'##NO_TEXT.
1259
1260     ELSE."IF LF_FIREF = 'X' and LF_FAID = 'X'.
1261         ls_final-zz_status = 'B'.
1262         ls_final-zz_reason = '2' ##NO_TEXT.
1263     ENDIF.
1264
1265
1266
1267     ELSE.
1268 *-Checking A012 relationship and age for Supervisor
1269 *           READ TABLE LT_HRP1001 INTO LS_HRP1001 WITH KEY OBJID = P000
1-PLANS.
1270     READ TABLE lt_position INTO ls_position WITH KEY pernr = lv_pernr.
1271     READ TABLE lt_hrp1001 INTO ls_hrp1001 WITH KEY objid = ls_positi
on-plans.
1272     IF sy-subrc = 0 AND lv_age LT '50'."For Supervisor
1273 *           if LV_AGE LT '50'.
1274     IF lf_firef = '' .
1275         ls_final-zz_status = 'B'##NO_TEXT.
1276         ls_final-zz_reason = '2'##NO_TEXT.
1277
1278     ELSE."IF LF_FIREF = 'X' and LF_FAID = 'X'.
1279         ls_final-zz_status = 'U'.
1280         ls_final-zz_reason = '9' ##NO_TEXT.
1281     ENDIF.
1282     ELSE.
1283         lf_del = 'X'. "If Age is >50
1284     ENDIF.
1285
1286     ENDIF.
1287
1288
1289     ENDIF.
1290
1291     MODIFY lt_final FROM ls_final TRANSPORTING zz_status zz_reason
1292         zz_60002065 zz_80010932 zz_60014576 zz_60002787
1293     WHERE zz_pernr = ls_final-zz_pernr
1294     AND zz_begda = ls_final-zz_begda
1295     AND zz_endda = ls_final-zz_endda
1296     AND zz_begtym = ls_final-zz_begtym

```

```

1297     AND zz_endtym = ls_final-zz_endtym.
1298
1299     ENDLOOP.
1300
1301     IF lf_del = 'X'. "Excluding if A012 relationship is not present or Age
is >50
1302         DELETE lt_final WHERE zz_status IS INITIAL.
1303     ENDIF.
1304     *   IF LF_DEL = 'X'.
1305     *       DELETE LT_FINAL WHERE ZZ_80010932 IS INITIAL.
1306     *   ENDIF.
1307     *   DELETE LT_FINAL WHERE ZZ_REASON IS INITIAL."deleting for Stage II
1308     *   DELETE LT_FINAL WHERE ZZ_STATUS EQ 'U' AND ZZ_REASON NE 'Qualificatio
n is expiring in less than 40 days' ##NO_TEXT."Stage II
1309
1310
1311     ENDFORM.
1312
1313     *&-----*
1314     *&       Form   CHECK_TIME
1315     *&-----*
1316     *       text
1317     *-----*
1318     *   ->  p1       text
1319     *   <-  p2       text
1320     *-----*
1321     FORM check_time .
1322
1323     DATA : lt_pers_avail TYPE TABLE OF  pdpsp.
1324     DATA : lt_day_before TYPE TABLE OF  pdpsp WITH HEADER LINE.
1325     DATA : lv_id TYPE objec-objid.
1326     DATA : lt_final_avail TYPE TABLE OF pdpsp.
1327     DATA : ls_final_avail1 TYPE pdpsp ." WITH HEADER LINE.
1328
1329     LOOP AT lt_list_pernr INTO ls_list_pernr WHERE status = 'U'.
1330
1331         REFRESH : lt_pers_avail .
1332         lv_id = ls_list_pernr-pernr.
1333         CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
1334             EXPORTING
1335                 begin_date      = lv_begda
1336                 end_date        = lv_enddda
1337                 person_id       = lv_id
1338             TABLES
1339                 pers_avail      = lt_pers_avail " STRUCTURE  PDPSP
1340             EXCEPTIONS
1341                 no_capacity_available = 1
1342                 no_integration_activ  = 2
1343                 timeinfo_error       = 3
1344                 OTHERS               = 4.
1345         IF sy-subrc <> 0.
1346     * Implement suitable error handling here
1347         ENDIF.
1348
1349         APPEND LINES OF lt_pers_avail TO lt_final_avail.
1350
1351     ENDLOOP.
1352
1353     IF lt_list_pernr IS NOT INITIAL.
1354     *
1355         SELECT * FROM pa2001 INTO TABLE lt_pa2001

```



```

1356     FOR ALL ENTRIES IN lt_list_pernr
1357     WHERE pernr = lt_list_pernr-pernr
1358     AND subty NE '1070'
1359     AND endda GE lv_begda
1360     AND begda LE lv_endda.
1361 *
1362     ENDIF.
1363 *****
1364 ** Attendance Quotas overtime
1365     IF lt_list_pernr IS NOT INITIAL.
1366         SELECT * FROM pa2007 INTO TABLE lt_pa2007
1367         FOR ALL ENTRIES IN lt_list_pernr
1368         WHERE pernr = lt_list_pernr-pernr
1369         AND subty = '01'
1370         AND endda GE lv_begda
1371         AND begda LE lv_endda.
1372     ENDIF.
1373 *****
1374     LOOP AT lt_final_avail INTO ls_final_avail.
1375
1376 *     READ TABLE LT_PA2007 into LS_PA2007 with key PERNR =
1377 *     SELECT SINGLE * FROM PA2007 "TABLE LT_PA2007
1378 **     FOR ALL ENTRIES IN LT_LIST_PERNR
1379 *     WHERE PERNR = LS_FINAL_AVAIL-PERNR
1380 *     AND SUBTY = '1070'
1381 *     AND ENDDA GE LS_FINAL_AVAIL-DATUM
1382 *     AND BEGDA LE LS_FINAL_AVAIL-DATUM
1383 *     INTO CORRESPONDING FIELDS OF ls_pa2007 .
1384
1385     READ TABLE lt_pa2001 INTO ls_pa2001 WITH KEY pernr = ls_final_avail-
1386     pernr.
1387
1388     IF ls_pa2001-begda GE ls_final_avail-datum AND ls_pa2001-endda LE ls
1389     _final_avail-datum .
1390
1391         ls_time_t-zz_eg = ls_final-zz_eg.
1392         ls_time_t-zz_psg = ls_final-zz_psg.
1393         IF ls_time_t-zz_eg IS INITIAL OR ls_time_t-zz_psg IS INITIAL.
1394             CLEAR: ls_list_pernr.
1395             READ TABLE lt_list_pernr INTO ls_list_pernr WITH KEY pernr = l
1396             s_final_avail-pernr.
1397             ls_time_t-zz_eg = ls_list_pernr-eg.
1398             ls_time_t-zz_psg = ls_list_pernr-psg.
1399         ENDIF.
1400
1401         ls_time_t-zz_pernr = ls_final_avail-pernr.
1402         ls_time_t-zz_begda = ls_final_avail-datum.
1403         ls_time_t-zz_endda = ls_final_avail-datum.
1404         ls_time_t-zz_begtym = lv_def_beg.
1405         ls_time_t-zz_endtym = lv_def_end.
1406         ls_time_t-zz_status = 'B'.
1407         ls_time_t-zz_reason = '8'.
1408         APPEND ls_time_t TO lt_time_t.
1409 *EXIT FROM
1410     CONTINUE.
1411 *     SKIP.
1412     ENDIF.
1413
1414     REFRESH : lt_temp_time .
1415

```

```

1414     READ TABLE lt_final INTO ls_final WITH KEY zz_pernr = ls_time_t-zz_p
      ernr.
1415     ls_time_t-zz_eg = ls_final-zz_eg.
1416     ls_time_t-zz_psg = ls_final-zz_psg.
1417     IF sy-subrc <> 0.
1418         CLEAR: ls_list_pernr.
1419         READ TABLE lt_list_pernr INTO ls_list_pernr WITH KEY pernr = ls_
      final_avail-pernr.
1420         ls_time_t-zz_eg = ls_list_pernr-eg.
1421         ls_time_t-zz_psg = ls_list_pernr-psg.
1422     ENDIF .
1423
1424     IF ls_final_avail-begzt GT ls_final_avail-endzt.
1425         "Night Shift
1426         ls_time_t-zz_pernr = ls_final_avail-pernr.
1427         ls_time_t-zz_begda = ls_final_avail-datum.
1428         ls_time_t-zz_endda = ls_final_avail-datum.
1429         ls_time_t-zz_status = 'B'.
1430
1431         lv_day_before = ls_final_avail-datum - 1.
1432
1433         READ TABLE lt_final_avail INTO ls_final_avail1
1434             WITH KEY pernr = ls_final_avail-pernr
1435                 datum = lv_day_before.
1436
1437         IF sy-subrc = 0.
1438             IF ls_final_avail1-begzt GT ls_final_avail1-endzt.
1439                 ls_time_t-zz_begtym = ls_final_avail1-endzt.
1440             ELSE.
1441                 ls_time_t-zz_begtym = lv_def_beg.
1442             ENDIF.
1443
1444         ELSE.
1445             CALL FUNCTION 'HR_READ_TIMEDATA_PSP' "fetch yesterday's WSR
1446                 EXPORTING
1447                     begin_date           = lv_day_before
1448                     end_date             = lv_day_before
1449                     person_id            = ls_final_avail-pernr
1450                 TABLES
1451                     pers_avail           = lt_day_before
1452                 EXCEPTIONS
1453                     no_capacity_available = 1
1454                     no_integration_activ = 2
1455                     timeinfo_error      = 3
1456                     OTHERS               = 4.
1457
1458             IF sy-subrc = 0.
1459                 READ TABLE lt_day_before INDEX 1.
1460                 IF lt_day_before-begzt GT lt_day_before-endzt.
1461                     ls_time_t-zz_begtym = lt_day_before-endzt.
1462                 ELSE.
1463                     ls_time_t-zz_begtym = lv_def_beg.
1464                 ENDIF.
1465             ENDIF.
1466         ENDIF.
1467     ENDIF.
1468 ENDIF.
1469
1470
1471     READ TABLE lt_pa2007 INTO ls_pa2007 WITH KEY pernr = ls_final_avai
      l-pernr."BEGDA .overtime

```

```

1472
1473     IF sy-subrc = 0 AND ls_pa2007-begda GE ls_final_avail-datum AND l
s_pa2007-endda LE ls_final_avail-datum ."Overtime
1474
1475         ls_temp_time-pernr = ls_final_avail-pernr.
1476         ls_temp_time-begda = ls_final_avail-datum.
1477         ls_temp_time-endda = ls_final_avail-datum.
1478
1479         ls_temp_time-begty = lv_def_beg.
1480         ls_temp_time-endty = ls_time_t-zz begtym.
1481         ls_time_t-zz_reason = '7'##NO TEXT.
1482         APPEND ls_temp_time TO lt_temp_time. "1st shift from Morning
1483         CLEAR : ls_time_t-zz_begtym , ls_time_t-zz_endtym.
1484
1485     LOOP AT lt_pa2007 INTO ls_pa2007 WHERE pernr = ls_final_avail-p
ernr."Overtimes " AND BEGDA = LS_FINAL_AVAIL-DATUM.
1486
1487         ls_temp_time-begty = ls_pa2007-beguz.
1488         ls_temp_time-endty = ls_pa2007-enduz.
1489         APPEND ls_temp_time TO lt_temp_time.
1490
1491         lv_def_endl = ls_temp_time-endty.
1492         CLEAR : ls_time_t-zz_begtym , ls_time_t-zz_endtym.
1493
1494     ENDLOOP.
1495
1496     lv_def_begl = ls_final_avail-begzt - lv_seconds .
1497     ls_temp_time-begty = lv_def_begl.
1498     ls_temp_time-endty = lv_def_end.
1499     ls_time_t-zz_reason = '7'##NO TEXT.
1500     APPEND ls_temp_time TO lt_temp_time. "last shift from Morning
1501     CLEAR : ls_time_t-zz_begtym , ls_time_t-zz_endtym.
1502
1503     SORT lt_temp_time BY begda endda begty endty .
1504     DESCRIBE TABLE lt_temp_time LINES lv_line.
1505
1506     LOOP AT lt_temp_time INTO ls_temp_time."Block between shifts bet
ween time
1507         IF sy-tabix NE lv_line.
1508
1509             ls_time_t-zz_begtym = ls_temp_time-endty.
1510             lv_index = sy-tabix + 1.
1511             READ TABLE lt_temp_time INTO ls_temp_time INDEX lv_index.
1512             ls_time_t-zz_endtym = ls_temp_time-begty.
1513
1514             APPEND ls_time_t TO lt_time_t.
1515             CLEAR : ls_time_t-zz_begtym , ls_time_t-zz_endtym.
1516
1517         ENDIF.
1518     ENDLOOP.
1519
1520     ELSE "No Overtime Night
1521
1522         lv_def_begl = ls_final_avail-begzt - lv_seconds "Before 2 hrs
1523         ls_time_t-zz_endtym = lv_def_begl.
1524         ls_time_t-zz_status = 'B'.
1525         ls_time_t-zz_reason = '7'##NO TEXT.
1526         APPEND ls_time_t TO lt_time_t.
1527         CLEAR :ls_time_t,lv_def_begl.
1528
1529     ENDIF.

```

```

1530
1531     ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt.
1532         "Day shift
1533         ls_time_t-zz_pernr = ls_final_avail-pernr.
1534         ls_time_t-zz_begda = ls_final_avail-datum.
1535         ls_time_t-zz_endda = ls_final_avail-datum.
1536         ls_time_t-zz_status = 'B'.
1537
1538     *****If yesterday is night
1539     *Check for day Before
1540
1541     lv_day_before = ls_final_avail-datum - 1.
1542     CALL FUNCTION 'HR_READ_TIMedata_PSP' "fetch yesterday's WSR
1543         EXPORTING
1544             begin_date           = lv_day_before
1545             end_date             = lv_day_before
1546             person_id            = ls_final_avail-pernr
1547         TABLES
1548             pers_avail           = lt_day_before
1549         EXCEPTIONS
1550             no_capacity_available = 1
1551             no_integration_activ  = 2
1552             timeinfo_error       = 3
1553             OTHERS               = 4.
1554
1555     IF sy-subrc = 0.
1556         READ TABLE lt_day_before INDEX 1.
1557         IF lt_day_before-begzt GT lt_day_before-endzt.
1558             lf_night = 'X'.
1559         ENDIF.
1560     ENDIF.
1561
1562     *****
1563     READ TABLE lt_pa2007 INTO ls_pa2007 WITH KEY pernr = ls_final_avai
1564     l-pernr ."overtime
1565     IF sy-subrc = 0 AND ls_pa2007-begda GE ls_final_avail-datum AND ls
1566     _pa2007-endda LE ls_final_avail-datum ." there is overtime
1567         CLEAR : ls_pa2007.
1568     ***** TR
1569     Y
1570
1571     lv_def_beg1 = ls_final_avail-begzt - lv_seconds .
1572
1573     ls_temp_time-pernr = ls_final_avail-pernr.
1574     ls_temp_time-begda = ls_final_avail-datum.
1575     ls_temp_time-endda = ls_final_avail-datum.
1576     ls_temp_time-begty = lv_def_beg1.
1577     ls_temp_time-endty = ls_final_avail-endzt.
1578     APPEND ls_temp_time TO lt_temp_time.
1579
1580     IF lf_night = 'X'."If yesterday was night shift
1581
1582         ls_temp_time-pernr = ls_final_avail-pernr.
1583         ls_temp_time-begda = ls_final_avail-datum.
1584         ls_temp_time-endda = ls_final_avail-datum.
1585         ls_temp_time-begty = lv_def_beg.
1586         ls_temp_time-endty = lt_day_before-endzt.
1587         APPEND ls_temp_time TO lt_temp_time.
1588
1589     ENDIF.
1590     CLEAR:lf_night.

```

```

1588
1589      LOOP AT lt_pa2007 INTO ls_pa2007 WHERE pernr = ls_final_avail-p
ernr."Overtimes "
1590          ls_temp_time-pernr = ls_pa2007-pernr.
1591          ls_temp_time-begda = ls_pa2007-begda.
1592          ls_temp_time-endda = ls_pa2007-endda.
1593          ls_temp_time-begty = ls_pa2007-beguz.
1594          ls_temp_time-endty = ls_pa2007-enduz.
1595          APPEND ls_temp_time TO lt_temp_time.
1596      ENDLOOP.
1597
1598      SORT lt_temp_time BY begda endda begty endty .
1599
1600      DESCRIBE TABLE lt_temp_time LINES lv_line.
1601
1602      LOOP AT lt_temp_time INTO ls_temp_time.
1603
1604          IF sy-tabix = 1.
1605              ls_time_t-zz_begty = lv_def_beg .
1606              ls_time_t-zz_endty = ls_temp_time-begty.
1607              ls_time_t-zz_reason = '7'##NO_TEXT.
1608
1609              APPEND ls_time_t TO lt_time_t.
1610              CLEAR : ls_time_t-zz_begty , ls_time_t-zz_endty.
1611
1612          ELSE.
1613
1614              lv_def_end1 = ls_temp_time-endty .
1615              ls_time_t-zz_endty = ls_temp_time-begty.
1616              lv_index = sy-tabix - 1.
1617              READ TABLE lt_temp_time INTO ls_temp_time INDEX lv_index.
1618              ls_time_t-zz_begty = ls_temp_time-endty.
1619              ls_time_t-zz_reason = '7'.
1620
1621              APPEND ls_time_t TO lt_time_t.
1622              CLEAR : ls_time_t-zz_begty , ls_time_t-zz_endty.
1623          ENDIF.
1624
1625      ENDLOOP.
1626
1627      ls_time_t-zz_begty = lv_def_end1 .
1628      ls_time_t-zz_endty = lv_def_end.
1629      ls_time_t-zz_reason = '7'##NO_TEXT.
1630      APPEND ls_time_t TO lt_time_t.
1631      CLEAR : ls_time_t-zz_begty , ls_time_t-zz_endty, lv_def_end1 .
1632      CLEAR : ls_time_t.
1633
1634      ELSE. "No overtime Day
1635
1636          ls_time_t-zz_begty = lv_def_beg .
1637          lv_def_beg1 = ls_final_avail-begzt - lv_seconds .
1638          ls_time_t-zz_endty = lv_def_beg1."Before 2 hours
1639          ls_time_t-zz_reason = '7'##NO_TEXT.
1640          APPEND ls_time_t TO lt_time_t."Before Shift
1641          CLEAR: ls_time_t-zz_begty,ls_time_t-zz_endty,ls_time_t-zz_reas
on, lv_def_beg1.
1642
1643          lv_def_end1 = ls_final_avail-endzt."
1644          ls_time_t-zz_begty = lv_def_end1.
1645          ls_time_t-zz_endty = lv_def_end.
1646          ls_time_t-zz_reason = '7'##NO_TEXT.

```

```

1647
1648     APPEND ls_time_t TO lt_time_t."After Shift
1649     CLEAR: ls_time_t.
1650
1651     ENDIF.
1652
1653 ELSE.
1654     "NO shift
1655 *     *****overtime
1656
1657     ls_time_t-zz_pernr = ls_final_avail-pernr.
1658     ls_time_t-zz_begda = ls_final_avail-datum.
1659     ls_time_t-zz_endda = ls_final_avail-datum.
1660     ls_time_t-zz_status = 'B'.
1661
1662 *****Check for day Before
1663     lv_day_before = ls_final_avail-datum - 1.
1664     CALL FUNCTION 'HR_READ_TIMedata_PSP' "fetch yesterday's WSR
1665     EXPORTING
1666         begin_date           = lv_day_before
1667         end_date             = lv_day_before
1668         person_id            = ls_final_avail-pernr
1669     TABLES
1670         pers_avail           = lt_day_before
1671     EXCEPTIONS
1672         no_capacity_available = 1
1673         no_integration_activ  = 2
1674         timeinfo_error       = 3
1675         OTHERS                = 4.
1676
1677     IF sy-subrc = 0.
1678         READ TABLE lt_day_before INDEX 1.
1679         IF lt_day_before-begzt GT lt_day_before-endzt.
1680             lf_night = 'X'.
1681         ENDIF.
1682     ENDIF.
1683
1684     READ TABLE lt_pa2007 INTO ls_pa2007 WITH KEY pernr = ls_final_avai
1685     l-pernr .
1686     IF sy-subrc = 0 AND ( ls_pa2007-begda GE ls_final_avail-datum OR l
1687     s_pa2007-endda LE ls_final_avail-datum ) .
1688         IF ls_pa2007-beguz EQ '' AND ls_pa2007-enduz EQ ''."initial ove
1689         rtime
1690         ELSE." there is overtime mentioned
1691             CLEAR : ls_pa2007.
1692
1693             lv_def_beg1 = ls_final_avail-begzt - lv_seconds .
1694
1695             ls_temp_time-pernr = ls_final_avail-pernr.
1696             ls_temp_time-begda = ls_final_avail-datum.
1697             ls_temp_time-endda = ls_final_avail-datum.
1698             ls_temp_time-begty = lv_def_beg1.
1699             ls_temp_time-endty = ls_final_avail-endzt.
1700             APPEND ls_temp_time TO lt_temp_time.
1701
1702             IF lf_night = 'X'."If yesterday was night shift
1703
1704                 ls_temp_time-pernr = ls_final_avail-pernr.
1705                 ls_temp_time-begda = ls_final_avail-datum.
1706                 ls_temp_time-endda = ls_final_avail-datum.

```

```

1705         ls_temp_time-begty = lv_def_beg.
1706         ls_temp_time-endty = lt_day_before-endzt.
1707         APPEND ls_temp_time TO lt_temp_time.
1708
1709     ENDIF.
1710     CLEAR:lf_night.
1711
1712     LOOP AT lt_pa2007 INTO ls_pa2007 WHERE pernr = ls_final_avail
-pernr."Overtimes "
1713         ls_temp_time-pernr = ls_pa2007-pernr.
1714         ls_temp_time-begda = ls_pa2007-begda.
1715         ls_temp_time-endda = ls_pa2007-endda.
1716         ls_temp_time-begty = ls_pa2007-beguz.
1717         ls_temp_time-endty = ls_pa2007-enduz.
1718         APPEND ls_temp_time TO lt_temp_time.
1719     ENDLOOP.
1720
1721     SORT lt_temp_time BY begda endda begty endty .
1722     DESCRIBE TABLE lt_temp_time LINES lv_line.
1723
1724     LOOP AT lt_temp_time INTO ls_temp_time.
1725
1726         IF sy-tabix = 1.
1727             ls_time_t-zz_begty = lv_def_beg .
1728             ls_time_t-zz_endty = ls_temp_time-begty.
1729             ls_time_t-zz_reason = '7'##NO_TEXT.
1730
1731             APPEND ls_time_t TO lt_time_t.
1732             CLEAR : ls_time_t-zz_begty, ls_time_t-zz_endty.
1733
1734         ELSE.
1735
1736             lv_def_endl = ls_temp_time-endty .
1737             ls_time_t-zz_endty = ls_temp_time-begty.
1738             lv_index = sy-tabix - 1.
1739             READ TABLE lt_temp_time INTO ls_temp_time INDEX lv_index.
1740             ls_time_t-zz_begty = ls_temp_time-endty.
1741             ls_time_t-zz_reason = '7'.
1742
1743             APPEND ls_time_t TO lt_time_t.
1744             CLEAR : ls_time_t-zz_begty, ls_time_t-zz_endty.
1745         ENDIF.
1746
1747     ENDLOOP.
1748     ENDIF."if initial overtime
1749 *****Bfro*****
1750     ELSE."no ovetime
1751         READ TABLE lt_day_before INDEX 1.
1752         IF lt_day_before-begzt GT lt_day_before-endzt."night before
1753
1754             ls_time_t-zz_begty = lt_day_before-endzt.
1755             ls_time_t-zz_endty = lv_def_end.
1756             ls_time_t-zz_reason = '7'##NO_TEXT.
1757             APPEND ls_time_t TO lt_time_t.""Shift
1758             CLEAR: ls_time_t-zz_begty,ls_time_t-zz_endty,ls_time_t-zz_re
ason, lv_def_begl.
1759             CLEAR :ls_time_t.
1760
1761         ELSE.
1762             ls_time_t-zz_begty = lv_def_beg.
1763             ls_time_t-zz_endty = lv_def_end.

```

```

1764         ls_time_t-zz_reason = '7'##NO_TEXT.
1765
1766         APPEND ls_time_t TO lt_time_t."After Shift
1767         CLEAR: ls_time_t.
1768
1769     ENDIF.
1770 ENDIF.
1771
1772 ENDIF.
1773
1774     CLEAR : ls_final_avail.
1775 ENDLOOP.
1776
1777 ENDFORM.
1778 *&-----*
1779 *&         Form   CHECK_2004
1780 *&-----*
1781 *         text
1782 *-----*
1783 *  —>  p1         text
1784 *  <—  p2         text
1785 *-----*
1786 FORM check 2004 .
1787 "Availability
1788 DATA : lt_pa2004 TYPE TABLE OF pa2004,
1789         ls_pa2004 TYPE pa2004.
1790
1791 IF lt_list_pernr IS NOT INITIAL.
1792
1793     SELECT * FROM pa2004 INTO TABLE lt_pa2004
1794     FOR ALL ENTRIES IN lt_list_pernr
1795     WHERE pernr = lt_list_pernr-pernr
1796     AND ( subty = '40' OR subty = '99' )
1797     AND endda GE lv_begda
1798     AND begda LE lv_endda.
1799
1800 ENDIF.
1801
1802 IF lt_pa2004 IS NOT INITIAL.
1803     LOOP AT lt_time_t INTO ls_time_t.
1804
1805         READ TABLE lt_pa2004 INTO ls_pa2004
1806         WITH KEY pernr = ls_time_t-zz_pernr
1807                begda = ls_time_t-zz_begda.
1808
1809         IF ls_pa2004-begda GE ls_time_t-zz_begda
1810             AND ls_pa2004-endda LE ls_time_t-zz_endda.
1811 **as employee is on standby it will be considered as Unblock for day
1812         DELETE lt_time_t WHERE zz_begda = ls_time_t-zz_begda."
1813
1814     ENDIF.
1815 ENDLOOP.
1816 ENDIF.
1817
1818 ENDFORM.
1819 *&-----*
1820 *&         Form   CHECK_2002
1821 *&-----*
1822 *         text
1823 *-----*
1824 *  —>  p1         text

```



```

1825 *  <—  p2          text
1826 *
1827 FORM check_2002 .
1828 "Attendances
1829 DATA : lt_pa2002 TYPE TABLE OF pa2002,
1830         ls_pa2002 TYPE pa2002.
1831
1832 IF lt_list_pernr IS NOT INITIAL.
1833
1834     SELECT * FROM pa2002 INTO TABLE lt_pa2002
1835     FOR ALL ENTRIES IN lt_list_pernr
1836     WHERE pernr = lt_list_pernr-pernr
1837     AND subty  = '0200'
1838     AND endda GE lv_begda
1839     AND begda LE lv_endda.
1840
1841 ENDIF.
1842
1843 IF lt_pa2002 IS NOT INITIAL.
1844     LOOP AT lt_time_t INTO ls_time_t.
1845
1846         READ TABLE lt_pa2002 INTO ls_pa2002
1847         WITH KEY pernr = ls_time_t-zz_pernr
1848                begda = ls_time_t-zz_begda.
1849
1850         IF ls_pa2002-begda GE ls_time_t-zz_begda
1851            AND ls_pa2002-endda LE ls_time_t-zz_endda.
1852 **as employee is on Call-out it will be considered as Unblock for day
1853     DELETE lt_time_t WHERE zz_begda = ls_time_t-zz_begda."
1854
1855     ENDIF.
1856 ENDLOOP.
1857 ENDIF.
1858
1859
1860 ENDFORM.
1861
1862 *&-----*
1863 *&      Form  ADD_QUAL_TIME
1864 *&-----*
1865 *      text
1866 *-----*
1867 *  —>  p1      text
1868 *  <—  p2      text
1869 *-----*
1870 FORM add_qual_time .
1871
1872 *****"Addinng for unblock
1873
1874 DATA : lt_time_u TYPE TABLE OF zppa0088,
1875         ls_time_u TYPE zppa0088.
1876
1877 DATA : lv_index TYPE sy-tabix.
1878 DATA : lv_line2 TYPE i.
1879 DATA : lv_def_beg TYPE sy-uzeit,
1880         lv_def_end TYPE sy-uzeit.
1881 lv_def_beg = '000000'.
1882 lv_def_end = '235959'.
1883
1884 DESCRIBE TABLE lt_time_t LINES lv_line2.
1885 SORT lt_time_t BY zz_pernr zz_begda zz_begtym.

```

```

1886
1887     LOOP AT lt_time_t INTO ls_time_t.
1888
1889         CHECK sy-tabix NE lv_line2.
1890         lv_index = sy-tabix + 1.
1891         READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
1892
1893         CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
1894
1895         IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
1896             ls_time_u-zz_begtym = lv_def_beg.
1897             ls_time_u-zz_endtym = ls_time_t-zz_begtym.
1898             ls_time_u-zz_reason = '9'.
1899             ls_time_u-zz_status = 'U'.
1900             APPEND ls_time_u TO lt_time_u.
1901             CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
1902
1903             ELSEIF ls_time_t-zz_begtym EQ '000000'.
1904                 ls_time_u-zz_endtym = ls_time_u-zz_begtym.
1905                 ls_time_u-zz_begtym = ls_time_t-zz_endtym.
1906                 ls_time_u-zz_reason = '9'.
1907                 ls_time_u-zz_status = 'U'.
1908                 APPEND ls_time_u TO lt_time_u.
1909                 CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
1910
1911             ENDIF.
1912
1913         ENDLOOP.
1914         CLEAR : ls_time_u.
1915         *****
1916
1917         LOOP AT lt_time_t INTO ls_time_t."adding timed block records
1918             MOVE ls_time_t TO ls_final .
1919             APPEND ls_final TO lt_final.
1920             CLEAR: ls_final, ls_time_t.
1921
1922         ENDLOOP.
1923
1924         LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
1925             MOVE ls_time_u TO ls_final .
1926             APPEND ls_final TO lt_final.
1927             CLEAR: ls_final, ls_time_u.
1928
1929         ENDLOOP.
1930
1931         DATA : lv_date_count TYPE i.
1932         DATA : lv_date2 TYPE sy-datum.
1933
1934         lv_date_count = lv_endda - lv_begda.
1935         lv_date_count = lv_date_count + 1.
1936         *
1937
1938         lv_date2 = lv_begda.
1939         DATA : ls_final2 TYPE zppa0088." if missed a day because of qualificat
ion
1940
1941         DO lv_date_count TIMES.

```

```

1942
1943     LOOP AT lt_final INTO ls_final.
1944
1945         READ TABLE lt_final INTO ls_final2
1946             WITH KEY zz_pernr = ls_final-zz_pernr
1947                 zz_begda = lv_date2.
1948
1949         IF sy-subrc <> 0.
1950             ls_final-zz_begda = ls_final-zz_endda = lv_date2.
1951             ls_final-zz_status = 'Ü'.
1952             ls_final-zz_reason = '9'.
1953             ls_final-zz_begty = lv_def_beg.
1954             ls_final-zz_endty = lv_def_end.
1955             APPEND ls_final TO lt_final.
1956         ENDIF.
1957     ENDLOOP.
1958
1959     lv_date2 = lv_date2 + 1.
1960 ENDDO.
1961
1962     SORT lt_final BY zz_pernr zz_begda zz_endda zz_begty zz_endty.
1963
1964 ENDFORM.
1965
1966 *&-----*
1967 *&      Form  BADI_0007
1968 *&-----*
1969 *      text
1970 *-----*
1971 *  →  p1      text
1972 *  ←  p2      text
1973 *-----*
1974 FORM badi_0007 .
1975 *
1976 *PERNR TO LT_EXP_0007
1977 *      BEGDA TO LT_EXP_0007
1978 *      ENDDA TO LT_EXP_0007
1979 *      data  to lt_exp_0007
1980 READ TABLE lt_exp_0007 INTO ls_exp_0007 INDEX 1.
1981
1982 TYPES : BEGIN OF ty_temp_time,
1983         pernr TYPE pernr-pernr,
1984         begda TYPE begda,
1985         endda TYPE endda,
1986         begty TYPE tims,
1987         endty TYPE tims,
1988         END OF ty_temp_time.
1989
1990 DATA : lt_temp_time TYPE TABLE OF ty_temp_time,
1991         ls_temp_time TYPE ty_temp_time,
1992         ls_temp_time1 TYPE ty_temp_time.
1993
1994
1995 DATA: l_tprog(14) TYPE c.
1996 DATA: it_0001 TYPE TABLE OF pa0001,
1997         wa_0001 TYPE pa0001,
1998         wa_t001p TYPE t001p,
1999         wa_t503 TYPE t503.
2000 DATA: it_2003 TYPE STANDARD TABLE OF p2003,
2001         wa_2003 TYPE p2003.
2002

```

```

2003 DATA: lv_begda_temp TYPE sy-datum.
2004 DATA: lv_endda_temp TYPE sy-datum.
2005
2006 DATA: wa_t552a      TYPE t552a,
2007         wa_t550a_su TYPE t550a,
2008         wa_t550a_pw TYPE t550a.
2009
2010 FIELD-SYMBOLS <fs_t552a> TYPE any.
2011 * *****
2012 *
2013 *   LS_EXP_0007-PERNR = LV_PERNR.
2014 *   LS_EXP_0007-BEGDA = LV_BEGDA."09.
2015 *   LS_EXP_0007-ENDDA = LV_ENDDA.
2016 **   IF LS_EXP_0007-DATA IS INITIAL.
2017 *     LS_EXP_0007-DATA = '1008-01D'.
2018 *   ENDIF.
2019
2020 *****
2021 *SELECT SINGLE *
2022
2023 SELECT SINGLE * FROM pa0001 INTO wa_0001
2024         WHERE pernr = ls_exp_0007-pernr
2025         AND begda < ls_exp_0007-begda
2026         AND endda > ls_exp_0007-endda.
2027
2028 SELECT SINGLE * FROM t001p INTO wa_t001p
2029         WHERE werks = wa_0001-werks AND
2030         btrtl = wa_0001-btrtl.
2031
2032 SELECT SINGLE * FROM t503 INTO wa_t503
2033         WHERE persg = wa_0001-persg AND
2034         persk = wa_0001-persk.
2035
2036 *   CALL FUNCTION 'HR_READ_INFOTYPE'
2037 *     EXPORTING
2038 *       TCLAS      = 'A'
2039 *       PERNR      = LS_EXP_0007-PERNR
2040 *       INFTY      = '2003'
2041 *       BEGDA      = LV_BEGDA
2042 *       ENDDA      = LV_ENDDA
2043 *     TABLES
2044 *       INFTY_TAB = IT_2003.
2045 *
2046 *   IF SY-SUBRC EQ 0 AND IT_2003 IS NOT INITIAL.
2047 *
2048 *     LV_BEGDA_TEMP = LV_BEGDA - 1 .
2049 *     LV_ENDDA_TEMP = LV_ENDDA + 1.
2050 *
2051 *     LOOP AT IT_2003 INTO WA_2003.
2052 *       LV_BEGDA = WA_2003-BEGDA.
2053 *       LV_ENDDA = WA_2003-ENDDA.
2054 *
2055 *       WHILE LV_BEGDA LE LV_ENDDA.
2056 *         IF ( LV_BEGDA LT LV_BEGDA_TEMP ) OR ( LV_ENDDA GT LV_ENDDA_TEMP
2057 * ) .
2058 *           LV_BEGDA = LV_BEGDA + 1.
2059 *           CONTINUE.
2060 *         ENDIF.
2061 *         CLEAR: L_TPROG,WA_T550A_SU,WA_T550A_PW .
2062 *
2063 *       IF WA_2003-TPROG IS NOT INITIAL.

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2063 *          SELECT SINGLE * FROM T550A INTO WA_T550A_SU WHERE MOTPR = WA
2064 *          2003-MOTPR AND TPROG = WA_2003-TPROG
2065 *          AND BEGDA LE WA_2003-BEGDA AND ENDDA GE WA_2003-BEGDA.
2066 *          LS_TEMP_TIME-PERNR = ls_exp_0007-PERNR.
2067 *          LS_TEMP_TIME-BEGDA = LV_BEGDA.
2068 *          LS_TEMP_TIME-ENDDA = LV_ENDDA.
2069 *          LS_TEMP_TIME-BEGTY = WA_T550A_SU-sOBEG.
2070 *          LS_TEMP_TIME-ENDTY = WA_T550A_SU-sOend.
2071 *          APPEND LS_TEMP_TIME to LT_TEMP_TIME.
2072 *
2073 *          ELSE.
2074 *          SELECT SINGLE * FROM T552A INTO WA_T552A WHERE ZEITY = WA_T50
2075 *          3-ZEITY AND MOFID = WA_2003-MOFID AND MOSID = WA_2003-MOSID
2076 *          AND SCHKZ = WA_2003-SCHKZ AND KJAHR EQ LV_
2077 *          BEGDA+0(4) AND MONAT = LV_BEGDA+4(2).
2078 *          CONCATENATE 'TPR' LV_BEGDA+6(2) INTO L_TPROG.
2079 *          ASSIGN COMPONENT L_TPROG OF STRUCTURE WA_T552A TO <FS_T552A>.
2080 *          SELECT SINGLE * FROM T550A INTO WA_T550A_SU WHERE MOTPR = WA
2081 *          2003-MOTPR
2082 *          AND TPROG = <FS_T552A> AND BEGDA LE LV_BEGDA AND ENDDA GE
2083 *          LV_BEGDA.
2084 *          ENDIF.
2085 *          LS_TEMP_TIME-PERNR = ls_exp_0007-PERNR.
2086 *          LS_TEMP_TIME-BEGDA = LV_BEGDA.
2087 *          LS_TEMP_TIME-ENDDA = LV_ENDDA.
2088 *          LS_TEMP_TIME-BEGTY = WA_T550A_SU-sOBEG.
2089 *          LS_TEMP_TIME-ENDTY = WA_T550A_SU-sOend.
2090 *          APPEND LS_TEMP_TIME to LT_TEMP_TIME.
2091 *          LV_BEGDA = LV_BEGDA + 1.
2092 *          ENDWHILE.
2093 *          ENDLOOP.
2094 *          ELSEIF IT_2003 IS INITIAL .
2095 *          LV_BEGDA_TEMP = LV_BEGDA - 1 .
2096 *          LV_ENDDA_TEMP = LV_ENDDA + 1.
2097 *          WHILE LV_BEGDA LE LV_ENDDA.
2098 *          IF ( LV_BEGDA LT LV_BEGDA_TEMP ) OR ( LV_ENDDA GT LV_ENDDA_TEMP )
2099 *          .
2100 *          LV_BEGDA = LV_BEGDA + 1.
2101 *          CONTINUE.
2102 *          ENDIF.
2103 *          CLEAR: L_TPROG,WA_T550A_SU,WA_T550A_PW .
2104 *          SELECT SINGLE * FROM T552A INTO WA_T552A WHERE ZEITY = WA_T503-ZE
2105 *          ITY AND MOFID = WA_T001P-mofid "WA_2003-MOFID
2106 *          AND MOSID = WA_T001P-mosid
2107 *          AND SCHKZ = LS_EXP_0007-data AND KJAHR EQ LV_B
2108 *          EGDA+0(4) AND MONAT = LV_BEGDA+4(2).
2109 *          CONCATENATE 'TPR' LV_BEGDA+6(2) INTO L_TPROG.
2110 *          ASSIGN COMPONENT L_TPROG OF STRUCTURE WA_T552A TO <FS_T552A>.
2111 *          SELECT SINGLE * FROM T550A INTO WA_T550A_SU WHERE MOTPR = WA T00
2112 *          1P-mosid AND TPROG = <FS_T552A> AND BEGDA LE LV_BEGDA AND ENDDA GE LV_B
2113 *          EGDA.

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2114 *          LS_TEMP_TIME-PERNR = ls_exp_0007-PERNR.
2115 *          LS_TEMP_TIME-BEGDA = LV_BEGDA.
2116 *          LS_TEMP_TIME-ENDDA = LV_BEGDA.
2117 *          LS_TEMP_TIME-BEGTY = WA_T550A_SU-SOBEG.
2118 *          LS_TEMP_TIME-ENDTY = WA_T550A_SU-SOEND.
2119 *          APPEND LS_TEMP_TIME to LT_TEMP_TIME.
2120 *
2121 *          LV_BEGDA = LV_BEGDA + 1.
2122 *          ENDWHILE.
2123 *
2124 *
2125 *          ENDIF.
2126
2127 DATA : lt_pa2007 TYPE TABLE OF pa2007,
2128         ls_pa2007 TYPE pa2007. "overtime
2129
2130 lv_begda1 = lv_begda - 1.
2131 lv_endda1 = lv_endda - 1.
2132 CONCATENATE 'TPR' lv_begda1+6(2) INTO l_tprog.
2133 ASSIGN COMPONENT 1 tprog OF STRUCTURE wa_t552a TO <fs_t552a>.
2134 SELECT SINGLE * FROM t550a INTO wa_t550a_su WHERE motpr = wa_t001p-mo
sid
2135 AND tprog = <fs_t552a> AND begda LE lv_begda1 AND endda GE lv_be
gdal."if one day before was night
2136
2137 IF wa_t550a_su-sobeg GT wa_t550a_su-soend AND sy-subrc = 0.
2138
2139     ls_temp_time-pernr = ls_exp_0007-pernr.
2140     ls_temp_time-begda = lv_begda1.
2141     ls_temp_time-endda = lv_begda1.
2142     ls_temp_time-begty = lv_def_beg.
2143     ls_temp_time-endty = wa_t550a_su-soend.
2144     APPEND ls_temp_time TO lt_temp_time.
2145     CLEAR : ls_temp_time, ls_pa2007.
2146
2147 ENDIF.
2148
2149 WHILE lv_begda LE lv_endda.
2150
2151     CLEAR: l_tprog, wa_t550a_su, wa_t550a_pw .
2152
2153     CALL FUNCTION 'HR_READ_INFOTYPE' "
2154         EXPORTING
2155             tclas      = 'A'
2156             pernr      = ls_exp_0007-pernr
2157             infy       = '2003'
2158             begda      = lv_begda
2159             endda      = lv_begda
2160     TABLES
2161         infy_tab = it_2003.
2162
2163     READ TABLE it_2003 INTO wa_2003 INDEX 1.
2164     IF sy-subrc = 0 AND wa_2003-tprog IS NOT INITIAL.
2165         SELECT SINGLE * FROM t550a INTO wa_t550a_su WHERE motpr = wa_2003
-motpr AND tprog = wa_2003-tprog
2166         AND begda LE wa_2003-begda AND endda GE wa_2003-begda.
2167
2168         ls_temp_time-pernr = ls_exp_0007-pernr.
2169         ls_temp_time-begda = lv_begda.
2170         ls_temp_time-endda = lv_begda.
2171         ls_temp_time-begty = wa_t550a_su-sobeg.

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```

2172      ls_temp_time-endty = wa_t550a_su-soend.
2173      APPEND ls_temp_time TO lt_temp_time.
2174
2175      ELSE.
2176      SELECT SINGLE * FROM t552a INTO wa_t552a WHERE zeity = wa_t503-zei
ty AND mofid = wa_t001p-mofid AND mosid = wa_t001p-mosid
2177      AND schkz = ls_exp_0007-data AND kjahr EQ lv_b
egda+0(4) AND monat = lv_begda+4(2).
2178
2179      CONCATENATE 'TPR' lv_begda+6(2) INTO l_tprog.
2180      ASSIGN COMPONENT l_tprog OF STRUCTURE wa_t552a TO <fs_t552a>.
2181      SELECT SINGLE * FROM t550a INTO wa_t550a_su WHERE motpr = wa_t001
p-mosid
2182      AND tprog = <fs_t552a> AND begda LE lv_begda AND endda GE lv_b
egda.
2183
2184      IF wa_t550a_su-sobeg GT wa_t550a_su-soend."night
2185
2186      ls_temp_time-pernr = ls_exp_0007-pernr.
2187      ls_temp_time-begda = lv_begda.
2188      ls_temp_time-endda = lv_begda.
2189      ls_temp_time-begty = wa_t550a_su-sobeg - lv_seconds.
2190      ls_temp_time-endty = lv_def_end.
2191      APPEND ls_temp_time TO lt_temp_time.
2192
2193      ls_temp_time-pernr = ls_exp_0007-pernr.
2194      ls_temp_time-begda = lv_begda + 1.
2195      ls_temp_time-endda = lv_begda + 1.
2196      ls_temp_time-begty = lv_def_beg.
2197      ls_temp_time-endty = wa_t550a_su-soend.
2198      APPEND ls_temp_time TO lt_temp_time.
2199
2200      ELSEIF wa_t550a_su-sobeg LT wa_t550a_su-soend."day
2201
2202      ls_temp_time-pernr = ls_exp_0007-pernr.
2203      ls_temp_time-begda = lv_begda.
2204      ls_temp_time-endda = lv_begda.
2205      ls_temp_time-begty = wa_t550a_su-sobeg - lv_seconds .
2206      ls_temp_time-endty = wa_t550a_su-soend.
2207      APPEND ls_temp_time TO lt_temp_time.
2208
2209      ELSE.
2210      *      LS_TEMP_TIME-PERNR = LS_EXP_0007-PERNR.
2211      *      LS_TEMP_TIME-BEGDA = LV_BEGDA.
2212      *      LS_TEMP_TIME-ENDDA = LV_BEGDA.
2213      *      LS_TEMP_TIME-BEGTY = WA_T550A_SU-SOBEG.
2214      *      LS_TEMP_TIME-ENDTY = WA_T550A_SU-SOEND.
2215      *      APPEND LS_TEMP_TIME TO LT_TEMP_TIME.
2216
2217      ls_time_t-zz_pernr = ls_exp_0007-pernr.
2218      ls_time_t-zz_begda = lv_begda.
2219      ls_time_t-zz_endda = lv_begda.
2220      ls_time_t-zz_status = 'B'.
2221      ls_time_t-zz_eg = wa_0001-persg.
2222      ls_time_t-zz_psg = wa_0001-persk.
2223      ls_time_t-zz_begty = lv_def_beg.
2224      ls_time_t-zz_endty = lv_def_end.
2225      ls_time_t-zz_reason = '7'.
2226
2227      APPEND ls_time_t TO lt_time_t.
2228      CLEAR: ls_time_t.

```

```

2229
2230     ENDIF.
2231 ENDIF.
2232
2233 """"overtime
2234 SELECT SINGLE * FROM pa2007 INTO ls_pa2007" if overtime
2235 WHERE pernr = ls_exp_0007-pernr
2236 AND ( subty = '01' OR subty = '02' OR subty = '10' )
2237 AND endda GE lv_begda
2238 AND begda LE lv_begda.
2239
2240 IF sy-subrc = 0.
2241
2242     DELETE lt_time t WHERE zz_begda = lv_begda.
2243     " and BEGTY = ENDTY." removing if off shift have overtime
2244
2245     ls_temp_time-pernr = ls_exp_0007-pernr.
2246     ls_temp_time-begda = lv_begda.
2247     ls_temp_time-endda = lv_begda.
2248     ls_temp_time-begty = ls_pa2007-beguz.
2249     ls_temp_time-endty = ls_pa2007-enduz.
2250     APPEND ls_temp_time TO lt_temp_time.
2251     CLEAR : ls_temp_time, ls_pa2007.
2252 ENDIF.
2253
2254 CLEAR: ls_temp_time.
2255 lv_begda = lv_begda + 1.
2256
2257 ENDWHILE.
2258 DATA:lv_datum      TYPE datum,
2259       lv_datum2     TYPE datum,
2260       total         TYPE i,
2261       count         TYPE i,
2262       lv_time2      TYPE tims,
2263       ls_temp_dummy TYPE ty_temp_time.
2264
2265 SORT lt_temp_time BY begda endda begty endty .
2266 *****begin of changes
2267
2268 CLEAR:lv_datum,lv_datum2,total,count.
2269 total = 0.count = 0.
2270 LOOP AT lt_temp_time INTO ls_temp_time."all shift including overtime
2271     lv_datum = ls_temp_time-begda.
2272     AT NEW begda.
2273         CLEAR : total, count.
2274         LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
2275 -begda.
2276             total = total + 1.
2277         ENDLOOP.
2278     ENDAT.
2279
2280 IF lv_datum = lv_datum2 AND count < total.
2281     ls_time_t-zz_pernr = ls_temp_time-pernr.
2282     ls_time_t-zz_begda = ls_temp_time-begda.
2283     ls_time_t-zz_endda = ls_temp_time-begda.
2284     ls_time_t-zz_status = 'B'.
2285     ls_time_t-zz_eg = wa_0001-persg.
2286     ls_time_t-zz_psg = wa_0001-persk.
2287     ls_time_t-zz_begtym = lv_time2.
2288     ls_time_t-zz_endtym = ls_temp_time-begty.
2289     ls_time_t-zz_reason = '7'.

```



```

2289         IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
2290             APPEND ls_time_t TO lt_time_t.
2291             CLEAR:ls_time_t.
2292         ENDIF.
2293         CLEAR : ls_time_t.
2294         count = count + 1.
2295
2296     ELSEIF count = 0.
2297     *         IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
2298         ls_time_t-zz_pernr = ls_temp_time-pernr.
2299         ls_time_t-zz_begda = ls_temp_time-begda.
2300         ls_time_t-zz_endda = ls_temp_time-begda.
2301         ls_time_t-zz_status = 'B'.
2302         ls_time_t-zz_eg = wa_0001-persg.
2303         ls_time_t-zz_psg = wa_0001-persk.
2304         ls_time_t-zz_begtym = '000000'.
2305         ls_time_t-zz_endtym = ls_temp_time-begty.
2306         ls_time_t-zz_reason = '7'.
2307         APPEND ls_time_t TO lt_time_t.
2308         CLEAR : ls_time_t.
2309
2310     *         ELSE.
2311     *             LS_TIME_T-ZZ_PERNR = LS_TEMP_TIME-PERNR.
2312     *             LS_TIME_T-ZZ_BEGDA = LS_TEMP_TIME-BEGDA.
2313     *             LS_TIME_T-ZZ_ENDDA = LS_TEMP_TIME-BEGDA.
2314     *             LS_TIME_T-ZZ_STATUS = 'B'.
2315     *             LS_TIME_T-ZZ_EG = WA_0001-PERSG.
2316     *             LS_TIME_T-ZZ_PSG = WA_0001-PERSK.
2317     *             LS_TIME_T-ZZ_BEGTYM = '000000'.
2318     *             LS_TIME_T-ZZ_ENDTYM = LV_DEF_END.
2319     *             LS_TIME_T-ZZ_REASON = '5'.
2320     *             APPEND LS_TIME_T TO LT_TIME_T.
2321     *             CLEAR : LS_TIME_T.
2322     *         ENDIF.
2323         count = count + 1.
2324
2325     ENDIF.
2326     IF count = total.
2327         ls_time_t-zz_pernr = ls_temp_time-pernr.
2328         ls_time_t-zz_begda = ls_temp_time-begda.
2329         ls_time_t-zz_endda = ls_temp_time-begda.
2330         ls_time_t-zz_status = 'B'.
2331         ls_time_t-zz_eg = wa_0001-persg.
2332         ls_time_t-zz_psg = wa_0001-persk.
2333         ls_time_t-zz_begtym = ls_temp_time-endty.
2334         ls_time_t-zz_endtym = '235959'.
2335         ls_time_t-zz_reason = '7'.
2336         APPEND ls_time_t TO lt_time_t.
2337         CLEAR : ls_time_t.
2338
2339     ENDIF.
2340
2341     lv_datum2 = lv_datum.
2342     lv_time2 = ls_temp_time-endty.
2343     ENDLÖOP.
2344     *****start      of end
2345
2346     DATA : lt_time_u TYPE TABLE OF zppa0088,
2347           ls_time_u TYPE zppa0088.
2348
2349     DATA :lv_index TYPE sy-tabix.

```

```

2350 DATA : lv_line2 TYPE i.
2351 DATA : lv_def_beg TYPE sy-uzeit,
2352         lv_def_end TYPE sy-uzeit.
2353 lv_def_beg = '000000'.
2354 lv_def_end = '235959'.
2355
2356 DESCRIBE TABLE lt_time_t LINES lv_line2.
2357 SORT lt_time_t BY zz_pernr zz_begda zz_begty.
2358
2359 LOOP AT lt_time_t INTO ls_time_t.
2360
2361     CHECK sy-tabix NE lv_line2.
2362     lv_index = sy-tabix + 1.
2363     READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
2364
2365     CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
2366
2367     IF ls_time_t-zz_begty GT '000000' AND ls_time_t-zz_endty NE '23595
9'.
2368         ls_time_u-zz_begty = lv_def_beg.
2369         ls_time_u-zz_endty = ls_time_t-zz_begty.
2370         ls_time_u-zz_reason = '9'.
2371         ls_time_u-zz_status = 'U'.
2372         APPEND ls_time_u TO lt_time_u.
2373         CLEAR : ls_time_u-zz_begty, ls_time_u-zz_endty, ls_time_u-zz_rea
son.
2374
2375     ELSEIF ls_time_t-zz_begty EQ '000000'.
2376         ls_time_u-zz_endty = ls_time_u-zz_begty.
2377         ls_time_u-zz_begty = ls_time_t-zz_endty.
2378         ls_time_u-zz_reason = '9'.
2379         ls_time_u-zz_status = 'U'.
2380         APPEND ls_time_u TO lt_time_u.
2381         CLEAR : ls_time_u-zz_begty, ls_time_u-zz_endty, ls_time_u-zz_rea
son.
2382
2383     ENDIF.
2384
2385 ENDLOOP.
2386 CLEAR : ls_time_u.
2387 *****
2388 DELETE lt_final WHERE zz_pernr = ls_exp_0007-pernr.
2389
2390
2391 LOOP AT lt_time_t INTO ls_time_t."adding timed block records
2392     MOVE ls_time_t TO ls_final .
2393     APPEND ls_final TO lt_final.
2394     CLEAR: ls_final, ls_time_t.
2395
2396 ENDLOOP.
2397
2398 LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
2399     MOVE ls_time_u TO ls_final .
2400     APPEND ls_final TO lt_final.
2401     CLEAR: ls_final, ls_time_u.
2402
2403 ENDLOOP.
2404
2405 DATA : lt_pa2002 TYPE TABLE OF pa2002,
2406         ls_pa2002 TYPE pa2002.

```

```

2407
2408     TYPES : BEGIN OF ty_date,
2409             begda TYPE begda,
2410             END OF ty_date.
2411     DATA : lt_date TYPE TABLE OF ty_date,
2412             ls_date TYPE ty_date.
2413
2414     SELECT * FROM pa2002 INTO TABLE lt_pa2002
2415 *       FOR ALL ENTRIES IN LT_LIST_PERNR
2416         WHERE pernr = ls_exp_0007-pernr
2417         AND subty = '0200'
2418         AND endda GE lv_begda
2419         AND begda LE lv_endda.
2420
2421     IF lt_pa2002 IS NOT INITIAL.
2422
2423         LOOP AT lt_pa2002 INTO ls_pa2002.
2424             DELETE lt_final WHERE zz_begda GE ls_pa2002-begda AND zz_endda LE
ls_pa2002-endda.
2425             MOVE ls_pa2002-begda TO ls_date-begda.
2426             APPEND ls_date TO lt_date.
2427             MOVE ls_pa2002-endda TO ls_date-begda.
2428             APPEND ls_date TO lt_date.
2429             ENDLOOP.
2430
2431     ENDIF.
2432
2433     DATA : lt_pa2004 TYPE TABLE OF pa2004,
2434             ls_pa2004 TYPE pa2004.
2435     SELECT * FROM pa2004 INTO TABLE lt_pa2004
2436 *       FOR ALL ENTRIES IN LT_LIST_PERNR
2437         WHERE pernr = ls_exp_0007-pernr
2438         AND ( subty = '40' OR subty = '99' )
2439         AND endda GE lv_begda
2440         AND begda LE lv_endda.
2441
2442     IF lt_pa2004 IS NOT INITIAL.
2443
2444         LOOP AT lt_pa2004 INTO ls_pa2004.
2445             DELETE lt_final WHERE zz_begda GE ls_pa2004-begda AND zz_endda LE
ls_pa2004-endda.
2446             MOVE ls_pa2004-begda TO ls_date-begda.
2447             APPEND ls_date TO lt_date.
2448             MOVE ls_pa2004-endda TO ls_date-begda.
2449             APPEND ls_date TO lt_date.
2450             ENDLOOP.
2451
2452     ENDIF.
2453
2454
2455     DELETE ADJACENT DUPLICATES FROM lt_date.
2456
2457     LOOP AT lt_date INTO ls_date.
2458
2459         ls_final-zz_pernr = ls_exp_0007-pernr.
2460         ls_final-zz_begda = ls_date-begda.
2461         ls_final-zz_endda = ls_date-begda.
2462         ls_final-zz_status = 'U'.
2463         ls_final-zz_eg = wa_0001-persg.
2464         ls_final-zz_psg = wa_0001-persk.
2465         ls_final-zz_begtym = '000000'.

```

```

2466     ls_final-zz_endtym = '235959'.
2467     ls_final-zz_reason = '7'.
2468     APPEND ls_final TO lt_final.
2469
2470 ENDLOOP.
2471
2472 SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
2473 REFRESH : lt_date.
2474
2475 DATA : lt_pa2001 TYPE TABLE OF pa2001,
2476         ls_pa2001 TYPE pa2001.
2477 SELECT * FROM pa2001 INTO TABLE lt_pa2001
2478 *      FOR ALL ENTRIES IN LT_LIST_PERNR
2479         WHERE pernr = ls_exp_0007-pernr
2480         AND subty EQ '1070'
2481         AND endda GE lv_begda
2482         AND begda LE lv_endda.
2483
2484 IF lt_pa2001 IS NOT INITIAL.
2485
2486     LOOP AT lt_pa2001 INTO ls_pa2001.
2487         DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_endda LE
ls_pa2001-endda.
2488         MOVE ls_pa2001-begda TO ls_date-begda.
2489         APPEND ls_date TO lt_date.
2490         MOVE ls_pa2001-endda TO ls_date-begda.
2491         APPEND ls_date TO lt_date.
2492     ENDLOOP.
2493
2494 ENDIF.
2495
2496 LOOP AT lt_date INTO ls_date.
2497     ls_final-zz_pernr = ls_exp_0007-pernr.
2498     ls_final-zz_begda = ls_date-begda.
2499     ls_final-zz_endda = ls_date-begda.
2500     ls_final-zz_status = 'B'.
2501     ls_final-zz_eg = wa_0001-persg.
2502     ls_final-zz_psg = wa_0001-persk.
2503     ls_final-zz_begtym = '000000'.
2504     ls_final-zz_endtym = '235959'.
2505     ls_final-zz_reason = '7'.
2506     APPEND ls_final TO lt_final.
2507 ENDLOOP.
2508 SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
2509
2510
2511 * LOOP AT LT_TEMP_TIME INTO LS_TEMP_TIME.
2512 *
2513 *     LS_TIME_T-ZZ_PERNR = LS_TEMP_TIME-PERNR.
2514 *     LS_TIME_T-ZZ_BEGDA = LS_TEMP_TIME-BEGDA.
2515 *     LS_TIME_T-ZZ_ENDDA = LS_TEMP_TIME-BEGDA.
2516 *     LS_TIME_T-ZZ_STATUS = 'B'.
2517 *     LS_TIME_T-ZZ_EG = WA_0001-PERSG.
2518 *     LS_TIME_T-ZZ_PSG = WA_0001-PERSK.
2519 *
2520 *     IF SY-TABIX = 1 AND LS_TIME_T-ZZ_BEGTYM NE '000000'.
2521 *         LS_TIME_T-ZZ_BEGTYM = LV_DEF_BEG .
2522 *         IF LS_TEMP_TIME-BEGTY = ' ' .
2523 *             LS_TEMP_TIME-BEGTY = LV_DEF_BEG.
2524 *         ELSE.
2525 *             LS_TIME_T-ZZ_ENDTYM = LS_TEMP_TIME-BEGTY.

```

```

2526 *      ENDIF.
2527 *      LS_TIME_T-ZZ_REASON = '5'##NO_TEXT.
2528 *
2529 *      APPEND LS_TIME_T TO LT_TIME_T.
2530 *      CLEAR : LS_TIME_T-ZZ_BEGTYM , LS_TIME_T-ZZ_ENDTYM.
2531 *
2532 *      ELSE.
2533 *
2534 *      LV_DEF_END1 = LS_TEMP_TIME-ENDTY .
2535 *      LS_TIME_T-ZZ_ENDTYM = LS_TEMP_TIME-BEGTY.
2536 *      LV_INDEX = SY-TABIX - 1.
2537 *
2538 *      READ TABLE LT_TEMP_TIME INTO LS_TEMP_TIME1 INDEX LV_INDEX.
2539 *      IF LS_TEMP_TIME1-BEGDA = LS_TEMP_TIME-BEGDA.
2540 *          LS_TIME_T-ZZ_BEGTYM = LS_TEMP_TIME-ENDTY.
2541 *      ELSE.
2542 *          LS_TIME_T-ZZ_BEGTYM = LV_DEF_BEG.
2543 *      ENDIF.
2544 *
2545 *      LS_TIME_T-ZZ_REASON = '5'.
2546 *
2547 *      APPEND LS_TIME_T TO LT_TIME_T.
2548 *      CLEAR : LS_TIME_T-ZZ_BEGTYM , LS_TIME_T-ZZ_ENDTYM.
2549 *      ENDIF.
2550 *
2551 *      IF SY-TABIX = 1.
2552 *          LS_TIME_T-ZZ_BEGTYM = LV_DEF_BEG .
2553 *          IF LS_TEMP_TIME-BEGTY = ' ' .
2554 *              LS_TEMP_TIME-BEGTY = LV_DEF_BEG.
2555 *          ELSE.
2556 *              LS_TIME_T-ZZ_ENDTYM = LS_TEMP_TIME-BEGTY.
2557 *          ENDIF.
2558 *          LS_TIME_T-ZZ_REASON = '5'##NO_TEXT.
2559 *
2560 *          APPEND LS_TIME_T TO LT_TIME_T.
2561 *          CLEAR : LS_TIME_T-ZZ_BEGTYM , LS_TIME_T-ZZ_ENDTYM.
2562 *
2563 *      ELSE.
2564 *
2565 *          LV_DEF_END1 = LS_TEMP_TIME-ENDTY .
2566 *
2567 *          IF LS_TEMP_TIME-BEGTY = ' ' .
2568 *              CONTINUE.
2569 *          ELSE.
2570 *              LS_TIME_T-ZZ_ENDTYM = LS_TEMP_TIME-BEGTY.
2571 *          ENDIF.
2572 *          LV_INDEX = SY-TABIX - 1.
2573 *
2574 *          READ TABLE LT_TEMP_TIME INTO LS_TEMP_TIME INDEX LV_INDEX.
2575 *          IF LS_TEMP_TIME-ENDTY = ' ' .
2576 *              LS_TIME_T-ZZ_BEGTYM = LV_DEF_BEG.
2577 *          ELSE.
2578 *              LS_TIME_T-ZZ_BEGTYM = LS_TEMP_TIME-ENDTY.
2579 *          ENDIF.
2580 *          LS_TIME_T-ZZ_REASON = '5'.
2581 *
2582 *          APPEND LS_TIME_T TO LT_TIME_T.
2583 *          CLEAR : LS_TIME_T-ZZ_BEGTYM , LS_TIME_T-ZZ_ENDTYM.
2584 *      ENDIF.
2585 *
2586 *      ENDLOOP.

```

```

2587
2588 ** WHILE .
2589 **
2590 ** ENDDWHILE.
2591 *
2592 ** LOOP at LT_PA2007 INTO LS_PA2007.
2593 **
2594 **     LS_TEMP_TIME-PERNR = LS_EXP_0007-PERNR.
2595 **     LS_TEMP_TIME-BEGDA = LS_PA2007-BEGDA.
2596 **     LS_TEMP_TIME-ENDDA = LS_PA2007-BEGDA.
2597 **     LS_TEMP_TIME-BEGTY = WA_T550A_SU-SOBEG.
2598 **     LS_TEMP_TIME-ENDTY = WA_T550A_SU-SOEND.
2599 **     APPEND LS_TEMP_TIME TO LT_TEMP_TIME.
2600 **
2601 **     CLEAR:LS_TEMP_TIME.
2602 ** ENDLOOP.
2603
2604 ENDFORM.
2605 *&-----*
2606 *&     Form  BADI_2003
2607 *&-----*
2608 *     text
2609 *-----*
2610 *  —> p1      text
2611 *  <— p2      text
2612 *-----*
2613 FORM badi_2003 .
2614
2615 READ TABLE lt_exp_2003 INTO ls_exp_2003 INDEX 1.
2616 DATA: it_0001 TYPE TABLE OF pa0001,
2617       wa_0001 TYPE pa0001.
2618 SELECT SINGLE * FROM pa0001 INTO wa_0001
2619       WHERE pernr = ls_exp_2003-pernr
2620       AND begda < ls_exp_2003-begda
2621       AND endda > ls_exp_2003-endda.
2622
2623 DATA : lt_final_avail TYPE TABLE OF pdpsp.
2624 DATA : ls_final_avail TYPE pdpsp ." WITH HEADER LINE.
2625
2626 CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
2627     EXPORTING
2628         begin_date           = lv_begda
2629         end_date             = lv_endda
2630         person_id            = ls_exp_2003-pernr
2631 *     PERSON_TYP              = 'P'
2632 *     RP_BUILD_PSP_ERROR      = ' '
2633     TABLES
2634         pers_avail           = lt_final_avail " STRUCTURE  PDPSP
2635     EXCEPTIONS
2636         no_capacity_available = 1
2637         no_integration_activ  = 2
2638         timeinfo_error       = 3
2639         OTHERS                = 4 ##FM_SUBRC_OK.
2640
2641 DATA : lv_day_before TYPE begda.
2642 DATA : lt_day_before TYPE TABLE OF  pdpsp WITH HEADER LINE.
2643
2644 LOOP AT lt_final_avail INTO ls_final_avail.
2645
2646     lv_day_before = ls_final_avail-datum - 1.
2647     CALL FUNCTION 'HR_READ_TIMEDATA_PSP' "fetch yesterday's WSR

```

```

2648 EXPORTING
2649     begin_date           = lv_day_before
2650     end_date             = lv_day_before
2651     person_id            = ls_final_avail-pernr
2652 TABLES
2653     pers_avail           = lt_day_before
2654 EXCEPTIONS
2655     no_capacity_available = 1
2656     no_integration_activ  = 2
2657     timeinfo_error       = 3
2658     OTHERS                = 4.
2659
2660 IF sy-subrc = 0.
2661     READ TABLE lt_day_before INDEX 1.
2662     IF lt_day_before-begzt GT lt_day_before-endzt." if yesterday was n
ight
2663         ls_temp_time-pernr = ls_final_avail-pernr.
2664         ls_temp_time-begda = ls_final_avail-datum.
2665         ls_temp_time-endda = ls_final_avail-datum.
2666         ls_temp_time-begty = lv_def_beg.
2667         ls_temp_time-endty = lt_day_before-endzt.
2668         APPEND ls_temp_time TO lt_temp_time.
2669     ENDIF.
2670 ENDIF.
2671
2672 IF    ls_final_avail-begzt GT ls_final_avail-endzt."night
2673
2674     lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
2675     ls_temp_time-pernr = ls_exp_2003-pernr."normal shifts
2676     ls_temp_time-begda = ls_final_avail-datum.
2677     ls_temp_time-endda = ls_final_avail-datum.
2678     ls_temp_time-begty = lv_def_beg1.
2679     ls_temp_time-endty = lv_def_end.
2680     APPEND ls_temp_time TO lt_temp_time.
2681
2682 ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt."day
2683
2684     lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
2685
2686     ls_temp_time-pernr = ls_exp_2003-pernr."normal shifts
2687     ls_temp_time-begda = ls_final_avail-datum.
2688     ls_temp_time-endda = ls_final_avail-datum.
2689     ls_temp_time-begty = lv_def_beg1.
2690     ls_temp_time-endty = ls_final_avail-endzt.
2691     APPEND ls_temp_time TO lt_temp_time.
2692
2693 ELSE.
2694
2695     ls_time_t-zz_pernr = ls_exp_2003-pernr.
2696     ls_time_t-zz_begda = lv_begda.
2697     ls_time_t-zz_endda = lv_begda.
2698     ls_time_t-zz_status = 'B'.
2699     ls_time_t-zz_eg     = wa_0001-persg.
2700     ls_time_t-zz_psg    = wa_0001-persk.
2701     ls_time_t-zz_begtym = lv_def_beg.
2702     ls_time_t-zz_endtym = lv_def_end.
2703     ls_time_t-zz_reason = '7'.
2704
2705     APPEND ls_time_t TO lt_time_t.
2706     CLEAR: ls_time_t.
2707

```

```

2708     ENDIF.
2709
2710
2711     SELECT SINGLE * FROM pa2007 INTO ls_pa2007" if overtime
2712     WHERE pernr = ls_exp_2003-pernr
2713     AND ( subty = '01' OR subty = '02' OR subty = '10' )
2714     AND endda GE ls_final_avail-datum
2715     AND begda LE ls_final_avail-datum.
2716
2717     IF sy-subrc = 0.
2718
2719         DELETE lt_time t WHERE zz_begda = ls_final_avail-datum.
2720         " and BEGTY = ENDTY." removing if off shift have overtime
2721         ls_temp_time-pernr = ls_exp_2003-pernr.
2722         ls_temp_time-begda = lv_begda.
2723         ls_temp_time-endda = lv_begda.
2724         ls_temp_time-begty = ls_pa2007-beguz.
2725         ls_temp_time-enty = ls_pa2007-enduz.
2726         APPEND ls_temp_time TO lt_temp_time.
2727         CLEAR : ls_temp_time, ls_pa2007.
2728
2729     ENDIF.
2730
2731     CLEAR: ls_temp_time.
2732
2733 ENDLOOP.
2734 * *****begin of changes
2735 DATA:lv_datum      TYPE datum,
2736       lv_datum2     TYPE datum,
2737       total         TYPE i,
2738       count         TYPE i,
2739       lv_time2      TYPE tims,
2740       ls_temp_dummy TYPE ty_temp_time.
2741
2742 SORT lt_temp_time BY begda endda begty endty .
2743
2744 CLEAR:lv_datum,lv_datum2,total,count.
2745 total = 0.count = 0.
2746 LOOP AT lt_temp_time INTO ls_temp_time.
2747     lv_datum = ls_temp_time-begda.
2748     AT NEW begda.
2749         CLEAR : total, count.
2750         LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
-begda.
2751             total = total + 1.
2752         ENDLOOP.
2753     ENDAT.
2754
2755     IF lv_datum = lv_datum2 AND count < total.
2756         ls_time_t-zz_pernr = ls_temp_time-pernr.
2757         ls_time_t-zz_begda = ls_temp_time-begda.
2758         ls_time_t-zz_endda = ls_temp_time-begda.
2759         ls_time_t-zz_status = 'B'.
2760         ls_time_t-zz_eg = wa_0001-persg.
2761         ls_time_t-zz_psg = wa_0001-persk.
2762         ls_time_t-zz_begtym = lv_time2.
2763         ls_time_t-zz_endtym = ls_temp_time-begty.
2764         ls_time_t-zz_reason = '7'.
2765         IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
2766             APPEND ls_time_t TO lt_time_t.
2767             CLEAR:ls_time_t.

```



```

2768         ENDIF.
2769         CLEAR : ls_time_t.
2770         count = count + 1.
2771
2772     ELSEIF count = 0.
2773 *         IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
2774             ls_time_t-zz_pernr = ls_temp_time-pernr.
2775             ls_time_t-zz_begda = ls_temp_time-begda.
2776             ls_time_t-zz_endda = ls_temp_time-begda.
2777             ls_time_t-zz_status = 'B'.
2778             ls_time_t-zz_eg = wa_0001-persg.
2779             ls_time_t-zz_psg = wa_0001-persk.
2780             ls_time_t-zz_begtym = '000000'.
2781             ls_time_t-zz_endtym = ls_temp_time-begty.
2782             ls_time_t-zz_reason = '7'.
2783             APPEND ls_time_t TO lt_time_t.
2784             CLEAR : ls_time_t.
2785
2786             count = count + 1.
2787
2788         ENDIF.
2789         IF count = total.
2790             ls_time_t-zz_pernr = ls_temp_time-pernr.
2791             ls_time_t-zz_begda = ls_temp_time-begda.
2792             ls_time_t-zz_endda = ls_temp_time-begda.
2793             ls_time_t-zz_status = 'B'.
2794             ls_time_t-zz_eg = wa_0001-persg.
2795             ls_time_t-zz_psg = wa_0001-persk.
2796             ls_time_t-zz_begtym = ls_temp_time-endty.
2797             ls_time_t-zz_endtym = '235959'.
2798             ls_time_t-zz_reason = '7'.
2799             APPEND ls_time_t TO lt_time_t.
2800             CLEAR : ls_time_t.
2801
2802         ENDIF.
2803
2804         lv_datum2 = lv_datum.
2805         lv_time2 = ls_temp_time-endty.
2806     ENDLÖOP.
2807
2808     DATA : lt_time_u TYPE TABLE OF zppa0088,
2809            ls_time_u TYPE zppa0088.
2810
2811     DATA : lv_index TYPE sy-tabix.
2812     DATA : lv_line2 TYPE i.
2813     DATA : lv_def_beg TYPE sy-uzeit,
2814            lv_def_end TYPE sy-uzeit.
2815     lv_def_beg = '000000'.
2816     lv_def_end = '235959'.
2817
2818     DESCRIBE TABLE lt_time_t LINES lv_line2.
2819     SORT lt_time_t BY zz_pernr zz_begda zz_begtym.
2820
2821     LOOP AT lt_time_t INTO ls_time_t.
2822
2823         CHECK sy-tabix NE lv_line2.
2824         lv_index = sy-tabix + 1.
2825         READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
2826
2827         CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.

```

```

2828
2829     IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
2830         ls_time_u-zz_begtym = lv_def_beg.
2831         ls_time_u-zz_endtym = ls_time_t-zz_begtym.
2832         ls_time_u-zz_reason = '9'.
2833         ls_time_u-zz_status = 'U'.
2834         APPEND ls_time_u TO lt_time_u.
2835         CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
2836
2837     ELSEIF ls_time_t-zz_begtym EQ '000000'.
2838         ls_time_u-zz_endtym = ls_time_u-zz_begtym.
2839         ls_time_u-zz_begtym = ls_time_t-zz_endtym.
2840         ls_time_u-zz_reason = '9'.
2841         ls_time_u-zz_status = 'U'.
2842         APPEND ls_time_u TO lt_time_u.
2843         CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
2844
2845     ENDIF.
2846
2847     ENDLOOP.
2848     CLEAR : ls_time_u.
2849     *****
2850     DELETE lt_final WHERE zz_pernr = ls_exp_2003-pernr.
2851     LOOP AT lt_time_t INTO ls_time_t."adding timed block records
2852         MOVE ls_time_t TO ls_final .
2853         APPEND ls_final TO lt_final.
2854         CLEAR: ls_final, ls_time_t.
2855
2856     ENDLOOP.
2857
2858     LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
2859         MOVE ls_time_u TO ls_final .
2860         APPEND ls_final TO lt_final.
2861         CLEAR: ls_final, ls_time_u.
2862
2863     ENDLOOP.
2864
2865     DATA : lt_pa2002 TYPE TABLE OF pa2002,
2866             ls_pa2002 TYPE pa2002.
2867
2868     TYPES : BEGIN OF ty_date,
2869             begda TYPE begda,
2870             END OF ty_date.
2871     DATA : lt_date TYPE TABLE OF ty_date,
2872             ls_date TYPE ty_date.
2873
2874     SELECT * FROM pa2002 INTO TABLE lt_pa2002
2875     *     IF 2002 present
2876         WHERE pernr = ls_exp_2003-pernr
2877         AND subty = '0200'
2878         AND endda GE lv_begda
2879         AND begda LE lv_endda.
2880
2881     IF lt_pa2002 IS NOT INITIAL.
2882
2883         LOOP AT lt_pa2002 INTO ls_pa2002.
2884             DELETE lt_final WHERE zz_begda GE ls_pa2002-begda AND zz_endda LE
ls_pa2002-endda.

```

```

2885         MOVE ls_pa2002-begda TO ls_date-begda.
2886         APPEND ls_date TO lt_date.
2887         MOVE ls_pa2002-enddda TO ls_date-begda.
2888         APPEND ls_date TO lt_date.
2889     ENDLOOP.
2890
2891 ENDIF.
2892
2893 DATA : lt_pa2004 TYPE TABLE OF pa2004,
2894         ls_pa2004 TYPE pa2004. " if 2004 present
2895 SELECT * FROM pa2004 INTO TABLE lt_pa2004
2896 *     FOR ALL ENTRIES IN LT_LIST_PERNR
2897     WHERE pernr = ls_exp_2003-pernr
2898     AND ( subty = '40' OR subty = '99' )
2899     AND enddda GE lv_begda
2900     AND begda LE lv_enddda.
2901
2902 IF lt_pa2004 IS NOT INITIAL.
2903
2904     LOOP AT lt_pa2004 INTO ls_pa2004.
2905         DELETE lt_final WHERE zz_begda GE ls_pa2004-begda AND zz_enddda LE
ls_pa2004-enddda.
2906         MOVE ls_pa2002-begda TO ls_date-begda.
2907         APPEND ls_date TO lt_date.
2908         MOVE ls_pa2002-enddda TO ls_date-begda.
2909         APPEND ls_date TO lt_date.
2910     ENDLOOP.
2911
2912 ENDIF.
2913
2914 DELETE ADJACENT DUPLICATES FROM lt_date.
2915
2916 LOOP AT lt_date INTO ls_date.
2917     ls_final-zz_pernr = ls_exp_2003-pernr.
2918     ls_final-zz_begda = ls_date-begda.
2919     ls_final-zz_enddda = ls_date-begda.
2920     ls_final-zz_status = 'U'.
2921     ls_final-zz_eg = wa_0001-persg.
2922     ls_final-zz_psg = wa_0001-persk.
2923     ls_final-zz_begtym = '000000'.
2924     ls_final-zz_endtym = '235959'.
2925     ls_final-zz_reason = '7'.
2926     APPEND ls_final TO lt_final.
2927 ENDLOOP.
2928
2929 SORT lt_final BY zz_begda zz_enddda zz_begtym zz_endtym.
2930
2931
2932 REFRESH : lt_date.
2933 DATA : lt_pa2001 TYPE TABLE OF pa2001,
2934         ls_pa2001 TYPE pa2001.
2935 SELECT * FROM pa2001 INTO TABLE lt_pa2001
2936 *     FOR ALL ENTRIES IN LT_LIST_PERNR
2937     WHERE pernr = ls_exp_2003-pernr
2938     AND subty EQ '1070'
2939     AND enddda GE lv_begda
2940     AND begda LE lv_enddda.
2941
2942 IF lt_pa2001 IS NOT INITIAL.
2943     LOOP AT lt_pa2001 INTO ls_pa2001.
2944         DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_enddda LE

```

```

ls_pa2001-enddda.
2945     MOVE ls_pa2001-begda TO ls_date-begda.
2946     APPEND ls_date TO lt_date.
2947     MOVE ls_pa2001-enddda TO ls_date-begda.
2948     APPEND ls_date TO lt_date.
2949     ENDLOOP.
2950     ENDIF.
2951
2952     LOOP AT lt_date INTO ls_date.
2953         ls_final-zz_pernr = ls_exp_2003-pernr.
2954         ls_final-zz_begda = ls_date-begda.
2955         ls_final-zz_endda = ls_date-begda.
2956         ls_final-zz_status = 'B'.
2957         ls_final-zz_eg = wa_0001-persg.
2958         ls_final-zz_psg = wa_0001-persk.
2959         ls_final-zz_begtym = '000000'.
2960         ls_final-zz_endtym = '235959'.
2961         ls_final-zz_reason = '7'.
2962         APPEND ls_final TO lt_final.
2963     ENDLOOP.
2964     SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
2965
2966     *****      Start of end
2967
2968
2969     ENDFORM.
2970     *&-----*
2971     *&      Form  BADI_2004*
2972     *&-----*
2973     *      text*
2974     *-----*
2975     *  ->  p1      text*
2976     *  <-  p2      text*
2977     *-----*
2978     FORM badi_2004 .
2979
2980     READ TABLE lt_exp_2004 INTO ls_exp_2004 INDEX 1.
2981     DATA: it_0001 TYPE TABLE OF pa0001,
2982           wa_0001 TYPE pa0001.
2983     SELECT SINGLE * FROM pa0001 INTO wa_0001
2984           WHERE pernr = ls_exp_2004-pernr
2985           AND begda < ls_exp_2004-begda
2986           AND endda > ls_exp_2004-endda.
2987
2988     DATA : lt_final_avail TYPE TABLE OF pdpsp.
2989     DATA : ls_final_avail TYPE pdpsp ." WITH HEADER LINE.
2990
2991     CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
2992         EXPORTING
2993             begin_date           = lv_begda
2994             end_date             = lv_endda
2995             person_id            = ls_exp_2004-pernr
2996     *     PERSON_TYP              = 'P'
2997     *     RP_BUILD_PSP_ERROR      = ' '
2998     TABLES
2999         pers_avail              = lt_final_avail " STRUCTURE  PDPSP
3000     EXCEPTIONS
3001         no_capacity_available   = 1
3002         no_integration_activ    = 2
3003         timeinfo_error         = 3
3004         OTHERS                  = 4 ##FM_SUBRC_OK.

```

```

3005
3006 DATA : lv_day_before TYPE begda.
3007 DATA : lt_day_before TYPE TABLE OF pdpsp WITH HEADER LINE.
3008
3009 LOOP AT lt_final_avail INTO ls_final_avail.
3010
3011     lv_day_before = ls_final_avail-datum - 1.
3012     CALL FUNCTION 'HR_READ_TIMedata_PSP' "fetch yesterday's WSR
3013         EXPORTING
3014             begin_date           = lv_day_before
3015             end_date             = lv_day_before
3016             person_id            = ls_final_avail-pernr
3017         TABLES
3018             pers_avail           = lt_day_before
3019         EXCEPTIONS
3020             no_capacity_available = 1
3021             no_integration_activ  = 2
3022             timeinfo_error       = 3
3023             OTHERS               = 4 ##FM_SUBRC_OK.
3024
3025     IF sy-subrc = 0.
3026         READ TABLE lt_day_before INDEX 1.
3027         IF lt_day_before-begzt GT lt_day_before-endzt." if yesterday was n
ight
3028             ls_temp_time-pernr = ls_final_avail-pernr.
3029             ls_temp_time-begda = ls_final_avail-datum.
3030             ls_temp_time-endda = ls_final_avail-datum.
3031             ls_temp_time-begty = lv_def_beg.
3032             ls_temp_time-endty = lt_day_before-endzt.
3033             APPEND ls_temp_time TO lt_temp_time.
3034         ENDIF.
3035     ENDIF.
3036
3037     IF ls_final_avail-begzt GT ls_final_avail-endzt."night
3038
3039         lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3040         ls_temp_time-pernr = ls_exp_2004-pernr."normal shifts
3041         ls_temp_time-begda = ls_final_avail-datum.
3042         ls_temp_time-endda = ls_final_avail-datum.
3043         ls_temp_time-begty = lv_def_beg1.
3044         ls_temp_time-endty = lv_def_end.
3045         APPEND ls_temp_time TO lt_temp_time.
3046
3047     ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt."day
3048
3049         lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3050
3051         ls_temp_time-pernr = ls_exp_2004-pernr."normal shifts
3052         ls_temp_time-begda = ls_final_avail-datum.
3053         ls_temp_time-endda = ls_final_avail-datum.
3054         ls_temp_time-begty = lv_def_beg1.
3055         ls_temp_time-endty = ls_final_avail-endzt.
3056         APPEND ls_temp_time TO lt_temp_time.
3057
3058     ELSE.
3059
3060         ls_time_t-zz_pernr = ls_exp_2004-pernr.
3061         ls_time_t-zz_begda = lv_begda.
3062         ls_time_t-zz_endda = lv_begda.
3063         ls_time_t-zz_status = 'B'.
3064         ls_time_t-zz_eg = wa_0001-persg.

```

```

3065      ls_time_t-zz_psg = wa_0001-persk.
3066      ls_time_t-zz_begtym = lv_def_beg.
3067      ls_time_t-zz_endtym = lv_def_end.
3068      ls_time_t-zz_reason = '7'.
3069
3070      APPEND ls_time_t TO lt_time_t.
3071      CLEAR: ls_time_t.
3072
3073  ENDIF.
3074
3075
3076  SELECT SINGLE * FROM pa2007 INTO ls_pa2007" if overtime
3077  WHERE pernr = ls_exp_2004-pernr
3078  AND ( subty = '01' OR subty = '02' OR subty = '10' )
3079  AND endda GE ls_final_avail-datum
3080  AND begda LE ls_final_avail-datum.
3081
3082  IF sy-subrc = 0.
3083
3084      DELETE lt_time_t WHERE zz_begda = ls_final_avail-datum.
3085      " and BEGTY = ENDTY." removing if off shift have overtime
3086      ls_temp_time-pernr = ls_exp_2004-pernr.
3087      ls_temp_time-begda = lv_begda.
3088      ls_temp_time-endda = lv_begda.
3089      ls_temp_time-begty = ls_pa2007-beguz.
3090      ls_temp_time-endty = ls_pa2007-enduz.
3091      APPEND ls_temp_time TO lt_temp_time.
3092      CLEAR : ls_temp_time, ls_pa2007.
3093
3094  ENDIF.
3095
3096  CLEAR: ls_temp_time.
3097
3098  ENDLOOP.
3099  * *****begin of changes
3100  DATA:lv_datum      TYPE datum,
3101        lv_datum2     TYPE datum,
3102        total         TYPE i,
3103        count         TYPE i,
3104        lv_time2       TYPE tims,
3105        ls_temp_dummy TYPE ty_temp_time.
3106
3107  SORT lt_temp_time BY begda endda begty endty .
3108
3109
3110  CLEAR:lv_datum,lv_datum2,total,count.
3111  total = 0.count = 0.
3112  LOOP AT lt_temp_time INTO ls_temp_time.
3113      lv_datum = ls_temp_time-begda.
3114      AT NEW begda.
3115          CLEAR : total, count.
3116          LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
3117          -begda.
3118              total = total + 1.
3119          ENDLOOP.
3120      ENDAT.
3121
3122  IF lv_datum = lv_datum2 AND count < total.
3123      ls_time_t-zz_pernr = ls_temp_time-pernr.
3124      ls_time_t-zz_begda = ls_temp_time-begda.
3125      ls_time_t-zz_endda = ls_temp_time-begda.

```

```

3125     ls_time_t-zz_status = 'B'.
3126     ls_time_t-zz_eg = wa_0001-persg.
3127     ls_time_t-zz_psg = wa_0001-persk.
3128     ls_time_t-zz_begtym = lv_time2.
3129     ls_time_t-zz_endtym = ls_temp_time-begty.
3130     ls_time_t-zz_reason = '7'.
3131     IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
3132         APPEND ls_time_t TO lt_time_t.
3133         CLEAR:ls_time_t.
3134     ENDIF.
3135     CLEAR : ls_time_t.
3136     count = count + 1.
3137
3138 ELSEIF count = 0.
3139 *     IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
3140         ls_time_t-zz_pernr = ls_temp_time-pernr.
3141         ls_time_t-zz_begda = ls_temp_time-begda.
3142         ls_time_t-zz_endda = ls_temp_time-begda.
3143         ls_time_t-zz_status = 'B'.
3144         ls_time_t-zz_eg = wa_0001-persg.
3145         ls_time_t-zz_psg = wa_0001-persk.
3146         ls_time_t-zz_begtym = '000000'.
3147         ls_time_t-zz_endtym = ls_temp_time-begty.
3148         ls_time_t-zz_reason = '7'.
3149         APPEND ls_time_t TO lt_time_t.
3150         CLEAR : ls_time_t.
3151
3152         count = count + 1.
3153
3154     ENDIF.
3155     IF count = total.
3156         ls_time_t-zz_pernr = ls_temp_time-pernr.
3157         ls_time_t-zz_begda = ls_temp_time-begda.
3158         ls_time_t-zz_endda = ls_temp_time-begda.
3159         ls_time_t-zz_status = 'B'.
3160         ls_time_t-zz_eg = wa_0001-persg.
3161         ls_time_t-zz_psg = wa_0001-persk.
3162         ls_time_t-zz_begtym = ls_temp_time-endty.
3163         ls_time_t-zz_endtym = '235959'.
3164         ls_time_t-zz_reason = '7'.
3165         APPEND ls_time_t TO lt_time_t.
3166         CLEAR : ls_time_t.
3167
3168     ENDIF.
3169
3170     lv_datum2 = lv_datum.
3171     lv_time2 = ls_temp_time-endty.
3172 ENDLÖOP.
3173
3174 DATA : lt_time_u TYPE TABLE OF zppa0088,
3175         ls_time_u TYPE zppa0088.
3176
3177 DATA : lv_index TYPE sy-tabix.
3178 DATA : lv_line2 TYPE i.
3179 DATA : lv_def_beg TYPE sy-uzeit,
3180         lv_def_end TYPE sy-uzeit.
3181 lv_def_beg = '000000'.
3182 lv_def_end = '235959'.
3183
3184 DESCRIBE TABLE lt_time_t LINES lv_line2.
3185 SORT lt_time_t BY zz_pernr zz_begda zz_begtym.

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```

3186
3187     LOOP AT lt_time_t INTO ls_time_t.
3188
3189         CHECK sy-tabix NE lv_line2.
3190         lv_index = sy-tabix + 1.
3191         READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
3192
3193         CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
3194
3195         IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
3196             ls_time_u-zz_begtym = lv_def_beg.
3197             ls_time_u-zz_endtym = ls_time_t-zz_begtym.
3198             ls_time_u-zz_reason = '9'.
3199             ls_time_u-zz_status = 'U'.
3200             APPEND ls_time_u TO lt_time_u.
3201             CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
3202
3203             ELSEIF ls_time_t-zz_begtym EQ '000000'.
3204                 ls_time_u-zz_endtym = ls_time_u-zz_begtym.
3205                 ls_time_u-zz_begtym = ls_time_t-zz_endtym.
3206                 ls_time_u-zz_reason = '9'.
3207                 ls_time_u-zz_status = 'U'.
3208                 APPEND ls_time_u TO lt_time_u.
3209                 CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
3210
3211             ENDIF.
3212
3213         ENDLOOP.
3214         CLEAR : ls_time_u.
3215         *****
3216         DELETE lt_final WHERE zz_pernr = ls_exp_2004-pernr.
3217         LOOP AT lt_time_t INTO ls_time_t."adding timed block records
3218             MOVE ls_time_t TO ls_final .
3219             APPEND ls_final TO lt_final.
3220             CLEAR: ls_final, ls_time_t.
3221
3222         ENDLOOP.
3223
3224         LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
3225             MOVE ls_time_u TO ls_final .
3226             APPEND ls_final TO lt_final.
3227             CLEAR: ls_final, ls_time_u.
3228
3229         ENDLOOP.
3230
3231         DATA : lt_pa2002 TYPE TABLE OF pa2002,
3232                 ls_pa2002 TYPE pa2002.
3233
3234         TYPES : BEGIN OF ty_date,
3235                   begda TYPE begda,
3236                   END OF ty_date.
3237         DATA : lt_date TYPE TABLE OF ty_date,
3238                 ls_date TYPE ty_date.
3239
3240         SELECT * FROM pa2002 INTO TABLE lt_pa2002
3241         *     IF 2002 present
3242             WHERE pernr = ls_exp_0007-pernr

```



```

3243     AND subty = '0200'
3244     AND endda GE lv_begda
3245     AND begda LE lv_endda.
3246
3247     IF lt_pa2002 IS NOT INITIAL.
3248
3249         LOOP AT lt_pa2002 INTO ls_pa2002.
3250             DELETE lt_final WHERE zz_begda GE ls_pa2002-begda AND zz_endda LE
ls_pa2002-endda.
3251             MOVE ls_pa2002-begda TO ls_date-begda.
3252             APPEND ls_date TO lt_date.
3253             MOVE ls_pa2002-endda TO ls_date-begda.
3254             APPEND ls_date TO lt_date.
3255         ENDLOOP.
3256
3257     ENDIF.
3258
3259 *   DATA : LT_PA2004 TYPE TABLE OF PA2004,
3260 *           LS_PA2004 TYPE PA2004." if 2004 present
3261 *   SELECT * FROM PA2004 INTO TABLE LT_PA2004
3262 **       FOR ALL ENTRIES IN LT_LIST_PERNR
3263 *       WHERE PERNR = LS_EXP_0007-PERNR
3264 *       AND ( SUBTY = '40' OR SUBTY = '99' )
3265 *       AND ENDDA GE LV_BEGDA
3266 *       AND BEGDA LE LV_ENDDA.
3267 *
3268 *   IF LT_PA2004 IS NOT INITIAL.
3269 *
3270 *       LOOP AT LT_PA2004 INTO LS_PA2004.
3271 *           DELETE LT_FINAL WHERE zz_BEGDA ge LS_PA2004-BEGDA and zz_ENDDA le
LS_PA2004-ENDDA.
3272 *           move ls_pa2002-BEGDA to ls_date-BEGDA.
3273 *           APPEND ls_date to lt_date.
3274 *           move LS_PA2002-ENDDA to ls_date-BEGDA.
3275 *           APPEND ls_date to lt_date.
3276 *           ENDLOOP.
3277 *
3278 *   ENDIF.
3279 lv_begda1 = ls_exp_2004-begda.
3280
3281 WHILE lv_begda1 LE ls_exp_2004-endda.
3282     MOVE lv_begda1 TO ls_date-begda.
3283     APPEND ls_date TO lt_date.
3284     lv_begda1 = lv_begda1 + 1.
3285 ENDWHILE.
3286
3287 DELETE ADJACENT DUPLICATES FROM lt_date.
3288
3289 LOOP AT lt_date INTO ls_date.
3290     DELETE lt_final WHERE zz_begda EQ ls_date-begda.
3291
3292     ls_final-zz_pernr = ls_exp_2004-pernr.
3293     ls_final-zz_begda = ls_date-begda.
3294     ls_final-zz_endda = ls_date-begda.
3295     ls_final-zz_status = 'U'.
3296     ls_final-zz_eg = wa_0001-persg.
3297     ls_final-zz_psg = wa_0001-persk.
3298     ls_final-zz_begty = '000000'.
3299     ls_final-zz_endty = '235959'.
3300     ls_final-zz_reason = '7'.
3301     APPEND ls_final TO lt_final.

```

```

3302     ENDLOOP.
3303
3304     SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
3305
3306     REFRESH : lt_date.
3307     DATA : lt_pa2001 TYPE TABLE OF pa2001,
3308             ls_pa2001 TYPE pa2001.
3309     SELECT * FROM pa2001 INTO TABLE lt_pa2001
3310 *       FOR ALL ENTRIES IN LT_LIST_PERNR
3311         WHERE pernr = ls_exp_2004-pernr
3312         AND subty EQ '1070'
3313         AND endda GE lv_begda
3314         AND begda LE lv_endda.
3315
3316     IF lt_pa2001 IS NOT INITIAL.
3317         LOOP AT lt_pa2001 INTO ls_pa2001.
3318             DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_endda LE
3319 ls_pa2001-endda.
3320             MOVE ls_pa2001-begda TO ls_date-begda.
3321             APPEND ls_date TO lt_date.
3322             MOVE ls_pa2001-endda TO ls_date-begda.
3323             APPEND ls_date TO lt_date.
3324         ENDLOOP.
3325     ENDIF.
3326
3327     LOOP AT lt_date INTO ls_date.
3328
3329         DELETE lt_final WHERE zz_begda EQ ls_date-begda.
3330
3331         ls_final-zz_pernr = ls_exp_2004-pernr.
3332         ls_final-zz_begda = ls_date-begda.
3333         ls_final-zz_endda = ls_date-begda.
3334         ls_final-zz_status = 'B'.
3335         ls_final-zz_eg = wa_0001-persg.
3336         ls_final-zz_psg = wa_0001-persk.
3337         ls_final-zz_begtym = '000000'.
3338         ls_final-zz_endtym = '235959'.
3339         ls_final-zz_reason = '7'.
3340         APPEND ls_final TO lt_final.
3341     ENDLOOP.
3342     SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
3343
3344     ***** Start of end
3345
3346     ENDFORM.
3347 *-----*
3348 *       Form BADI_2002
3349 *-----*
3350 *       text
3351 *-----*
3352 *   →  p1      text
3353 *   ←  p2      text
3354 *-----*
3355     FORM badi_2002 .
3356
3357         READ TABLE lt_exp_2002 INTO ls_exp_2002 INDEX 1.
3358         DATA: it_0001 TYPE TABLE OF pa0001,
3359                 wa_0001 TYPE pa0001.
3360         SELECT SINGLE * FROM pa0001 INTO wa_0001
3361             WHERE pernr = ls_exp_2002-pernr
3362             AND begda < ls_exp_2002-begda

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"IT2003

```

3362         AND endda > ls_exp_2002-enda.
3363
3364 DATA : lt_final_avail TYPE TABLE OF pdpsp.
3365 DATA : ls_final_avail TYPE pdpsp ." WITH HEADER LINE.
3366
3367 CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
3368     EXPORTING
3369         begin_date           = lv_begda
3370         end_date             = lv_endda
3371         person_id            = ls_exp_2002-pernr
3372 *         PERSON_TYP          = 'P'
3373 *         RP_BUILD_PSP_ERROR  = ' '
3374     TABLES
3375         pers_avail           = lt_final_avail " STRUCTURE PDPSP
3376     EXCEPTIONS
3377         no_capacity_available = 1
3378         no_integration_activ  = 2
3379         timeinfo_error       = 3
3380         OTHERS               = 4 ##FM_SUBRC_OK.
3381
3382 DATA : lv_day_before TYPE begda.
3383 DATA : lt_day_before TYPE TABLE OF pdpsp WITH HEADER LINE.
3384
3385 LOOP AT lt_final_avail INTO ls_final_avail.
3386
3387     lv_day_before = ls_final_avail-datum - 1.
3388     CALL FUNCTION 'HR_READ_TIMEDATA_PSP' "fetch yesterday's WSR
3389         EXPORTING
3390             begin_date           = lv_day_before
3391             end_date             = lv_day_before
3392             person_id            = ls_final_avail-pernr
3393         TABLES
3394             pers_avail           = lt_day_before
3395         EXCEPTIONS
3396             no_capacity_available = 1
3397             no_integration_activ  = 2
3398             timeinfo_error       = 3
3399             OTHERS               = 4.
3400
3401     IF sy-subrc = 0.
3402         READ TABLE lt_day_before INDEX 1.
3403         IF lt_day_before-begzt GT lt_day_before-endzt." if yesterday was n
ight
3404             ls_temp_time-pernr = ls_final_avail-pernr.
3405             ls_temp_time-begda = ls_final_avail-datum.
3406             ls_temp_time-enda = ls_final_avail-datum.
3407             ls_temp_time-begty = lv_def_beg.
3408             ls_temp_time-endty = lt_day_before-endzt.
3409             APPEND ls_temp_time TO lt_temp_time.
3410         ENDIF.
3411     ENDIF.
3412
3413     IF ls_final_avail-begzt GT ls_final_avail-endzt."night
3414
3415         lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3416         ls_temp_time-pernr = ls_exp_2002-pernr."normal shifts
3417         ls_temp_time-begda = ls_final_avail-datum.
3418         ls_temp_time-enda = ls_final_avail-datum.
3419         ls_temp_time-begty = lv_def_beg1.
3420         ls_temp_time-endty = lv_def_end.
3421         APPEND ls_temp_time TO lt_temp_time.

```

```

3422
3423 ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt."day
3424
3425     lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3426
3427     ls_temp_time-pernr = ls_exp_2002-pernr."normal shifts
3428     ls_temp_time-begda = ls_final_avail-datum.
3429     ls_temp_time-endda = ls_final_avail-datum.
3430     ls_temp_time-begty = lv_def_beg1.
3431     ls_temp_time-endty = ls_final_avail-endzt.
3432     APPEND ls_temp_time TO lt_temp_time.
3433
3434 ELSE.
3435
3436     ls_time_t-zz_pernr = ls_exp_2002-pernr.
3437     ls_time_t-zz_begda = lv_begda.
3438     ls_time_t-zz_endda = lv_begda.
3439     ls_time_t-zz_status = 'B'.
3440     ls_time_t-zz_eg = wa_0001-persg.
3441     ls_time_t-zz_psg = wa_0001-persk.
3442     ls_time_t-zz_begtym = lv_def_beg.
3443     ls_time_t-zz_endtym = lv_def_end.
3444     ls_time_t-zz_reason = '7'.
3445
3446     APPEND ls_time_t TO lt_time_t.
3447     CLEAR: ls_time_t.
3448
3449 ENDIF.
3450
3451
3452 SELECT SINGLE * FROM pa2007 INTO ls_pa2007" if overtime
3453 WHERE pernr = ls_exp_2002-pernr
3454 AND ( subty = '01' OR subty = '02' OR subty = '10' )
3455 AND endda GE ls_final_avail-datum
3456 AND begda LE ls_final_avail-datum.
3457
3458 IF sy-subrc = 0.
3459
3460     DELETE lt_time_t WHERE zz_begda = ls_final_avail-datum.
3461     " and BEGTY = ENDTY." removing if off shift have overtime
3462     ls_temp_time-pernr = ls_exp_2002-pernr.
3463     ls_temp_time-begda = lv_begda.
3464     ls_temp_time-endda = lv_begda.
3465     ls_temp_time-begty = ls_pa2007-beguz.
3466     ls_temp_time-endty = ls_pa2007-enduz.
3467     APPEND ls_temp_time TO lt_temp_time.
3468     CLEAR : ls_temp_time, ls_pa2007.
3469
3470 ENDIF.
3471
3472 CLEAR: ls_temp_time.
3473
3474 ENDLOOP.
3475 * *****begin of changes
3476 DATA:lv_datum      TYPE datum,
3477       lv_datum2     TYPE datum,
3478       total         TYPE i,
3479       count         TYPE i,
3480       lv_time2      TYPE tims,
3481       ls_temp_dummy TYPE ty_temp_time.
3482

```

```

3483 SORT lt_temp_time BY begda endda begty endty .
3484
3485
3486 CLEAR:lv_datum,lv_datum2,total,count.
3487 total = 0.count = 0.
3488 LOOP AT lt_temp_time INTO ls_temp_time.
3489     lv_datum = ls_temp_time-begda.
3490     AT NEW begda.
3491         CLEAR : total, count.
3492         LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
-begda.
3493             total = total + 1.
3494         ENDLOOP.
3495     ENDAT.
3496
3497 IF lv_datum = lv_datum2 AND count < total.
3498     ls_time_t-zz_pernr = ls_temp_time-pernr.
3499     ls_time_t-zz_begda = ls_temp_time-begda.
3500     ls_time_t-zz_endda = ls_temp_time-begda.
3501     ls_time_t-zz_status = 'B'.
3502     ls_time_t-zz_eg = wa_0001-persg.
3503     ls_time_t-zz_psg = wa_0001-persk.
3504     ls_time_t-zz_begtym = lv_time2.
3505     ls_time_t-zz_endtym = ls_temp_time-begty.
3506     ls_time_t-zz_reason = '7'.
3507     IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
3508         APPEND ls_time_t TO lt_time_t.
3509         CLEAR:ls_time_t.
3510     ENDIF.
3511     CLEAR : ls_time_t.
3512     count = count + 1.
3513
3514 ELSEIF count = 0.
3515 *     IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
3516         ls_time_t-zz_pernr = ls_temp_time-pernr.
3517         ls_time_t-zz_begda = ls_temp_time-begda.
3518         ls_time_t-zz_endda = ls_temp_time-begda.
3519         ls_time_t-zz_status = 'B'.
3520         ls_time_t-zz_eg = wa_0001-persg.
3521         ls_time_t-zz_psg = wa_0001-persk.
3522         ls_time_t-zz_begtym = '000000'.
3523         ls_time_t-zz_endtym = ls_temp_time-begty.
3524         ls_time_t-zz_reason = '7'.
3525         APPEND ls_time_t TO lt_time_t.
3526         CLEAR : ls_time_t.
3527
3528         count = count + 1.
3529
3530     ENDIF.
3531 IF count = total.
3532     ls_time_t-zz_pernr = ls_temp_time-pernr.
3533     ls_time_t-zz_begda = ls_temp_time-begda.
3534     ls_time_t-zz_endda = ls_temp_time-begda.
3535     ls_time_t-zz_status = 'B'.
3536     ls_time_t-zz_eg = wa_0001-persg.
3537     ls_time_t-zz_psg = wa_0001-persk.
3538     ls_time_t-zz_begtym = ls_temp_time-endty.
3539     ls_time_t-zz_endtym = '235959'.
3540     ls_time_t-zz_reason = '7'.
3541     APPEND ls_time_t TO lt_time_t.
3542     CLEAR : ls_time_t.

```

```

3543
3544     ENDIF.
3545
3546     lv_datum2 = lv_datum.
3547     lv_time2 = ls_temp_time-endty.
3548 ENDLOOP.
3549
3550 DATA : lt_time_u TYPE TABLE OF zppa0088,
3551         ls_time_u TYPE zppa0088.
3552
3553 DATA : lv_index TYPE sy-tabix.
3554 DATA : lv_line2 TYPE i.
3555 DATA : lv_def_beg TYPE sy-uzeit,
3556         lv_def_end TYPE sy-uzeit.
3557 lv_def_beg = '000000'.
3558 lv_def_end = '235959'.
3559
3560 DESCRIBE TABLE lt_time_t LINES lv_line2.
3561 SORT lt_time_t BY zz_pernr zz_begda zz_begtym.
3562
3563 LOOP AT lt_time_t INTO ls_time_t.
3564
3565     CHECK sy-tabix NE lv_line2.
3566     lv_index = sy-tabix + 1.
3567     READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
3568
3569     CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
3570
3571     IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
3572         ls_time_u-zz_begtym = lv_def_beg.
3573         ls_time_u-zz_endtym = ls_time_t-zz_begtym.
3574         ls_time_u-zz_reason = '9'.
3575         ls_time_u-zz_status = 'U'.
3576         APPEND ls_time_u TO lt_time_u.
3577         CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
3578
3579     ELSEIF ls_time_t-zz_begda EQ '000000'.
3580         ls_time_u-zz_endtym = ls_time_u-zz_begtym.
3581         ls_time_u-zz_begtym = ls_time_t-zz_endtym.
3582         ls_time_u-zz_reason = '9'.
3583         ls_time_u-zz_status = 'U'.
3584         APPEND ls_time_u TO lt_time_u.
3585         CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
3586
3587     ENDIF.
3588
3589 ENDLOOP.
3590 CLEAR : ls_time_u.
3591 *****
3592 DELETE lt_final WHERE zz_pernr = ls_exp_2002-pernr.
3593 LOOP AT lt_time_t INTO ls_time_t."adding timed block records
3594     MOVE ls_time_t TO ls_final .
3595     APPEND ls_final TO lt_final.
3596     CLEAR: ls_final, ls_time_t.
3597
3598 ENDLOOP.
3599

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```

3600 LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
3601 MOVE ls_time_u TO ls_final.
3602 APPEND ls_final TO lt_final.
3603 CLEAR: ls_final, ls_time_u.
3604
3605 ENDLOOP.
3606
3607 DATA : lt_pa2002 TYPE TABLE OF pa2002,
3608         ls_pa2002 TYPE pa2002.
3609
3610 TYPES : BEGIN OF ty_date,
3611         begda TYPE begda,
3612         END OF ty_date.
3613 DATA : lt_date TYPE TABLE OF ty_date,
3614         ls_date TYPE ty_date.
3615
3616 lv_begda1 = ls_exp_2002-begda.
3617
3618 WHILE lv_begda1 LE ls_exp_2002-endda.
3619 MOVE lv_begda1 TO ls_date-begda.
3620 APPEND ls_date TO lt_date.
3621 lv_begda1 = lv_begda1 + 1.
3622 ENDWHILE.
3623
3624 * SELECT * FROM PA2002 INTO TABLE LT_PA2002
3625 ** if 2002 present
3626 * WHERE PERNR = LS_EXP_0007-PERNR
3627 * AND SUBTY = '0200'
3628 * AND ENDDA GE LV_BEGDA
3629 * AND BEGDA LE LV_ENDDA.
3630 *
3631 * IF LT_PA2002 IS NOT INITIAL.
3632 *
3633 * LOOP AT LT_PA2002 INTO LS_PA2002.
3634 * DELETE LT_FINAL WHERE zz_BEGDA ge LS_PA2002-BEGDA and zz_ENDDA le
LS_PA2002-ENDDA.
3635 * move ls_pa2002-BEGDA to ls_date-BEGDA.
3636 * APPEND ls_date to lt_date.
3637 * move LS_PA2002-ENDDA to ls_date-BEGDA.
3638 * APPEND ls_date to lt_date.
3639 * ENDLOOP.
3640 *
3641 * ENDIF.
3642
3643 DATA : lt_pa2004 TYPE TABLE OF pa2004,
3644         ls_pa2004 TYPE pa2004. " if 2004 present
3645 SELECT * FROM pa2004 INTO TABLE lt_pa2004
3646 * FOR ALL ENTRIES IN LT_LIST_PERNR
3647 WHERE pernr = ls_exp_2002-pernr
3648 AND ( subty = '40' OR subty = '99' )
3649 AND endda GE lv_begda
3650 AND begda LE lv_endda.
3651
3652 IF lt_pa2004 IS NOT INITIAL.
3653
3654 LOOP AT lt_pa2004 INTO ls_pa2004.
3655 DELETE lt_final WHERE zz_begda GE ls_pa2004-begda AND zz_endda LE
ls_pa2004-endda.
3656 MOVE ls_pa2002-begda TO ls_date-begda.
3657 APPEND ls_date TO lt_date.
3658 MOVE ls_pa2002-endda TO ls_date-begda.

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```

3659         APPEND ls_date TO lt_date.
3660     ENDLOOP.
3661
3662 ENDIF.
3663
3664 DELETE ADJACENT DUPLICATES FROM lt_date.
3665
3666 LOOP AT lt_date INTO ls_date.
3667     DELETE lt_final WHERE zz_begda EQ ls_date-begda.
3668
3669     ls_final-zz_pernr = ls_exp_2002-pernr.
3670     ls_final-zz_begda = ls_date-begda.
3671     ls_final-zz_endda = ls_date-begda.
3672     ls_final-zz_status = 'U'.
3673     ls_final-zz_eg = wa_0001-persg.
3674     ls_final-zz_psg = wa_0001-persk.
3675     ls_final-zz_begtym = '000000'.
3676     ls_final-zz_endtym = '235959'.
3677     ls_final-zz_reason = '9'.
3678     APPEND ls_final TO lt_final.
3679 ENDLOOP.
3680
3681 SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
3682
3683 REFRESH : lt_date.
3684 DATA : lt_pa2001 TYPE TABLE OF pa2001,
3685         ls_pa2001 TYPE pa2001.
3686 SELECT * FROM pa2001 INTO TABLE lt_pa2001
3687 *     FOR ALL ENTRIES IN LT_LIST_PERNR
3688     WHERE pernr = ls_exp_2002-pernr
3689     AND subty EQ '1070'
3690     AND endda GE lv_begda
3691     AND begda LE lv_endda.
3692
3693 IF lt_pa2001 IS NOT INITIAL.
3694     LOOP AT lt_pa2001 INTO ls_pa2001.
3695         DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_endda LE
3696 ls_pa2001-endda.
3697         MOVE ls_pa2001-begda TO ls_date-begda.
3698         APPEND ls_date TO lt_date.
3699         MOVE ls_pa2001-endda TO ls_date-begda.
3700         APPEND ls_date TO lt_date.
3701     ENDLOOP.
3702 ENDIF.
3703
3704 LOOP AT lt_date INTO ls_date.
3705
3706     DELETE lt_final WHERE zz_begda EQ ls_date-begda.
3707
3708     ls_final-zz_pernr = ls_exp_2002-pernr.
3709     ls_final-zz_begda = ls_date-begda.
3710     ls_final-zz_endda = ls_date-begda.
3711     ls_final-zz_status = 'B'.
3712     ls_final-zz_eg = wa_0001-persg.
3713     ls_final-zz_psg = wa_0001-persk.
3714     ls_final-zz_begtym = '000000'.
3715     ls_final-zz_endtym = '235959'.
3716     ls_final-zz_reason = '7'.
3717     APPEND ls_final TO lt_final.
3718 ENDLOOP.
3719 SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.

```



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3719
3720 ENDFORM.
3721 *&-----*
3722 *&      Form  BADI_2007
3723 *&-----*
3724 *      text
3725 *-----*
3726 *  →  p1      text
3727 *  ←  p2      text
3728 *-----*
3729 FORM badi_2007 .
3730
3731 READ TABLE lt_exp_2007 INTO ls_exp_2007 INDEX 1.
3732 DATA: it_0001 TYPE TABLE OF pa0001,
3733        wa_0001 TYPE pa0001.
3734 SELECT SINGLE * FROM pa0001 INTO wa_0001
3735        WHERE pernr = ls_exp_2007-pernr
3736        AND begda < ls_exp_2007-begda
3737        AND endda > ls_exp_2007-endda.
3738
3739 DATA : lt_final_avail TYPE TABLE OF pdpsp.
3740 DATA : ls_final_avail TYPE pdpsp ." WITH HEADER LINE.
3741
3742 CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
3743     EXPORTING
3744         begin_date      = lv_begda
3745         end_date        = lv_endda
3746         person_id       = ls_exp_2007-pernr
3747 *      PERSON_TYP      = 'P'
3748 *      RP_BUILD_PSP_ERROR = ' '
3749     TABLES
3750         pers_avail      = lt_final_avail " STRUCTURE PDPSP
3751     EXCEPTIONS
3752         no_capacity_available = 1
3753         no_integration_activ  = 2
3754         timeinfo_error       = 3
3755         OTHERS               = 4 ##FM_SUBRC_OK.
3756
3757 DATA : lv_day_before TYPE begda.
3758 DATA : lt_day_before TYPE TABLE OF pdpsp WITH HEADER LINE.
3759
3760 LOOP AT lt_final_avail INTO ls_final_avail.
3761
3762     lv_day_before = ls_final_avail-datum - 1.
3763     CALL FUNCTION 'HR_READ_TIMEDATA_PSP' "fetch yesterday's WSR
3764         EXPORTING
3765             begin_date      = lv_day_before
3766             end_date        = lv_day_before
3767             person_id       = ls_final_avail-pernr
3768         TABLES
3769             pers_avail      = lt_day_before
3770         EXCEPTIONS
3771             no_capacity_available = 1
3772             no_integration_activ  = 2
3773             timeinfo_error       = 3
3774             OTHERS               = 4.
3775
3776     IF sy-subrc = 0.
3777         READ TABLE lt_day_before INDEX 1.
3778         IF lt_day_before-begzt GT lt_day_before-endzt." if yesterday was n
ight

```

```

3779         ls_temp_time-pernr = ls_final_avail-pernr.
3780         ls_temp_time-begda = ls_final_avail-datum.
3781         ls_temp_time-endda = ls_final_avail-datum.
3782         ls_temp_time-begty = lv_def_beg.
3783         ls_temp_time-endty = lt_day_before-endzt.
3784         APPEND ls_temp_time TO lt_temp_time.
3785     ENDIF.
3786 ENDIF.
3787
3788 IF     ls_final_avail-begzt GT ls_final_avail-endzt."night
3789
3790     lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3791     ls_temp_time-pernr = ls_exp_2007-pernr."normal shifts
3792     ls_temp_time-begda = ls_final_avail-datum.
3793     ls_temp_time-endda = ls_final_avail-datum.
3794     ls_temp_time-begty = lv_def_beg1.
3795     ls_temp_time-endty = lv_def_end.
3796     APPEND ls_temp_time TO lt_temp_time.
3797
3798 ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt."day
3799
3800     lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
3801
3802     ls_temp_time-pernr = ls_exp_2007-pernr."normal shifts
3803     ls_temp_time-begda = ls_final_avail-datum.
3804     ls_temp_time-endda = ls_final_avail-datum.
3805     ls_temp_time-begty = lv_def_beg1.
3806     ls_temp_time-endty = ls_final_avail-endzt.
3807     APPEND ls_temp_time TO lt_temp_time.
3808
3809 ELSE.
3810
3811     ls_time_t-zz_pernr = ls_exp_2007-pernr.
3812     ls_time_t-zz_begda = lv_begda.
3813     ls_time_t-zz_endda = lv_begda.
3814     ls_time_t-zz_status = 'B'.
3815     ls_time_t-zz_eg = wa_0001-persg.
3816     ls_time_t-zz_psg = wa_0001-persk.
3817     ls_time_t-zz_begtym = lv_def_beg.
3818     ls_time_t-zz_endtym = lv_def_end.
3819     ls_time_t-zz_reason = '7'.
3820
3821     APPEND ls_time_t TO lt_time_t.
3822     CLEAR: ls_time_t.
3823
3824 ENDIF.
3825
3826
3827 SELECT * FROM pa2007 INTO TABLE lt_pa2007" if overtime
3828 WHERE pernr = ls_exp_2007-pernr
3829 AND ( subty = '01' OR subty = '02' OR subty = '10' )
3830 AND endda GE ls_final_avail-datum
3831 AND begda LE ls_final_avail-datum.
3832
3833 IF sy-subrc = 0.
3834
3835     DELETE lt_time_t WHERE zz_begda GE ls_final_avail-datum AND zz_en
3836     dda LE ls_final_avail-datum.
3837     " and BEGTY = ENDTY." removing if off shift have overtime
3838     LOOP AT lt_pa2007 INTO ls_pa2007.

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3839
3840     ls_temp_time-pernr = ls_exp_2007-pernr.
3841     ls_temp_time-begda = lv_begda.
3842     ls_temp_time-enddda = lv_begda.
3843     ls_temp_time-begty = ls_pa2007-beguz.
3844     ls_temp_time-endty = ls_pa2007-enduz.
3845     APPEND ls_temp_time TO lt_temp_time."working times
3846     CLEAR : ls_temp_time, ls_pa2007.
3847
3848     ENDLOOP.
3849
3850     ENDIF.
3851
3852     CLEAR: ls_temp_time.
3853
3854     ENDLOOP.
3855
3856     lv_begda1 = ls_exp_2007-begda.
3857
3858     * while LV BEGDA1 LE ls_exp_2007-ENDDDA.
3859     **     move LV BEGDA1 to ls_date-BEGDA.
3860     **     APPEND ls_date to LT_Date.
3861     **
3862     *     LS_TEMP_TIME-PERNR = ls_exp_2007-PERNR.
3863     *     LS_TEMP_TIME-BEGDA = LV_BEGDA1.
3864     *     LS_TEMP_TIME-ENDDDA = LV_BEGDA1.
3865     *     LS_TEMP_TIME-BEGTY = ls_exp_2007-DATA(6).
3866     *     LS_TEMP_TIME-ENDTY = ls_exp_2007-DATA+6(6).
3867     *     APPEND LS_TEMP_TIME TO LT_TEMP_TIME.
3868     *     CLEAR : LS_TEMP_TIME.
3869     *
3870     *     LV_BEGDA1 = LV_BEGDA1 + 1.
3871     * ENDWHILE.
3872
3873     * *****begin of changes
3874     DATA:lv_datum      TYPE datum,
3875           lv_datum2     TYPE datum,
3876           total         TYPE i,
3877           count         TYPE i,
3878           lv_time2      TYPE tims,
3879           ls_temp_dummy TYPE ty_temp_time.
3880
3881     SORT lt_temp_time BY begda enddda begty endty .
3882
3883
3884     CLEAR:lv_datum,lv_datum2,total,count.
3885     total = 0.count = 0.
3886     LOOP AT lt_temp_time INTO ls_temp_time.
3887         lv_datum = ls_temp_time-begda.
3888         AT NEW begda.
3889             CLEAR : total, count.
3890             LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
3891             -begda.
3892                 total = total + 1.
3893             ENDLOOP.
3894         ENDAT.
3895
3896     IF lv_datum = lv_datum2 AND count < total.
3897         ls_time_t-zz_pernr = ls_temp_time-pernr.
3898         ls_time_t-zz_begda = ls_temp_time-begda.
3899         ls_time_t-zz_enddda = ls_temp_time-begda.

```

```

3899     ls_time_t-zz_status = 'B'.
3900     ls_time_t-zz_eg = wa_0001-persg.
3901     ls_time_t-zz_psg = wa_0001-persk.
3902     ls_time_t-zz_begtym = lv_time2.
3903     ls_time_t-zz_endtym = ls_temp_time-begty.
3904     ls_time_t-zz_reason = '7'.
3905     IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
3906         APPEND ls_time_t TO lt_time_t.
3907         CLEAR:ls_time_t.
3908     ENDIF.
3909     CLEAR : ls_time_t.
3910     count = count + 1.
3911
3912 ELSEIF count = 0.
3913 *     IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
3914     ls_time_t-zz_pernr = ls_temp_time-pernr.
3915     ls_time_t-zz_begda = ls_temp_time-begda.
3916     ls_time_t-zz_endda = ls_temp_time-begda.
3917     ls_time_t-zz_status = 'B'.
3918     ls_time_t-zz_eg = wa_0001-persg.
3919     ls_time_t-zz_psg = wa_0001-persk.
3920     ls_time_t-zz_begtym = '000000'.
3921     ls_time_t-zz_endtym = ls_temp_time-begty.
3922     ls_time_t-zz_reason = '7'.
3923     IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
3924         APPEND ls_time_t TO lt_time_t.
3925         CLEAR : ls_time_t.
3926     ENDIF.
3927     count = count + 1.
3928
3929 ENDIF.
3930 IF count = total.
3931     ls_time_t-zz_pernr = ls_temp_time-pernr.
3932     ls_time_t-zz_begda = ls_temp_time-begda.
3933     ls_time_t-zz_endda = ls_temp_time-begda.
3934     ls_time_t-zz_status = 'B'.
3935     ls_time_t-zz_eg = wa_0001-persg.
3936     ls_time_t-zz_psg = wa_0001-persk.
3937     ls_time_t-zz_begtym = ls_temp_time-endty.
3938     ls_time_t-zz_endtym = '235959'.
3939     ls_time_t-zz_reason = '7'.
3940     IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
3941         APPEND ls_time_t TO lt_time_t.
3942         CLEAR : ls_time_t.
3943     ENDIF.
3944 ENDIF.
3945
3946     lv_datum2 = lv_datum.
3947     lv_time2 = ls_temp_time-endty.
3948 ENDLÖOP.
3949
3950 DATA : lt_time_u TYPE TABLE OF zppa0088,
3951        ls_time_u TYPE zppa0088.
3952
3953 DATA : lv_index TYPE sy-tabix.
3954 DATA : lv_line2 TYPE i.
3955 DATA : lv_def_beg TYPE sy-uzeit,
3956        lv_def_end TYPE sy-uzeit.
3957 DATA : lv_time TYPE sy-uzeit.
3958
3959 lv_def_beg = '000000'.

```

```

3960     lv_def_end = '235959'.
3961
3962     DESCRIBE TABLE lt_time_t LINES lv_line2.
3963     SORT lt_time_t BY zz_pernr zz_begda zz_begtym.
3964
3965     LOOP AT lt_time_t INTO ls_time_t.
3966
3967 *       CHECK SY-TABIX NE LV_LINE2.
3968         lv_index = sy-tabix + 1.
3969         READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
3970
3971         CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
3972
3973         IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
3974             IF lv_time IS INITIAL.
3975                 ls_time_u-zz_begtym = lv_def_beg.
3976             ELSE.
3977                 ls_time_u-zz_begtym = lv_time.
3978             ENDIF.
3979             ls_time_u-zz_endtym = ls_time_t-zz_begtym.
3980             ls_time_u-zz_reason = '9'.
3981             ls_time_u-zz_status = 'U'.
3982             APPEND ls_time_u TO lt_time_u.
3983 *       CLEAR : LS_TIME_U-ZZ_BEGTYM, LS_TIME_U-ZZ_ENDTYM, LS_TIME_U-ZZ_RE
ASON.
3984
3985         ELSEIF ls_time_t-zz_begtym EQ '000000'.
3986             ls_time_u-zz_endtym = ls_time_u-zz_begtym.
3987             ls_time_u-zz_begtym = ls_time_t-zz_endtym.
3988             ls_time_u-zz_reason = '9'.
3989             ls_time_u-zz_status = 'U'.
3990             APPEND ls_time_u TO lt_time_u.
3991 *       CLEAR : LS_TIME_U-ZZ_BEGTYM, LS_TIME_U-ZZ_ENDTYM, LS_TIME_U-ZZ_RE
ASON.
3992
3993         ELSEIF ls_time_t-zz_endtym EQ '235959'.
3994 *           if lv_time is INITIAL.
3995 *             LS_TIME_U-ZZ_BEGTYM = LV_DEF_BEG.
3996 *           ELSE.
3997             ls_time_u-zz_begtym = lv_time.
3998 *           ENDif.
3999             ls_time_u-zz_endtym = ls_time_t-zz_begtym.
4000 *           LS_TIME_U-ZZ_ENDTYM = LS_TIME_U-ZZ_BEGTYM.
4001 *           LS_TIME_U-ZZ_BEGTYM = LS_TIME_T-ZZ_ENDTYM.
4002             ls_time_u-zz_reason = '9'.
4003             ls_time_u-zz_status = 'U'.
4004             APPEND ls_time_u TO lt_time_u.
4005 *           CLEAR : LS_TIME_U-ZZ_BEGTYM, LS_TIME_U-ZZ_ENDTYM, LS_TIME_U-ZZ_RE
ASON.
4006
4007         ENDIF.
4008
4009         CLEAR: lv_time.
4010         lv_time = ls_time_t-zz_endtym.
4011
4012         CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_reaso
n.
4013
4014     ENDLOOP.

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4015     CLEAR : ls_time_u.
4016 *****
4017     DELETE lt_final WHERE zz_pernr = ls_exp_2007-pernr.
4018     LOOP AT lt_time_t INTO ls_time_t."adding timed block records
4019         MOVE ls_time_t TO ls_final .
4020         APPEND ls_final TO lt_final.
4021         CLEAR: ls_final, ls_time_t.
4022
4023     ENDLOOP.
4024
4025     LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
4026         MOVE ls_time_u TO ls_final .
4027         APPEND ls_final TO lt_final.
4028         CLEAR: ls_final, ls_time_u.
4029
4030     ENDLOOP.
4031
4032     DATA : lt_pa2002 TYPE TABLE OF pa2002,
4033             ls_pa2002 TYPE pa2002.
4034
4035     TYPES : BEGIN OF ty_date,
4036             begda TYPE begda,
4037             END OF ty_date.
4038     DATA : lt_date TYPE TABLE OF ty_date,
4039             ls_date TYPE ty_date.
4040
4041     *   lv_begda1 = ls_exp_2002-BEGDA.
4042     *
4043     *   while LV_BEGDA1 LE ls_exp_2002-ENDDA.
4044     *       move LV_BEGDA1 to ls_date-BEGDA.
4045     *       APPEND ls_date to LT_DATE.
4046     *       LV_BEGDA1 = LV_BEGDA1 + 1.
4047     *   ENDWHILE.
4048
4049     SELECT * FROM pa2002 INTO TABLE lt_pa2002
4050     *   if 2002 present
4051         WHERE pernr = ls_exp_2007-pernr
4052         AND subty = '0200'
4053         AND endda GE lv_begda
4054         AND begda LE lv_endda.
4055
4056     IF lt_pa2002 IS NOT INITIAL.
4057
4058         LOOP AT lt_pa2002 INTO ls_pa2002.
4059             DELETE lt_final WHERE zz_begda GE ls_pa2002-begda AND zz_endda LE
4060 ls_pa2002-endda.
4061             MOVE ls_pa2002-begda TO ls_date-begda.
4062             APPEND ls_date TO lt_date.
4063             MOVE ls_pa2002-endda TO ls_date-begda.
4064             APPEND ls_date TO lt_date.
4065         ENDLOOP.
4066     ENDIF.
4067
4068     DATA : lt_pa2004 TYPE TABLE OF pa2004,
4069             ls_pa2004 TYPE pa2004. " if 2004 present
4070     SELECT * FROM pa2004 INTO TABLE lt_pa2004
4071     *   FOR ALL ENTRIES IN LT_LIST_PERNR
4072         WHERE pernr = ls_exp_2007-pernr
4073         AND ( subty = '40' OR subty = '99' )
4074         AND endda GE lv_begda

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4075         AND begda LE lv_endda.
4076
4077     IF lt_pa2004 IS NOT INITIAL.
4078
4079         LOOP AT lt_pa2004 INTO ls_pa2004.
4080             DELETE lt_final WHERE zz_begda GE ls_pa2004-begda AND zz_endda LE
ls_pa2004-endda.
4081             MOVE ls_pa2002-begda TO ls_date-begda.
4082             APPEND ls_date TO lt_date.
4083             MOVE ls_pa2002-endda TO ls_date-begda.
4084             APPEND ls_date TO lt_date.
4085         ENDLOOP.
4086
4087     ENDIF.
4088
4089     DELETE ADJACENT DUPLICATES FROM lt_date.
4090
4091     LOOP AT lt_date INTO ls_date.
4092         DELETE lt_final WHERE zz_begda EQ ls_date-begda.
4093
4094         ls_final-zz_pernr = ls_exp_2007-pernr.
4095         ls_final-zz_begda = ls_date-begda.
4096         ls_final-zz_endda = ls_date-begda.
4097         ls_final-zz_status = 'U'.
4098         ls_final-zz_eg = wa_0001-persg.
4099         ls_final-zz_psg = wa_0001-persk.
4100         ls_final-zz_begtyp = '000000'.
4101         ls_final-zz_endtyp = '235959'.
4102         ls_final-zz_reason = '9'.
4103         APPEND ls_final TO lt_final.
4104     ENDLOOP.
4105
4106     SORT lt_final BY zz_begda zz_endda zz_begtyp zz_endtyp.
4107
4108     REFRESH : lt_date.
4109     DATA : lt_pa2001 TYPE TABLE OF pa2001,
4110             ls_pa2001 TYPE pa2001.
4111     SELECT * FROM pa2001 INTO TABLE lt_pa2001
4112 *      FOR ALL ENTRIES IN LT_LIST_PERNR
4113         WHERE pernr = ls_exp_2007-pernr
4114         AND subty EQ '1070'
4115         AND endda GE lv_begda
4116         AND begda LE lv_endda.
4117
4118     IF lt_pa2001 IS NOT INITIAL.
4119         LOOP AT lt_pa2001 INTO ls_pa2001.
4120             DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_endda LE
ls_pa2001-endda.
4121             MOVE ls_pa2001-begda TO ls_date-begda.
4122             APPEND ls_date TO lt_date.
4123             MOVE ls_pa2001-endda TO ls_date-begda.
4124             APPEND ls_date TO lt_date.
4125         ENDLOOP.
4126     ENDIF.
4127
4128     LOOP AT lt_date INTO ls_date.
4129
4130         DELETE lt_final WHERE zz_begda EQ ls_date-begda.
4131
4132         ls_final-zz_pernr = ls_exp_2007-pernr.
4133         ls_final-zz_begda = ls_date-begda.

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4134     ls_final-zz_endda = ls_date-begda.
4135     ls_final-zz_status = 'B'.
4136     ls_final-zz_eg = wa_0001-persg.
4137     ls_final-zz_psg = wa_0001-persk.
4138     ls_final-zz_begtym = '000000'.
4139     ls_final-zz_endtym = '235959'.
4140     ls_final-zz_reason = '7'.
4141     APPEND ls_final TO lt_final.
4142 ENDLOOP.
4143 SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
4144
4145 ENDFORM.
4146 *&-----*
4147 *&      Form  BADI_2001
4148 *&-----*
4149 *      text
4150 *-----*
4151 *  →  p1      text
4152 *  ←  p2      text
4153 *-----*
4154 FORM badi_2001 .
4155
4156 READ TABLE lt_exp_2001 INTO ls_exp_2001 INDEX 1.
4157 DATA: it_0001 TYPE TABLE OF pa0001,
4158       wa_0001 TYPE pa0001.
4159 SELECT SINGLE * FROM pa0001 INTO wa_0001
4160       WHERE pernr = ls_exp_2001-pernr
4161       AND begda < ls_exp_2001-begda
4162       AND endda > ls_exp_2001-enda.
4163
4164 DATA : lt_final_avail TYPE TABLE OF pdpsp.
4165 DATA : ls_final_avail TYPE pdpsp ." WITH HEADER LINE.
4166
4167 CALL FUNCTION 'HR_READ_TIMEDATA_PSP'
4168     EXPORTING
4169         begin_date           = lv_begda
4170         end_date             = lv_endda
4171         person_id            = ls_exp_2001-pernr
4172 *      PERSON_TYP           = 'P'
4173 *      RP_BUILD_PSP_ERROR   = ' '
4174     TABLES
4175         pers_avail           = lt_final_avail " STRUCTURE  PDPSP
4176     EXCEPTIONS
4177         no_capacity_available = 1
4178         no_integration_activ  = 2
4179         timeinfo_error       = 3
4180         OTHERS               = 4 ##FM_SUBRC_OK.
4181
4182 DATA : lv_day_before TYPE begda.
4183 DATA : lt_day_before TYPE TABLE OF  pdpsp WITH HEADER LINE.
4184
4185 LOOP AT lt_final_avail INTO ls_final_avail.
4186
4187     lv_day_before = ls_final_avail-datum - 1.
4188     CALL FUNCTION 'HR_READ_TIMEDATA_PSP' "fetch yesterday's WSR
4189         EXPORTING
4190             begin_date           = lv_day_before
4191             end_date             = lv_day_before
4192             person_id            = ls_final_avail-pernr
4193         TABLES
4194             pers_avail           = lt_day_before

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4195      EXCEPTIONS
4196          no_capacity_available = 1
4197          no_integration_activ  = 2
4198          timeinfo_error       = 3
4199          OTHERS                = 4.
4200
4201      IF sy-subrc = 0.
4202          READ TABLE lt_day_before INDEX 1.
4203          IF lt_day_before-begzt GT lt_day_before-endzt." if yesterday was n
ight
4204              ls_temp_time-pernr = ls_final_avail-pernr.
4205              ls_temp_time-begda = ls_final_avail-datum.
4206              ls_temp_time-endda = ls_final_avail-datum.
4207              ls_temp_time-begty = lv_def_beg.
4208              ls_temp_time-endty = lt_day_before-endzt.
4209              APPEND ls_temp_time TO lt_temp_time.
4210          ENDIF.
4211      ENDIF.
4212
4213      IF    ls_final_avail-begzt GT ls_final_avail-endzt."night
4214
4215          lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
4216          ls_temp_time-pernr = ls_exp_2001-pernr."normal shifts
4217          ls_temp_time-begda = ls_final_avail-datum.
4218          ls_temp_time-endda = ls_final_avail-datum.
4219          ls_temp_time-begty = lv_def_beg1.
4220          ls_temp_time-endty = lv_def_end.
4221          APPEND ls_temp_time TO lt_temp_time.
4222
4223      ELSEIF ls_final_avail-begzt LT ls_final_avail-endzt."day
4224
4225          lv_def_beg1 = ls_final_avail-begzt - lv_seconds.
4226
4227          ls_temp_time-pernr = ls_exp_2001-pernr."normal shifts
4228          ls_temp_time-begda = ls_final_avail-datum.
4229          ls_temp_time-endda = ls_final_avail-datum.
4230          ls_temp_time-begty = lv_def_beg1.
4231          ls_temp_time-endty = ls_final_avail-endzt.
4232          APPEND ls_temp_time TO lt_temp_time.
4233
4234      ELSE.
4235
4236          ls_time_t-zz_pernr = ls_exp_2001-pernr.
4237          ls_time_t-zz_begda = lv_begda.
4238          ls_time_t-zz_endda = lv_begda.
4239          ls_time_t-zz_status = 'B'.
4240          ls_time_t-zz_eg     = wa_0001-persg.
4241          ls_time_t-zz_psg    = wa_0001-persk.
4242          ls_time_t-zz_begtym = lv_def_beg.
4243          ls_time_t-zz_endtym = lv_def_end.
4244          ls_time_t-zz_reason = '7'.
4245
4246          APPEND ls_time_t TO lt_time_t.
4247          CLEAR: ls_time_t.
4248
4249      ENDIF.
4250
4251
4252      SELECT SINGLE * FROM pa2007 INTO ls_pa2007" if overtime
4253      WHERE pernr = ls_exp_2001-pernr
4254      AND ( subty = '01' OR subty = '02' OR subty = '10' )

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4255     AND endda GE ls_final_avail-datum
4256     AND begda LE ls_final_avail-datum.
4257
4258     IF sy-subrc = 0.
4259
4260         DELETE lt_time t WHERE zz_begda = ls_final_avail-datum.
4261         " and BEGTY = ENDTY." removing if off shift have overtime
4262         ls_temp_time-pernr = ls_exp_2001-pernr.
4263         ls_temp_time-begda = lv_begda.
4264         ls_temp_time-endda = lv_begda.
4265         ls_temp_time-begty = ls_pa2007-beguz.
4266         ls_temp_time-endty = ls_pa2007-enduz.
4267         APPEND ls_temp_time TO lt_temp_time.
4268         CLEAR : ls_temp_time, ls_pa2007.
4269
4270     ENDIF.
4271
4272     CLEAR: ls_temp_time.
4273
4274     ENDLOOP.
4275 * *****begin of changes
4276     DATA:lv_datum      TYPE datum,
4277           lv_datum2     TYPE datum,
4278           total         TYPE i,
4279           count         TYPE i,
4280           lv_time2      TYPE tims,
4281           ls_temp_dummy TYPE ty_temp_time.
4282
4283     SORT lt_temp_time BY begda endda begty endty .
4284
4285
4286     CLEAR:lv_datum,lv_datum2,total,count.
4287     total = 0.count = 0.
4288     LOOP AT lt_temp_time INTO ls_temp_time."all shifts including overlappi
ng
4289         lv_datum = ls_temp_time-begda.
4290         AT NEW begda.
4291             CLEAR : total, count.
4292             LOOP AT lt_temp_time INTO ls_temp_dummy WHERE begda = ls_temp_time
-begda.
4293                 total = total + 1.
4294             ENDLOOP.
4295         ENDAT.
4296
4297     IF lv_datum = lv_datum2 AND count < total.
4298         ls_time_t-zz_pernr = ls_temp_time-pernr.
4299         ls_time_t-zz_begda = ls_temp_time-begda.
4300         ls_time_t-zz_endda = ls_temp_time-begda.
4301         ls_time_t-zz_status = 'B'.
4302         ls_time_t-zz_eg = wa_0001-persg.
4303         ls_time_t-zz_psg = wa_0001-persk.
4304         ls_time_t-zz_begtym = lv_time2.
4305         ls_time_t-zz_endtym = ls_temp_time-begty.
4306         ls_time_t-zz_reason = '7'.
4307         IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
4308             APPEND ls_time_t TO lt_time_t.
4309             CLEAR:ls_time_t.
4310         ENDIF.
4311         CLEAR : ls_time_t.
4312         count = count + 1.
4313

```

```

4314 ELSEIF count = 0.
4315 *      IF LS_TEMP_TIME-BEGTY <> LS_TEMP_TIME-ENDTY.
4316         ls_time_t-zz_pernr = ls_temp_time-pernr.
4317         ls_time_t-zz_begda = ls_temp_time-begda.
4318         ls_time_t-zz_endda = ls_temp_time-begda.
4319         ls_time_t-zz_status = 'B'.
4320         ls_time_t-zz_eg = wa_0001-persg.
4321         ls_time_t-zz_psg = wa_0001-persk.
4322         ls_time_t-zz_begtym = '000000'.
4323         ls_time_t-zz_endtym = ls_temp_time-begty.
4324         ls_time_t-zz_reason = '7'.
4325         IF ls_time_t-zz_begtym <> ls_time_t-zz_endtym.
4326             APPEND ls_time_t TO lt_time_t.
4327             CLEAR : ls_time_t.
4328         ENDIF.
4329         count = count + 1.
4330
4331     ENDIF.
4332     IF count = total.
4333         ls_time_t-zz_pernr = ls_temp_time-pernr.
4334         ls_time_t-zz_begda = ls_temp_time-begda.
4335         ls_time_t-zz_endda = ls_temp_time-begda.
4336         ls_time_t-zz_status = 'B'.
4337         ls_time_t-zz_eg = wa_0001-persg.
4338         ls_time_t-zz_psg = wa_0001-persk.
4339         ls_time_t-zz_begtym = ls_temp_time-endty.
4340         ls_time_t-zz_endtym = '235959'.
4341         ls_time_t-zz_reason = '7'.
4342         APPEND ls_time_t TO lt_time_t.
4343         CLEAR : ls_time_t.
4344
4345     ENDIF.
4346
4347     lv_datum2 = lv_datum.
4348     lv_time2 = ls_temp_time-endty.
4349 ENDLOOP.
4350
4351 * DELETE LT_TIME_T WHERE
4352
4353 DATA : lt_time_u TYPE TABLE OF zppa0088,
4354         ls_time_u TYPE zppa0088.
4355
4356 DATA : lv_index TYPE sy-tabix.
4357 DATA : lv_line2 TYPE i.
4358 DATA : lv_def_beg TYPE sy-uzeit,
4359         lv_def_end TYPE sy-uzeit.
4360 lv_def_beg = '000000'.
4361 lv_def_end = '235959'.
4362
4363 DESCRIBE TABLE lt_time_t LINES lv_line2.
4364 SORT lt_time_t BY zz_pernr zz_begda zz_begtym.
4365
4366 LOOP AT lt_time_t INTO ls_time_t.
4367
4368     CHECK sy-tabix NE lv_line2.
4369     lv_index = sy-tabix + 1.
4370     READ TABLE lt_time_t INTO ls_time_u INDEX lv_index.
4371
4372     CHECK ls_time_t-zz_pernr = ls_time_u-zz_pernr AND ls_time_t-zz_begda
= ls_time_u-zz_begda.
4373

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```

4374      IF ls_time_t-zz_begtym GT '000000' AND ls_time_t-zz_endtym NE '23595
9'.
4375          ls_time_u-zz_begtym = lv_def_beg.
4376          ls_time_u-zz_endtym = ls_time_t-zz_begtym.
4377          ls_time_u-zz_reason = '9'.
4378          ls_time_u-zz_status = 'U'.
4379          APPEND ls_time_u TO lt_time_u.
4380          CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
4381
4382      ELSEIF ls_time_t-zz_begtym EQ '000000'.
4383          ls_time_u-zz_endtym = ls_time_u-zz_begtym.
4384          ls_time_u-zz_begtym = ls_time_t-zz_endtym.
4385          ls_time_u-zz_reason = '9'.
4386          ls_time_u-zz_status = 'U'.
4387          APPEND ls_time_u TO lt_time_u.
4388          CLEAR : ls_time_u-zz_begtym, ls_time_u-zz_endtym, ls_time_u-zz_rea
son.
4389
4390      ENDIF.
4391
4392      ENDLOOP.
4393      CLEAR : ls_time_u.
4394      *****
4395      DELETE lt_final WHERE zz_pernr = ls_exp_2001-pernr.
4396      LOOP AT lt_time_t INTO ls_time_t."adding timed block records
4397          MOVE ls_time_t TO ls_final.
4398          APPEND ls_final TO lt_final.
4399          CLEAR: ls_final, ls_time_t.
4400
4401      ENDLOOP.
4402
4403      LOOP AT lt_time_u INTO ls_time_u."adding timed unblock records
4404          MOVE ls_time_u TO ls_final.
4405          APPEND ls_final TO lt_final.
4406          CLEAR: ls_final, ls_time_u.
4407
4408      ENDLOOP.
4409
4410      DATA : lt_pa2002 TYPE TABLE OF pa2002,
4411              ls_pa2002 TYPE pa2002.
4412
4413      TYPES : BEGIN OF ty_date,
4414                begda TYPE begda,
4415            END OF ty_date.
4416      DATA : lt_date TYPE TABLE OF ty_date,
4417              ls_date TYPE ty_date.
4418
4419      lv_begda1 = ls_exp_2001-begda.
4420
4421      WHILE lv_begda1 LE ls_exp_2001-endda.
4422          MOVE lv_begda1 TO ls_date-begda.
4423          APPEND ls_date TO lt_date.
4424          lv_begda1 = lv_begda1 + 1.
4425      ENDWHILE.
4426
4427      SELECT * FROM pa2002 INTO TABLE lt_pa2002
4428      *      IF 2002 present
4429          WHERE pernr = ls_exp_0007-pernr
4430          AND subty = '0200'
4431          AND endda GE lv_begda

```

```

4432     AND begda LE lv_endda.
4433
4434     IF lt_pa2002 IS NOT INITIAL.
4435
4436         LOOP AT lt_pa2002 INTO ls_pa2002.
4437             DELETE lt_final WHERE zz_begda GE ls_pa2002-begda AND zz_endda LE
ls_pa2002-endda.
4438             MOVE ls_pa2002-begda TO ls_date-begda.
4439             APPEND ls_date TO lt_date.
4440             MOVE ls_pa2002-endda TO ls_date-begda.
4441             APPEND ls_date TO lt_date.
4442         ENDLOOP.
4443
4444     ENDIF.
4445
4446     DATA : lt_pa2004 TYPE TABLE OF pa2004,
4447             ls_pa2004 TYPE pa2004. " if 2004 present
4448     SELECT * FROM pa2004 INTO TABLE lt_pa2004
4449 *     FOR ALL ENTRIES IN LT_LIST_PERNR
4450     WHERE pernr = ls_exp_2001-pernr
4451     AND ( subty = '40' OR subty = '99' )
4452     AND endda GE lv_begda
4453     AND begda LE lv_endda.
4454
4455     IF lt_pa2004 IS NOT INITIAL.
4456
4457         LOOP AT lt_pa2004 INTO ls_pa2004.
4458             DELETE lt_final WHERE zz_begda GE ls_pa2004-begda AND zz_endda LE
ls_pa2004-endda.
4459             MOVE ls_pa2004-begda TO ls_date-begda.
4460             APPEND ls_date TO lt_date.
4461             MOVE ls_pa2004-endda TO ls_date-begda.
4462             APPEND ls_date TO lt_date.
4463         ENDLOOP.
4464
4465     ENDIF.
4466
4467 *     delete ADJACENT DUPLICATES FROM lt_date.
4468 *
4469 *     LOOP AT lt_date INTO ls_date.
4470 *         delete LT_FINAL where ZZ_BEGDA eq ls_date-BEGDA.
4471 *
4472 *         LS_FINAL-ZZ_PERNR = ls_exp_2001-PERNR.
4473 *         LS_FINAL-ZZ_BEGDA = ls_date-BEGDA.
4474 *         LS_FINAL-ZZ_ENDDA = ls_date-BEGDA.
4475 *         LS_FINAL-ZZ_STATUS = 'U'.
4476 *         LS_FINAL-ZZ_EG = WA_0001-PERSG.
4477 *         LS_FINAL-ZZ_PSG = WA_0001-PERSK.
4478 *         LS_FINAL-ZZ_BEGTYM = '000000'.
4479 *         LS_FINAL-ZZ_ENDTYM = '235959'.
4480 *         LS_FINAL-ZZ_REASON = '9'.
4481 *         APPEND LS_FINAL to LT_FINAL.
4482 *     ENDLOOP.
4483
4484     SORT lt_final BY zz_begda zz_endda zz_begtym zz_endtym.
4485
4486 *     refresh : lt_date.
4487     DATA : lt_pa2001 TYPE TABLE OF pa2001,
4488             ls_pa2001 TYPE pa2001.
4489     SELECT * FROM pa2001 INTO TABLE lt_pa2001
4490 *     FOR ALL ENTRIES IN LT_LIST_PERNR

```

```
4491      WHERE pernr = ls_exp_2001-pernr
4492      AND subty EQ '1070'
4493      AND endda GE lv_begda
4494      AND begda LE lv_endda.
4495
4496      IF lt_pa2001 IS NOT INITIAL.
4497          LOOP AT lt_pa2001 INTO ls_pa2001.
4498              DELETE lt_final WHERE zz_begda GE ls_pa2001-begda AND zz_endda LE
ls_pa2001-endda.
4499              MOVE ls_pa2001-begda TO ls_date-begda.
4500              APPEND ls_date TO lt_date.
4501              MOVE ls_pa2001-endda TO ls_date-begda.
4502              APPEND ls_date TO lt_date.
4503          ENDLOOP.
4504      ENDIF.
4505
4506      SORT lt_date BY begda.
4507      DELETE ADJACENT DUPLICATES FROM lt_date.
4508
4509      LOOP AT lt_date INTO ls_date.
4510
4511          DELETE lt_final WHERE zz_begda EQ ls_date-begda.
4512
4513          ls_final-zz_pernr = ls_exp_2001-pernr.
4514          ls_final-zz_begda = ls_date-begda.
4515          ls_final-zz_endda = ls_date-begda.
4516          ls_final-zz_status = 'B'.
4517          ls_final-zz_eg = wa_0001-persg.
4518          ls_final-zz_psg = wa_0001-persk.
4519          ls_final-zz_begty = '000000'.
4520          ls_final-zz_endty = '235959'.
4521          ls_final-zz_reason = '7'.
4522          APPEND ls_final TO lt_final.
4523      ENDLOOP.
4524      SORT lt_final BY zz_begda zz_endda zz_begty zz_endty.
4525
4526      ENDFORM.
```

Title:
Honeywell Interface Block employees with invalid Certificate

Selection texts:

P_CHECK	Test Run
P_RAD1	ALV display
P_RAD2	Send Idocs

Subroutines:	Line	Page
ADD_QUAL_TIME.....	ZRPPA0088	1870 33
BADI_0007.....	ZRPPA0088	1974 35
BADI_2001.....	ZRPPA0088	4154 72
BADI_2002.....	ZRPPA0088	3354 58
BADI_2003.....	ZRPPA0088	2613 46
BADI_2004.....	ZRPPA0088	2978 52
BADI_2007.....	ZRPPA0088	3729 65
BADI_CALL.....	ZRPPA0088	266 6
	ZRPPA0088	983 18
CHECK_2002.....	ZRPPA0088	1827 33
CHECK_2004.....	ZRPPA0088	1786 32
CHECK_TIME.....	ZRPPA0088	1321 24
DISP_DATA.....	ZRPPA0088	310 7
	ZRPPA0088	788 15
FIELD_CAT.....	ZRPPA0088	309 7
	ZRPPA0088	818 15
FILL_BADI_DATA.....	ZRPPA0088	283 6
	ZRPPA0088	1073 20
FILL_DATA.....	ZRPPA0088	269 6
	ZRPPA0088	448 9
SELECT_DATA.....	ZRPPA0088	262 6
	ZRPPA0088	334 7
SELECT_QUERIES.....	ZRPPA0088	258 6
	ZRPPA0088	751 14
SEND_IDOC.....	ZRPPA0088	314 7
	ZRPPA0088	908 17

External calls

Function modules:	Line	Page
DB_COMMIT.....	ZRPPA0088	941 17
DEQUEUE_ALL.....	ZRPPA0088	942 17
HRCM_TIME_PERIOD_CALCULATE....	ZRPPA0088	628 12
	ZRPPA0088	1227 22
HR_99S_INTERVAL_BETWEEN_DATES.	ZRPPA0088	413 8
HR_READ_INFOTYPE.....	ZRPPA0088	2153 38
HR_READ_TIMEDATA_PSP.....	ZRPPA0088	1333 24
	ZRPPA0088	1445 26
	ZRPPA0088	1542 28
	ZRPPA0088	1664 30
	ZRPPA0088	2626 46
	ZRPPA0088	2647 46
	ZRPPA0088	2991 52
	ZRPPA0088	3012 53
	ZRPPA0088	3367 59
	ZRPPA0088	3388 59
	ZRPPA0088	3742 65
	ZRPPA0088	3763 65
	ZRPPA0088	4167 72
	ZRPPA0088	4188 72
MASTER_IDOC_DISTRIBUTE.....	ZRPPA0088	926 17
REUSE_ALV_GRID_DISPLAY.....	ZRPPA0088	792 15
RP_CALC_DATE_IN_INTERVAL.....	ZRPPA0088	450 9
	ZRPPA0088	754 14
	ZRPPA0088	1075 20
ZRHPP_Q_PROFILE_READ.....	ZRPPA0088	473 9

DATABASE OPERATIONS

SELECT:		Line	Page
HRP1001.....	ZRPPA0088	766	14
PA0001.....	ZRPPA0088	2023	36
	ZRPPA0088	2618	46
	ZRPPA0088	2983	52
	ZRPPA0088	3359	58
	ZRPPA0088	3734	65
	ZRPPA0088	4159	72
PA0002.....	ZRPPA0088	623	12
	ZRPPA0088	1222	22
PA2001.....	ZRPPA0088	1355	24
	ZRPPA0088	2477	44
	ZRPPA0088	2935	51
	ZRPPA0088	3309	58
	ZRPPA0088	3686	64
	ZRPPA0088	4111	71
	ZRPPA0088	4489	77
PA2002.....	ZRPPA0088	1834	33
	ZRPPA0088	2414	43
	ZRPPA0088	2874	50
	ZRPPA0088	3240	56
	ZRPPA0088	4049	70
	ZRPPA0088	4427	76
PA2004.....	ZRPPA0088	1793	32
	ZRPPA0088	2435	43
	ZRPPA0088	2895	51
	ZRPPA0088	3645	63
	ZRPPA0088	4070	70
	ZRPPA0088	4448	77
PA2007.....	ZRPPA0088	1366	25
	ZRPPA0088	2234	40
	ZRPPA0088	2711	48
	ZRPPA0088	3076	54
	ZRPPA0088	3452	60
	ZRPPA0088	3827	66
	ZRPPA0088	4252	73
T001P.....	ZRPPA0088	2028	36
T503.....	ZRPPA0088	2032	36
T550A.....	ZRPPA0088	2134	38
	ZRPPA0088	2165	38
	ZRPPA0088	2181	39
T552A.....	ZRPPA0088	2176	39

DELETE:		Line	Page
ZPPA0088.....	ZRPPA0088	948	17

INTERNAL TABLE OPERATIONS

READ TABLE:

		Line	Page
IT_2003.....	ZRPPA0088	2163	38
LT_DAY_BEFORE.....	ZRPPA0088	1459	26
	ZRPPA0088	1556	28
	ZRPPA0088	1678	30
	ZRPPA0088	1751	31
	ZRPPA0088	2661	47
	ZRPPA0088	3026	53
	ZRPPA0088	3402	59
	ZRPPA0088	3777	65
	ZRPPA0088	4202	73
LT_EXP_0007.....	ZRPPA0088	1980	35
LT_EXP_2001.....	ZRPPA0088	4156	72
LT_EXP_2002.....	ZRPPA0088	3356	58
LT_EXP_2003.....	ZRPPA0088	2615	46
LT_EXP_2004.....	ZRPPA0088	2980	52
LT_EXP_2007.....	ZRPPA0088	3731	65
LT_EXP_NEW.....	ZRPPA0088	1090	20
	ZRPPA0088	1105	20
	ZRPPA0088	1120	20
	ZRPPA0088	1136	21
	ZRPPA0088	1153	21
	ZRPPA0088	1164	21
LT_FINAL.....	ZRPPA0088	1414	26
	ZRPPA0088	1945	35
LT_FINAL_AVAIL.....	ZRPPA0088	1433	26
LT_HRP1001.....	ZRPPA0088	673	13
	ZRPPA0088	1271	23
LT_LIST_PERNR.....	ZRPPA0088	1393	25
	ZRPPA0088	1419	26
LT_PA2001.....	ZRPPA0088	1385	25
LT_PA2002.....	ZRPPA0088	1846	33
LT_PA2004.....	ZRPPA0088	1805	32
LT_PA2007.....	ZRPPA0088	1471	26
	ZRPPA0088	1563	28
	ZRPPA0088	1684	30
LT_POSITION.....	ZRPPA0088	672	13
	ZRPPA0088	1270	23
LT_PROF.....	ZRPPA0088	491	10
	ZRPPA0088	507	10
	ZRPPA0088	523	10
	ZRPPA0088	538	11
	ZRPPA0088	555	11
	ZRPPA0088	566	11
LT_TEMP_TIME.....	ZRPPA0088	1511	27
	ZRPPA0088	1617	29
	ZRPPA0088	1739	31
LT_TIME_T.....	ZRPPA0088	1891	34
	ZRPPA0088	2363	42
	ZRPPA0088	2825	49
	ZRPPA0088	3191	56
	ZRPPA0088	3567	62
	ZRPPA0088	3969	69
	ZRPPA0088	4370	75
P0001.....	RP-PROVIDE-FROM-LAST	0	
P0008.....	RP-PROVIDE-FROM-LAST	0	

MODIFY:		Line	Page
LT_FINAL.....	ZRPPA0088	710	14
	ZRPPA0088	1291	23

DELETE:		Line	Page
LT_DATE.....	ZRPPA0088	2455	43
	ZRPPA0088	2914	51
	ZRPPA0088	3287	57
	ZRPPA0088	3664	64
	ZRPPA0088	4089	71
LT_FINAL.....	ZRPPA0088	4507	78
	ZRPPA0088	562	11
	ZRPPA0088	573	11
	ZRPPA0088	736	14
	ZRPPA0088	1160	21
	ZRPPA0088	1170	21
	ZRPPA0088	1302	24
	ZRPPA0088	2388	42
	ZRPPA0088	2424	43
	ZRPPA0088	2445	43
	ZRPPA0088	2487	44
	ZRPPA0088	2850	50
	ZRPPA0088	2884	50
	ZRPPA0088	2905	51
	ZRPPA0088	2944	52
	ZRPPA0088	3216	56
	ZRPPA0088	3250	57
	ZRPPA0088	3290	57
	ZRPPA0088	3318	58
	ZRPPA0088	3328	58
	ZRPPA0088	3592	62
	ZRPPA0088	3655	63
	ZRPPA0088	3667	64
	ZRPPA0088	3695	64
	ZRPPA0088	3705	64
	ZRPPA0088	4017	70
	ZRPPA0088	4059	70
	ZRPPA0088	4080	71
	ZRPPA0088	4092	71
	ZRPPA0088	4120	71
	ZRPPA0088	4130	71
	ZRPPA0088	4395	76
	ZRPPA0088	4437	77
	ZRPPA0088	4458	77
	ZRPPA0088	4498	78
	ZRPPA0088	4511	78
LT_HRP1001.....	ZRPPA0088	775	15
LT_LIST_PERNR.....	ZRPPA0088	738	14
LT_TIME_T.....	ZRPPA0088	1812	32
	ZRPPA0088	1853	33
	ZRPPA0088	2242	40
	ZRPPA0088	2719	48
	ZRPPA0088	3084	54
	ZRPPA0088	3460	60
	ZRPPA0088	3835	66
	ZRPPA0088	4260	74